

A Comparative Analysis of Hydraulic Stability Methods for Rock Protection of Subsea Cables in Nearshore Environments

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Preface

Summary

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1. Introduction

1.1 Background

Subsea power cables are a critical part of the modern energy system. They connect offshore wind farms to shore, link national grids across straits, and bring power to islands and coastal communities. As the energy transition drives a rapid growth in offshore wind and interconnectors, the number and the length of these cables increase every year, and so does the need to protect them.

A cable is most vulnerable where it approaches the coast. In this nearshore zone the water is shallow, the waves and the currents are strong, and the seabed is dynamic. Where possible the cable is buried, because burial is the most reliable and often the cheapest form of protection. Burial is not always possible. At a crossing of a third party cable or pipeline the cable must pass over the obstacle, and on a rocky seabed there is nothing to trench into. In these cases the cable rests on the seabed and is protected by a rock berm, a mound of rock built over the cable. These are among the most difficult situations to design for, because the cable is fully exposed to the hydrodynamic loading.

1.2 Problem statement

Designing a rock berm means choosing a rock size and a berm geometry that stay stable under the waves and the currents at that location. Several stability methods exist for this, but each was developed for a different type of structure and for a different range of conditions. Some describe the armour on a breakwater slope, some the toe of a breakwater, and some the protection of a flat bed under current and waves. None of them was developed specifically for a free standing rock berm over a cable in the nearshore zone.

This leads to two problems in practice. First, a single method is often applied to every situation, even when that situation lies outside the range for which the method is valid. The result cannot be trusted, and the error is not visible in the number that comes out. Second, to stay on the safe side, designers stack worst case assumptions on top of each other, for example by assuming that the waves and the current always act from the same direction. This produces rock berms that are larger than necessary, which means more rock, more vessel trips, higher cost and a larger environmental footprint. At the same time the knowledge needed to make these choices well rests with a small and shrinking group of specialists.

A further difficulty is that the methods do not describe damage in the same way. One method returns a rock size, another a number of displaced rocks, and another a deformation class. Because these damage measures are not the same physical quantity, the results of different methods cannot be compared directly, and there is no agreed way to translate one into another.

1.3 Objective

The objective of this thesis is to develop a design framework, and a computational tool that carries it out, for sizing rock berms that protect unbursed subsea cables in the nearshore zone. The framework selects the appropriate stability method for each situation and keeps every method within its own validity range. It sizes the rock for the smallest feasible rock at a controlled level of damage rather than for a stacked worst case. And it turns that rock size into a buildable berm geometry, presenting a small set of design alternatives so that the final choice can be made together with the client. In this way the design becomes faster, more consistent, more transparent and less wasteful of material.

1.4 Research questions

The main research question of this thesis is:

How can a rock berm for an unbursed subsea cable in the nearshore zone be designed so that the appropriate stability method is applied within its validity range, giving the best options for the rock size and buildable geometry at a controlled and acceptable level of damage?

This is supported by the following sub questions:

1. Which hydraulic stability methods are relevant for a free-standing rock berm on the seabed in the nearshore zone, and what are the validity ranges of each method?
2. How can we combine these different type of formulas to obtain the minimum rock size (one for breakwater, other one for 1 rock on the bottom of the sea bed, etc) and choose the right damage numbers.
3. How can the rock size and the berm geometry be coupled, with an accepted level of movement combined with a geometry that still covers the cable after it reshapes? And how do the rock density and the relative direction of the waves and the current affect the required rock berm?

1.5 Definition and area of interest

This thesis focuses on the nearshore zone, from the shoaling zone up to just past the breaker line, for the case where the cable rests on the seabed rather than being buried. Two seabed types are considered, sand at a crossing and bedrock. The very shallow inner surf zone with fully broken heavy waves is outside scope, because it lies beyond the validity of the methods used here. The scope and its boundaries are set out in more detail in Chapter 2 and Chapter 7.

1.6 Thesis outline

Chapter 2 describes the nearshore environment. Chapter 3 sets out the hydrodynamic forcing from currents and waves. Chapter 4 introduces the stability parameters and criteria. Chapter 5 reviews the rock stability assessment methods and compares them. Chapter 6 deals with scour, winnowing and other checks. Chapter 7 presents the design framework and the computational tool that form the core contribution of this thesis. Chapter 8 discusses the remaining knowledge gaps and open points.

Part A – Literature Study

2. The Nearshore Environment

2.1 Definition and area of interest

The coastal region is not uniform but a sequence of zones, each with its own conditions and its own degree of interaction between the waves and the seabed: the offshore zone, the nearshore (shoaling) zone, and the surf zone (Figure 2-1). The area of interest of this thesis is the nearshore zone, where many cable routes are laid and where the loading on a rock protection is at its most complex.

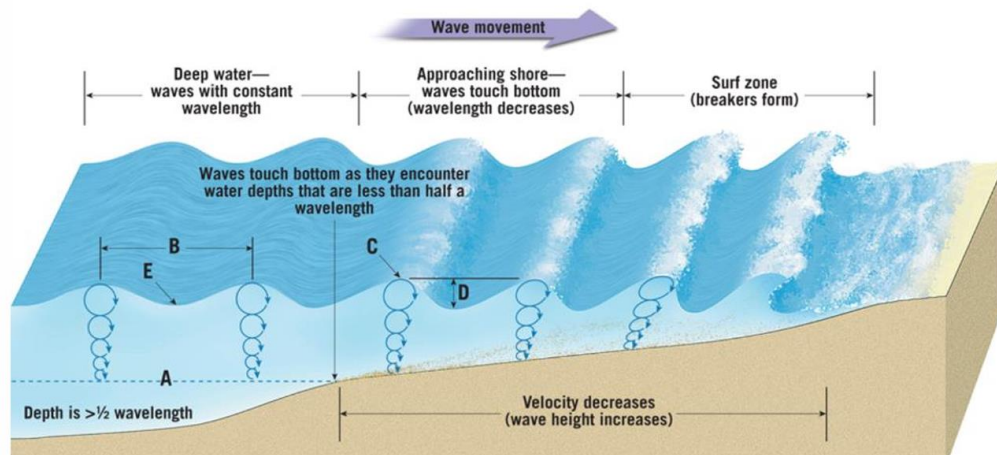


Figure 2.1: Nearshore Environment (Lecture Notes, Romano)

Offshore, the water is deep enough that the surface waves are hardly felt at the seabed. The flow near the bed is governed mainly by tidal and large-scale currents, and protection design there is comparatively straightforward (Schierack and Verhagen, 2019).

The nearshore, or shoaling, zone begins where the waves start to feel the seabed and extends to the point where they break consistently. As the waves move into shallower water they change shape, and their height first grows before it is limited by the depth (Schierack and Verhagen, 2019). Landward of this lies the surf zone, from the breaker line to the shoreline, where the waves have broken and the conditions are dominated by turbulence and by longshore currents, undertow and swash. Cable routes cannot always avoid the surf zone, for example at the landfall, but its extreme and highly variable loading would require a dedicated treatment and is therefore excluded from the scope. This study is limited to the nearshore zone, with attention to the transition near the breaker line where breaking can already begin.

What makes the nearshore the governing zone for design is that the water depth largely controls how strongly the waves load the seabed. In shallower water the waves are felt more strongly at the bed, and the depth also sets the largest wave that can exist before it breaks. Because of this, the required protection can differ strongly from one point to another along the same route, and even at a single point between high and low water. For this reason the assessment in this thesis is carried out point by point along the route, rather than with a single representative condition.

2.2 Bed Morphology and Composition

Burial is always the preferred protection, but this thesis deals with cables that lie on the seabed, where a rock berm provides the protection instead. Two situations govern the design: a mobile sandy seabed at crossings, and bedrock (Figures 2-2 and 2-3).

Sandy seabed (crossings). Where the cable crosses an existing third-party cable or pipeline, trenching is not permitted, so the cable is laid on the seabed and a rock berm provides the required cover. Fine to medium sand (D50 of 0.1 to 0.5 mm) is highly mobile under waves and current. DNV-RP-F109 (2021, Section 8.4) calls a bed mobile, or "live", when the bed shear stress from the flow exceeds the threshold for erosion of the grains present (Soulsby, 1997). This is the same combined wave-and-current loading that later sets the required rock size (Section 4.3), so a single physical reasoning governs both whether the sand moves and how large the protection rock must be. In this thesis the rock berm itself is sized first, for its own stability. The surrounding seabed, around and beneath the berm, is then checked separately in a later chapter, because the finer sand can be washed out from between and beneath the protection (winnowing) and eroded at its edges (scour) where the particles cannot withstand the applied shear. A correctly graded filter or underlayer beneath the armour limits this winnowing (Schierreck and Verhagen, 2019, Ch. 6; Chapter 6).

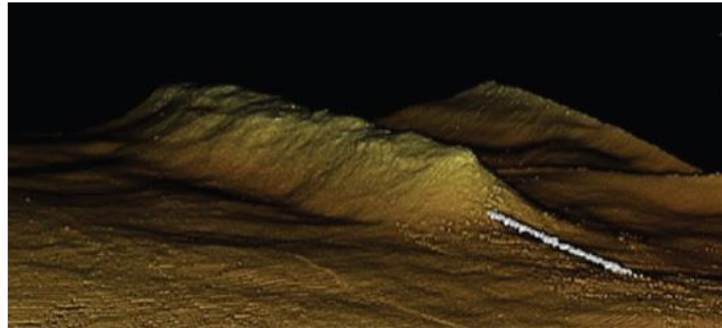


Figure 2.2

Bedrock. Where the seabed is bedrock, trenching is impossible and a berm is always required. Rocky, consolidated beds are common along steep coastlines, fjords and rocky headlands. There is little loose sediment to be lost, so winnowing and scour of the surrounding bed are only a minor concern, but the filter can settle into cracks and fissures over time and the uneven rock surface can leave unsupported gaps beneath the berm, both interface-stability concerns. Herrera and Medina (2015) show that steep rocky coasts, with slopes of 1/10 or steeper, behave very differently from gentle sandy slopes, so formulae developed for sand do not apply directly here.

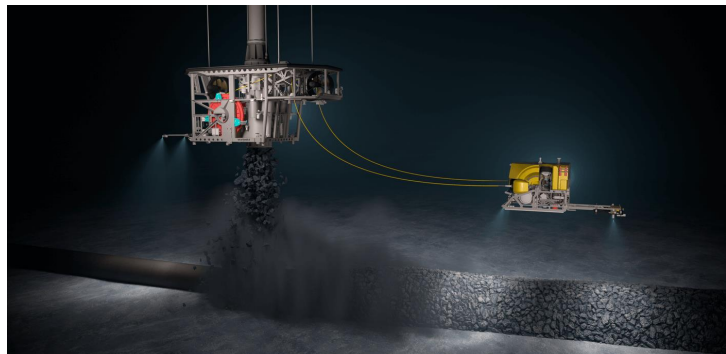


Figure 2.3: *Scour and Erosion IX – Keh-Chia (Ed.)*

3. Hydrodynamic Forcing

The hydraulic forcing is the set of hydrodynamic loads that act on the rock berm and that a stable rock must resist. Along a nearshore cable route this forcing comes from two sources, the currents and the waves, which together drive the flow close to the seabed. This near-bed flow exerts a drag force on each rock in the direction of the flow and a lift force directed away from the bed, and these destabilising forces are resisted by the submerged weight of the rock and the support of its neighbours. A rock stays in place as long as its weight and interlocking are enough to withstand the peak of the combined wave and current loading (Schierreck and Verhagen, 2019).

The waves are usually the dominant contribution in the nearshore, but a current, where present, adds to the load and its direction relative to the waves matters (Section 4.3). This chapter therefore describes both sources of forcing: the waves in Section 3.1 and the currents in Section 3.2. How the two are combined into a single near-bed loading, and how that loading is translated into a stability criterion, is set out in Chapter 4. Figure 3-1 illustrates this combined loading.

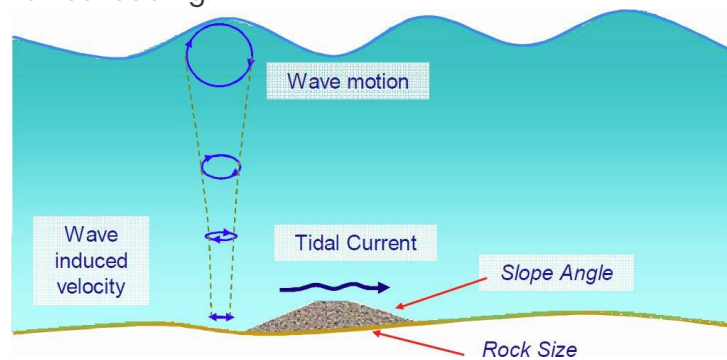


Figure 3.1: [Rock berm stability analysis cable crossing – Svasek Hydraulics Svasek Hydraulics](#)

3.1 Waves

Waves are usually the largest single hydrodynamic load in this thesis, because they generate an oscillatory motion throughout the water column and, in particular, at the seabed. Where a current is also present, the combined loading and the angle between the two govern stability (Section 3.3), but the wave motion is the starting point. The wave field is described with linear wave theory, from which both the near-bed velocity and the near-bed pressure follow, and its behaviour depends strongly on the water depth.

3.1.1 Wave data and propagation

The design wave conditions along a route are derived from measured, hindcast or reanalysis data at an offshore point and propagated to the site with a dedicated wave model. In this thesis these values are taken directly from the NKT route data, which give the significant wave height, the peak period and the wave direction per location, usually already chosen to represent a design return period (DNV-RP-F109, May 2021, Section 2.6). The offshore data represent deep-water conditions, while the site itself lies in intermediate to shallow water, so the derivation and propagation of the wave climate are not repeated here. This section focuses on translating the given wave conditions into the local loading at the seabed.

3.1.2 Wave parameters

A sea state is irregular and is described by bulk parameters. The significant wave height H_s is classically the mean of the highest one third of the waves ($H/3$), while spectral models report $H_{m0} = 4\sqrt{m_0}$, with m_0 the zeroth spectral moment. The two are nearly equal for a narrow-banded sea state in deep water, but they diverge on a shallow foreshore, where the highest waves break and the wave-height distribution changes shape (Schierack and Verhagen, 2019, p.149 and Appendix 7.7.2). That is exactly the nearshore condition of interest here. The time scale is given by the peak period T_p , at the peak of the spectrum, and by the spectral mean period $T_{m-1,0} = \frac{m-1}{m_0}$, which is more stable than T_p when the spectrum is broad. The wave direction is retained because it decides whether the wave attack is aligned with or oblique to the route, and how the waves interact with a current (Section 3.3).

3.1.3 Linear wave theory

Linear (Airy) wave theory gives a first-order description of the wave field and is the basis for the near-bed kinematics and pressure used throughout this thesis. The wave motion is represented by a velocity potential (Romano, Lecture Notes Coastal and Harbour Engineering, Section 2.3),

$$\varphi = \frac{a g}{\omega} \frac{\cosh(k(z+h))}{\cosh(kh)} \sin(kx - \omega t) \quad (3.1)$$

with $a = H_s/2$ the wave amplitude and z measured upward from the still water level, so $z = 0$ at the surface and $z = -h$ at the bed. The velocity field (Section 3.1.4) and the pressure (Section 3.1.5) follow directly from this potential. The wave number $k = 2\pi/L$ and the angular frequency $\omega = 2\pi/T$ are linked by the dispersion relation

$$\omega^2 = g k \tanh(kh) \quad (3.2)$$

which is solved iteratively for the local wavelength L at depth h (Schierack and Verhagen, 2019, p.151 and Table 7.1).

The behaviour of the wave depends on the relative depth h/L , or equivalently kh , and linear theory distinguishes three regimes (Schierack and Verhagen, 2019, Table 7.1),

- deep water, or offshore, for $h/L > 1/2$ ($kh > \pi$), where $\tanh(kh) \approx 1$ and the wave does not feel the bed,
- intermediate, or transitional, water for $1/20 < h/L < 1/2$, where the full hyperbolic functions are needed, and where most nearshore cable routes lie,
- shallow water for $h/L < 1/20$ ($kh < \pi/10$), where $\tanh(kh) \approx \sinh(kh) \approx kh$ and $\cosh(kh) \approx 1$, and the whole water column feels the wave.

The wave celerity $c = L/T$ follows from these limits, approaching $c = gT/(2\pi)$ in deep water and $c \approx \sqrt{gh}$ in shallow water, so the waves slow down and shorten as they shoal into intermediate and shallow water. Figure 3-2 shows the pressure field and the orbital motion in a periodic non-breaking wave that follow from this theory.

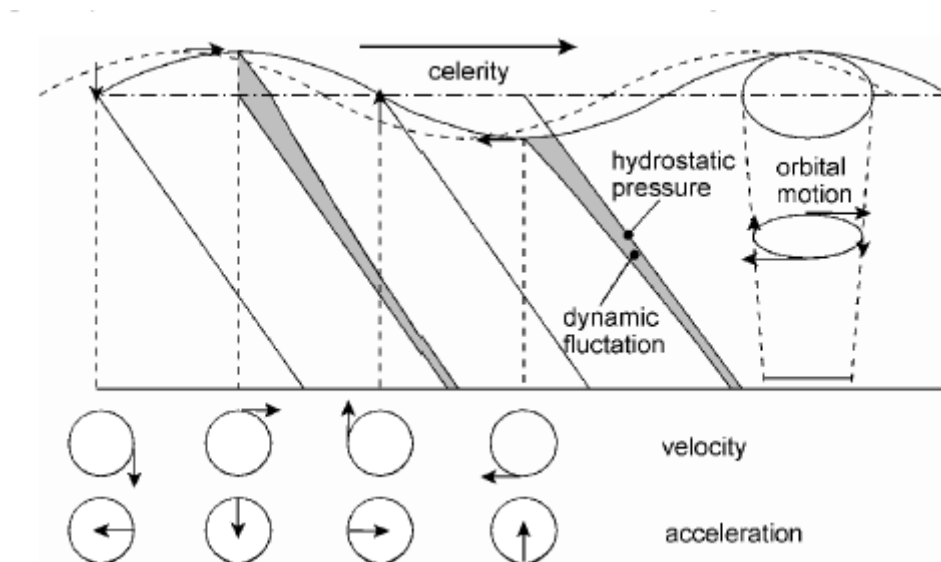


Figure 3.2: Water motion in a periodic non-breaking wave at a fixed instant, showing the celerity, the pressure as a hydrostatic part plus a dynamic fluctuation, and the orbital motion, which is circular near the surface and flattens to a horizontal back-and-forth motion at the bed (reproduced from Schierack and Verhagen, 2019, Fig. 7-3).

A fair question is whether linear theory may be applied in the nearshore at all, since the shallow and steep waves there fall outside its formal range of validity (Schierack and Verhagen, 2019, Fig. 7-4). Schierack and Verhagen (2019, p.152, Fig. 7-5) address this directly: tested on a mild-slope case well outside the permissible range, linear theory still reproduces the measured maximum bottom velocities remarkably well, although it does not reproduce surface quantities such as the surface profile and the surface velocity. Because this thesis uses linear theory only for the near-bed velocity and the near-bed pressure that load the protection, and not for any surface quantity, it is applied exactly in the regime where it holds. The theory does break down at and inside breaking, where the flow becomes turbulent and strongly nonlinear, so points in that regime are flagged

(Section 3.1.6) and assessed with the breaking-wave methods of Chapter 5. It also does not capture the onshore-offshore asymmetry of the near-bed velocity under steep, near-breaking waves, a known limitation that a nonlinear or CFD model would resolve.

Linear wave theory is used throughout this thesis because it gives a transparent, analytical link between the sea state and the near-bed loading, which is well suited to a comparative study of stability methods. For the detailed design of a specific project the nonlinear and turbulent effects would be resolved with a numerical model, for example a computational fluid dynamics solver such as OpenFOAM, which is beyond the scope of this thesis and is recommended as a follow-up step.

3.1.4 Near-bed orbital velocity

The velocity field follows from the potential by differentiation, $u = \partial\phi/\partial x$ and $w = \partial\phi/\partial z$ (Romano, Lecture Notes Coastal and Harbour Engineering, Section 2.3). The horizontal orbital velocity is

$$u(z) = \frac{a g k}{\omega} \frac{\cosh(k(z+h))}{\cosh(kh)} \cos(kx - \omega t) \quad (3.3)$$

and using the dispersion relation this is the form used in the rest of this thesis,

$$u(z) = \omega \frac{H_s}{2} \frac{\cosh(k(z+h))}{\sinh(kh)} \cos(kx - \omega t) \quad (3.4)$$

The horizontal orbital velocity is largest at the surface and decays with depth. At the bed ($z = -h$) the amplitude reduces to the peak near-bed orbital velocity.

$$\hat{u}_b = \omega \frac{H_s}{2 \sinh(kh)} = \frac{\pi H_s}{T_p \sinh(kh)} \quad (3.5)$$

and the orbital excursion, the back-and-forth distance a particle travels at the bed, is

$$a_b = \frac{H_s}{2 \sinh(kh)} \quad (3.6)$$

(Schierreck and Verhagen, 2019, Table 7.1; Romano, Section 2.3). This near-bed orbital velocity is the central wave quantity in this thesis. In the stability methods of Chapter 5 it appears as the near-bed peak orbital velocity U_w of Van Rijn (2019) and as the characteristic orbital velocity of Van Gent and Van der Werf (2014), and the excursion a_b is the orbital amplitude A_w of Van Rijn. Figure 3-3 shows a schematic of the near-bed orbital motion.

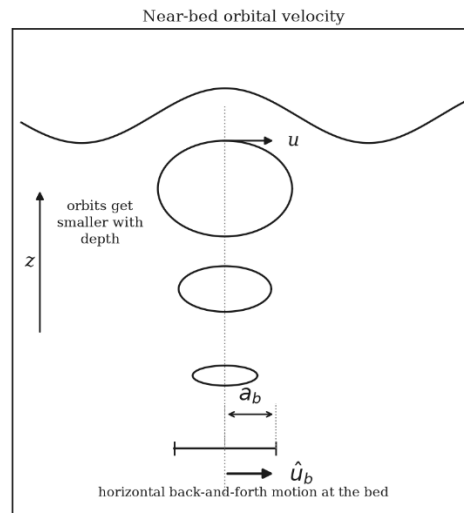


Figure 3.3: Near-bed orbital motion under a wave. The orbits flatten with depth, from nearly circular near the surface to a horizontal back-and-forth motion at the bed, with excursion amplitude a_b and peak near-bed orbital velocity $\hat{u}_b$. Own figure, after Schierreck and Verhagen (2019).

Its behaviour with depth follows the same three regimes. In deep water $\sinh(kh)$ is large, so $\bar{u}_b$ tends to zero and the bed hardly feels the wave. In intermediate water $\bar{u}_b$ grows as the wave shoals. In the shallow-water limit $\sinh(kh) \approx kh$, so $\bar{u}_b$ is large and almost uniform over the depth, and the entire column including the seabed feels the passing wave. For a given wave the near-bed velocity therefore increases as the wave moves into shallower water, which raises the load on the protection

The orbital velocity $\bar{u}_b$ acts just outside a thin near-bed boundary layer. Within this layer the flow is rotational and carries a high wave shear stress, while above it the flow follows the irrotational wave theory (Figure 3-3). Because this layer is thin compared with the water depth, the current velocity that is combined with the waves is taken at one metre above the bed, above this layer (Schierack and Verhagen, 2019, p.153-154, Fig. 7-7; Section 3.2.3).

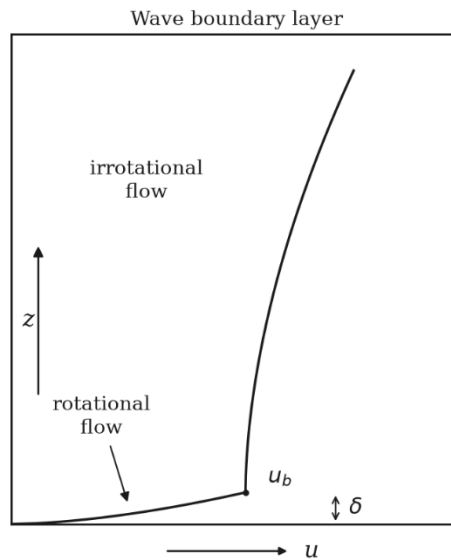


Figure 3.4: Velocity profile above the bed under a wave-crest. Outside a thin near-bed layer of thickness δ the flow is irrotational and follows the wave theory, while inside δ the flow is rotational and carries the high wave shear stress. u_b is the orbital velocity just outside the boundary layer (reproduced from Schierack and Verhagen, 2019, Fig. 7-7).

3.1.5 Excess dynamic pressure and effective near-bed loading

The same potential gives the pressure under a wave. The total pressure is the sum of the hydrostatic pressure $p^A = -\rho g z$ and an excess, or dynamic, pressure created by the wave motion. The excess pressure follows from the linearised unsteady Bernoulli equation (Romano, Lecture Notes Coastal and Harbour Engineering, Section 2.3),

$$p^+ = -\rho \frac{\partial \varphi}{\partial t} = \rho g a \frac{\cosh(k(z+h))}{\cosh(kh)} \cos(kx - \omega t) = \rho g \eta K_p \quad (3.7)$$

where $\eta = a \cos(kx - \omega t)$ is the surface elevation and K_p is the pressure response factor,

$$K_p(z) = \frac{\cosh(k(z+h))}{\cosh(kh)} \quad (3.8)$$

which sets how much of the surface wave signal reaches the level z . It is one at the surface ($K_p = 1$ at $z = 0$) and decreases towards the bed, where

$$K_{p,b} = \frac{1}{\cosh(kh)} \quad (3.9)$$

The two limits match the regimes and the figures. In deep water K_p decays exponentially away from the surface, so $K_{p,b}$ tends to zero and the bed hardly feels the wave. In shallow water K_p tends to one at every depth, so the excess pressure is almost constant over the depth, $p^+ \approx \pm \rho g a$, and the whole column including the bed feels almost the full wave (Figures 12 and 13 of the report). Because the excess pressure is proportional to the amplitude $a = H_s/2$, the near-bed response factor scales the surface wave height down to the value the bed experiences,

$$H_{s,b} = K_{p,b} H_s = \frac{H_s}{\cosh(kh)} \quad (3.10)$$

The velocity potential, the excess pressure and the response factor are standard linear wave theory (Romano, Lecture Notes Coastal and Harbour Engineering, Section 2.3; the same pressure appears in Schiereck and Verhagen, 2019, Table 7.1). The step of using the near-bed response factor as an effective wave height, and its application, are specific to this thesis (Chapter 7). The reduction $H_{s,b}$ is applied only to methods that use a surface wave height with no depth attenuation, such as Hudson. Methods that use the near-bed orbital velocity, such as Van Rijn and Van Gent and Van der Werf (2014), already contain the same attenuation through the $1/\sinh(kh)$ term and are not reduced again, and methods that use a depth-limited wave height at the structure, such as the shallow-water Van der Meer form and Herrera and Medina (2015), are used with the local wave height for which they were calibrated.

3.1.6 Depth-induced breaking

As the water shallows into the shallow-water regime the wave height eventually becomes limited by the depth and the wave breaks. For the significant wave height the depth limit is about $H_s/h \approx 0.4$ to 0.5 , lower than the individual-wave limit of $H_b/h \approx 0.78$ to 0.88 , because the highest waves in the sea state, roughly 1.5 to 2 times H_s , break first (Schierbeck and Verhagen, 2019, p.156, Section 7.3.1, Eq. 7.8). Breaking marks a change of loading regime, from smooth oscillatory motion to turbulent, impulsive forcing, and it therefore decides which stability method applies (Chapter 5). A point is flagged as breaking, or close to breaking, when H_s exceeds about half the local depth.

3.1.7 Relevance for the stability assessment

For the stability assessment the wave field is summarised by H_s , T_p and $T_{m-1,0}$, the wave direction, and, above all, the near-bed orbital velocity derived from them at the local depth. These feed the stability parameters of Chapter 4 and the methods of Chapter 5, and they are combined with the current wherever one is present (Section 3.3).

Linear wave theory (Section 3.1)	Symbol in the stability methods (Chapter 5)
near-bed orbital velocity $\hat{u}_b$	U_w (Van Rijn), characteristic orbital velocity (Van Gent and Van der Werf)
orbital excursion a_b	A_w (Van Rijn)
effective near-bed wave height $H_{s,b}$	H_s input to Hudson

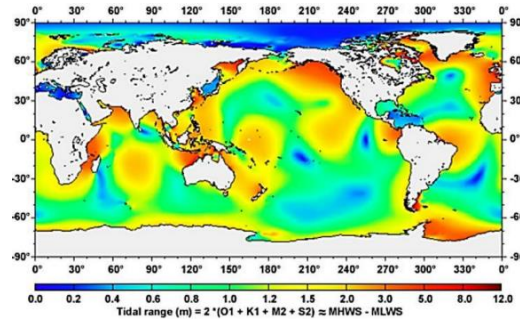
Table 3.1:

3.2 Currents

Currents are the steady component of the flow in the nearshore. In this thesis they matter for two reasons. Together with the waves they raise the load on the protection, and the angle between the current and the waves decides how severe that combined load is. Waves are generally the largest single load along the cable routes considered here (Section 3.1), but wherever a current is present in the NKT route data it is included. Of the six stability methods compared in this thesis, only the Van Rijn method uses a current, since the other methods were developed for wave loading alone. The tidal current is the governing steady contribution and is described first, followed by a short note on the secondary contributions and on how the current velocity is defined for use in the formulas.

3.2.1 Tidal currents

Tidal currents are generated by the periodic rise and fall of the water level under the gravitational attraction of the Moon and the Sun. The horizontal movement of water that follows this rise and fall is the tidal current, and its defining feature is that it reverses each tidal cycle, moving landward during the flood and seaward during the ebb. For a rock berm this reversal means the load arrives from both directions in turn. When the flood and ebb are unequal the tidal asymmetry produces a net sediment transport over time that can slowly rework the seabed around the berm, which is revisited as an extra stability check in a later chapter. The magnitude of the tidal current depends on the local tidal range, the water depth and the coastal geometry, and it ranges from modest at some sites to one of the governing loads at others. The tidal signal can be semidiurnal, diurnal or mixed, which sets how often the current reverses, but for the stability assessment it is the current magnitude and its direction relative to the waves that matter.



3.2.2 Other current contributions

Wind-driven currents can reinforce the tidal current during storms but are generally a secondary contribution in the nearshore. Density-driven currents, caused by differences in salinity or temperature, are a minor and site-dependent contribution. Where waves break at an angle to the shoreline they drive a longshore current in the surf zone (Battjes, 1974, in Schiereck and Verhagen, 2019), but the surf zone itself is outside the scope of this thesis (Chapter 2), so this mechanism is only relevant in the transition region near the breaker line. In the assessment these contributions are not treated separately, and the steady current is represented by the tidal current from the route data.

3.2.3 Velocity profile and the reference levels

A current has no single velocity. It follows a logarithmic profile that is zero at the bed and grows towards the surface (Figure 3-3),

$$u(z) = \frac{u_*}{\kappa} \ln\left(\frac{z}{z_0}\right) \quad (3.11)$$

where u_* is the shear velocity, κ the Von Karman constant and z_0 a roughness length set by the bed. Schiereck and Verhagen (2019) stress that rock stability depends on the near-bed velocity, the shape of this profile and the turbulence close to the bed, not on one bulk number alone, so any velocity used in a formula has to be tied to a stated height.

Two levels of the profile matter here. The NKT route data give the current at one meter above the bed. This is not a height above the current boundary layer, because for a steady current that boundary layer fills the whole water depth, so at one meter the flow is still well inside the sheared near-bed region. It is instead a fixed and repeatable datum that sits above the thin wave boundary layer and above the ill-defined roughness sublayer between the rocks. The stability methods do not all ask for the velocity at the same height. The Van Rijn method uses the depth-averaged current velocity in its current shear stress (Van Rijn, 2019), which for a logarithmic profile occurs at about 0.37 times the water depth above the bed and follows from

$$\bar{u} = \frac{u_*}{\kappa} \left[\ln\left(\frac{h}{z_0}\right) - 1 \right] \quad (3.12)$$

The one-meter datum and the depth-averaged velocity are linked through this same profile and the bed roughness, so the value each method requires is obtained from the one-meter datum standard by the logarithmic profile. This conversion is applied in the assessment (Chapter 5).

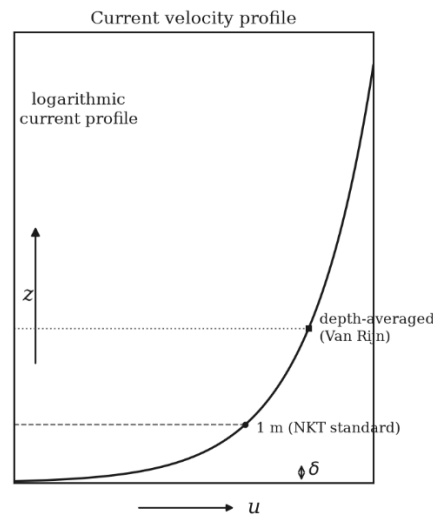


Figure 3.5: Logarithmic velocity profile of a steady current, with the reference levels used in this thesis: the NKT standard at one meter above the bed and the depth-averaged velocity at about 0.37 h that the Van Rijn method requires. The one-meter standard sits above the thin wave boundary layer and

the roughness sublayer, but still inside the current profile, which fills the whole depth. Own figure, based on Schiereck and Verhagen (2019).

The near-bed current defined here is combined with the wave motion in Section 3.3, and the selection of the design wave and current condition, including their return periods, is set out in Section 3.4.

- Schiereck & Verhagen (2019), Ch. 7 (Waves), Ch. 2 (Flow loads)
- Van Rijn (2019), Section 2 (Critical bed-shear stress, combined current and wave conditions, Eq. 5)
- DNV-RP-F109 (2021), Section 2.6 (Load combinations), Section 3.3 (Wave directionality and spreading)
- Deltares Handbook (2023), Ch. 2 (Scour prediction, general description of the scour process)

3.3 Combined waves and currents

In the nearshore a wave field and a steady current are usually present at the same time, so the bed feels both at once. The wave is represented by its near-bed orbital velocity $\hat{u}_b$ (Section 3.1) and the current by its velocity u_c (Section 3.2). This section describes how the two act together at the bed.

The velocities are combined, not the loads. When a wave and a current act together, their influence is combined by adding the current and the wave contributions vectorially at the level of the bed shear velocity, the velocity scale of the friction the flow exerts on the rocks (Bijker, 1967; Schiereck and Verhagen, 2019, p.155). Each flow defines its own bed shear velocity, from the current velocity u_c and from the orbital velocity $\hat{u}_b$, with C the Chezy coefficient and c_f the wave friction factor,

$$u_{*c} = \frac{\sqrt{g}}{C} u_c \quad (3.13)$$

$$u_{*w} = \sqrt{\frac{c_f}{2}} \hat{u}_b \sin(\omega t) \quad (3.14)$$

Added vectorially, the resulting bed shear velocity at the bed is

$$u_{*r} = \sqrt{\frac{g}{C^2} u_c^2 + \frac{c_f}{2} \hat{u}_b^2 \sin^2(\omega t) + 2 \frac{\sqrt{g}}{C} u_c \sqrt{\frac{c_f}{2}} \hat{u}_b \sin(\omega t) \sin(\varphi)} \quad (3.15)$$

and the resulting bed shear stress is

$$\tau_r = \rho u_{*r}^2 \quad (3.16)$$

where φ is the angle between the wave and the current direction, defined here as Schiereck does, so that $\varphi = 90^\circ$ corresponds to waves parallel to the current. The combination is made on the shear velocities and not on the shear stresses, because the shear stress grows with the square of the velocity and the thin wave boundary layer makes a wave produce a disproportionately large shear stress, so adding the two shear stresses directly is only correct when the wave and the current are perpendicular.

Working form for a bottom protection. For a bottom protection the maximum over the wave cycle governs the start of movement, because Schiereck and Verhagen (2019, p.155) state that the maximum value determines incipient motion. Setting $\sin(\omega t) = 1$, writing u_c and u_w for the two shear velocities of Equations 3.13 and 3.14, and using the ordinary geometric angle θ between the two directions ($\theta = 0$ for aligned, with $\theta = 90^\circ - \varphi$), Equation 3.15 reduces to the working form used in this thesis,

$$u_{*cw} = \sqrt{u_{*c}^2 + u_{*w}^2 + 2 u_{*c} u_{*w} \cos \theta} \quad (3.17)$$

with the combined bed shear stress $\tau_r = \rho u_{*cw}^2$.

The angle decides the result. The cross term $2 u^*c u^*w \cos \theta$ shows that the angle between the wave and the current controls the combined load. When the two are aligned ($\theta = 0$) the combined shear velocity is largest, $u^*c + u^*w$, and the shear stress reaches its maximum. When they are perpendicular ($\theta = 90^\circ$) the cross term vanishes and only the two separate contributions remain. Schiereck and Verhagen (2019, Figure 7-8) show the same result, and it reproduces their worked example on p.155, with a current-only shear of about 4 N/m^2 , a perpendicular sum of 51.4 N/m^2 , and an aligned maximum of 78.6 N/m^2 .

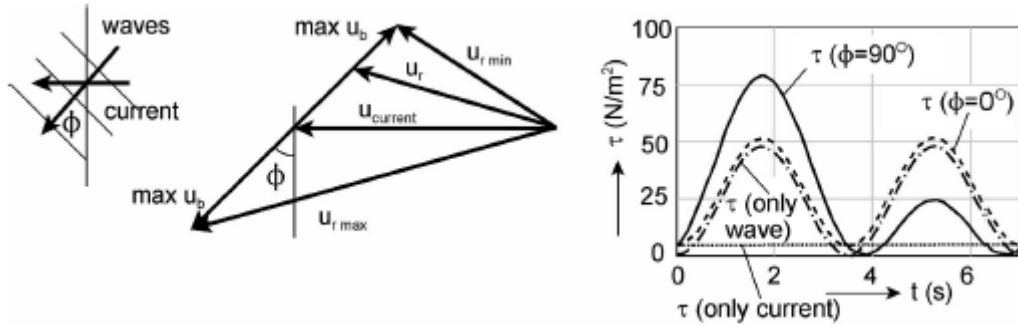


Figure 3.6: Combined wave-current action (Schiereck & Verhagen, 2019, Figure 7-8).

Consistency and use in this thesis. The two velocities are exactly the quantities used elsewhere in the thesis. The orbital velocity $\hat{u}_b$ is the near-bed peak orbital velocity of Section 3.1, which the Van Rijn method calls U_w , and the current velocity u_c is the depth-averaged current of Section 3.2. The bed friction is the same as well. Schiereck writes it with the Chezy coefficient C , which is linked to the Van Rijn current friction factor by $f_c = 8g/C^2$ (Van Rijn, 2019). In the Van Rijn method (Chapter 5) the current and wave contributions are added directly, which is exactly the perpendicular case of the combination above.

Because the NKT route data give both the wave and the current direction, this thesis keeps the angle term and uses the actual angle θ , which is treated as an optimisation lever. The combination will be implemented in the Van Rijn method, the only method of the six that uses a current.

3.4 Design conditions and return periods

The stability assessment is based on a single design condition defined by its return period. DNV-RP-F109 states that the characteristic load condition must reflect the most probable extreme response over the design period (DNV-RP-F109, May 2021, Section 2.6). For a permanent installation, this means using a 100-year event. This event has an annual exceedance probability of 1%, meaning there is a 1-in-100 chance of it happening in any given year. The extreme-value analysis behind these values is summarised in Appendix A, and the specific return-period values for the assessment are taken directly from the NKT route data.

Because an extreme wave and an extreme current rarely happen at the exact same time, their joint probability is usually unknown. To solve this, DNV-RP-F109 approximates the design condition by checking the most severe of two standard combinations:

- The 100-year wave combined with the 10-year current.
- The 10-year wave combined with the 100-year current.

Because the combined load depends non-linearly on the ratio between the wave and current velocities, a smaller current can sometimes be the governing factor. Therefore, lower current velocities are checked as well.

DNV-RP-F109 allows engineers to use the actual wave and current directions when they are available. If no direction data is available, the guideline forces a conservative assumption where waves and currents are omni-directional and perpendicular to the cable. This specific rule is the exact basis for treating the wave-current angle as a design variable in this thesis (Section 3.3), allowing us to avoid the over-design caused by the standard perpendicular default.

For each point along the route, both combinations are calculated, and the combination that requires the largest rock diameter is selected as the design condition for the methods in Chapter 5. Shorter, temporary installation phases follow the same approach but use lower return periods, such as seasonal 10-year and 1-year events or rely on a weather forecast for windows shorter than three days (DNV-RP-F109, May 2021, Section 2.6).

4. Stability Parameters & Criteria

4.1 Introduction

Chapter 3 described the hydraulic forcing as a hydrodynamic load at the bed, concluding with the combined near-bed shear velocity and the near-bed orbital velocity acting on the protection. This chapter turns that near-bed flow into a bed shear stress, establishes how a stationary rock responds to it, and defines the threshold at which rock movement is initiated. Rock stability is governed by the flow conditions experienced directly at the bed, rather than by depth-averaged flow characteristics.

In uniform, stationary flow a fully developed boundary layer occupies the entire water depth, producing a characteristic logarithmic velocity profile (Figure 4-1). Tidal currents, because of their long period relative to the boundary-layer development time, can reasonably be treated as uniform flow for baseline design purposes. Under oscillatory wave motion, in contrast, the flow direction reverses rapidly over short wave periods. Because the flow does not have time to propagate upwards into the water column, the wave boundary layer remains very thin, often only a few millimetres to centimetres above the bed. This confinement produces very high velocity gradients near the bed, so that for coarse rock the boundary is classified as fully rough ($Re^* > 70$). The viscous sublayer then disappears, exposing individual rocks directly to strong turbulent fluctuations and high local shear stresses.

This chapter sets out the theoretical flow-stability framework following Schiereck and Verhagen (2019, Chapter 3). Two core elements are established, the physical balance between the hydrodynamic load and the rock resistance, and the engineering choice of how much rock movement is accepted. Together these determine whether a rock protection is stable. The parameters and criteria reviewed below are the baseline from the literature, while their specific application in this study, and the values adopted, are set out in the methodology.

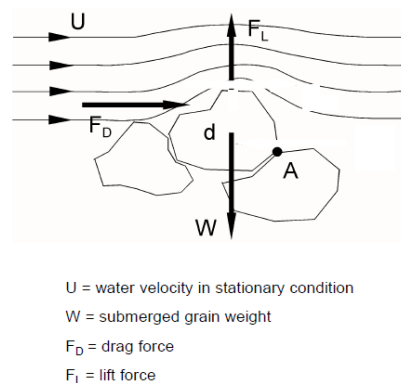


Figure 4.1: Steady-flow boundary layer with the logarithmic velocity profile (left) and the distinction between hydraulically smooth and rough boundaries (right). For coarse rock the boundary is fully rough, exposing rocks directly to turbulent eddies. Adapted from Romano (lecture notes).

4.2 Forces on a rock and the shields parameter

A rock resting on the seabed experiences the moving water as a combination of three hydrodynamic forces, a drag force acting parallel to the flow, a lift force generated by the pressure drop as the flow curves over the rock, and a shear force distributed across the surrounding grain bed (Schiereck and Verhagen, 2019). All three forces share the same fundamental form,

$$F = \frac{1}{2} C \rho_w u^2 A \quad (4.1)$$

so the total hydrodynamic load scales with the square of the near-bed velocity and with the exposed area, that is with $u^2 d^2$.

To find the threshold of instability, a moment equilibrium is evaluated around the rocks pivot point A (Figure 4-2). At incipient motion the overturning moment of the drag (F_D) and lift (F_L) forces is balanced by the restoring moment of the submerged weight (W),

$$(F_D + F_L) a = W b \quad (4.2)$$

where a and b are the hydrodynamic and gravitational lever arms set by the packing and orientation of the grains. Substituting the force expressions gives the classical dimensionless stability relation,

$$\frac{U^2}{\Delta g d} = \frac{2b}{a c} = f(Re) \quad (4.3)$$

where $c = \sqrt{(c_D^2 + c_L^2)}$ is the combined force coefficient. For the coarse rock considered in this thesis the particle Reynolds number lies in the fully turbulent Newton range (about $10^3 < Re < 2 \cdot 10^5$), where the drag coefficient is roughly constant. Treating c_D as constant then shows that the critical velocity for the start of motion scales with $\sqrt{(\Delta g d)}$, which is the empirical grain-stability form of Izbash (1932).

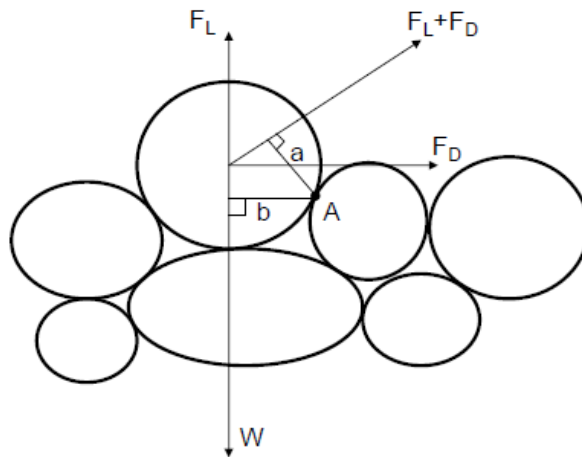


Figure 4.2: Moment equilibrium around the grain pivot point A. The overturning torque from the drag (F_D) and lift (F_L) forces with lever arm a is balanced by the restoring torque of the submerged weight (W) with lever arm b . Adapted from Romano (lecture notes).

Shields (1936) expressed the load as a bed shear stress rather than a velocity, and made it dimensionless with the submerged weight of the grain, giving the Shields parameter θ (Schierck and Verhagen, 2019),

$$\theta = \frac{\tau_b}{(\rho_s - \rho_w) g d} = \frac{\tau_b}{\rho_w g \Delta d} \quad (4.4)$$

The bed shear stress τ_b is the combined wave-current shear of Section 3.3, built from the near-bed orbital velocity of Section 3.1 and the current of Section 3.2, $\tau_b = \rho u^* c w^2$. On the empirical Shields curve (Figure 4-3) the threshold of motion becomes independent of the boundary Reynolds number under fully turbulent conditions ($Re^* > 400$). For coarse material this critical value is constant and is conventionally taken as $\theta_{cr} \approx 0.055$ (Schierck and Verhagen, 2019). The slightly lower baseline $\theta_{cr,shields} = 0.05$ used in the Van Rijn calibration is discussed in Section 4.4.

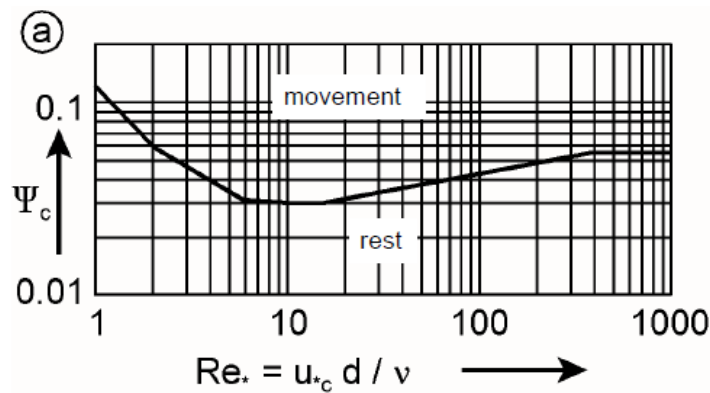


Figure 4.3: The classical Shields diagram, the critical dimensionless bed-shear stress as a function of the boundary (particle) Reynolds number. For coarse material at high Reynolds number the threshold settles at a constant value between 0.05 and 0.055. Adapted from Romano (lecture notes).

4.3 Stability criterion

The Shields parameter is used in two ways in design (Schiereck and Verhagen, 2019). Computed with a threshold shear stress it is the critical value (θ_{cr}) that marks the start of motion. Computed with the actual design load it is a mobility parameter, the acting value (θ) produced by the design condition,

$$\theta = \frac{\tau_b}{\rho_w g \Delta D_{50}} \quad (4.5)$$

where the characteristic grain size is the median rock diameter D_{50} . A rock protection is stable when the acting Shields parameter does not exceed the critical threshold,

$$\theta \leq \theta_{cr} \quad (4.6)$$

and rearranging fixes the smallest stable rock that resists the design load,

$$D_{50} \geq \frac{\tau_b}{\theta_{cr} \rho_w g \Delta} \quad (4.7)$$

In reality there is no sharp boundary between stable and unstable. Because the near-bed flow is highly turbulent and individual rocks differ in shape, nesting and orientation, both the acting load and the resistance behave as statistical distributions rather than single values, so some movement can occur well below the mean critical value. The critical Shields value is an empirically fitted number with a scatter of about fifteen per cent (Van der Wal, 1990).

For this reason θ_{cr} is best read as a damage marker, a lower value meaning a strict state of no movement and a higher value accepting limited, controlled transport (Schiereck and Verhagen, 2019).

4.4 Failure spectrum and the damage factor r

Because a strict threshold is absent, the failure of a rock protection occurs across a continuous spectrum rather than as a single fracture event. Delft Hydraulics described this as a progression from no movement, through occasional and frequent grain displacement, to general transport (Schiereck and Verhagen, 2019). For an offshore rock berm protecting a subsea cable three regimes matter,

1. threshold of motion, where individual rocks begin to rock or vibrate in their pockets, or occasionally leave their position (θ close to θ_{cr}),
2. transport and reshaping, where moderate movement lets the profile deform into a flatter, hydrodynamically smoother and more stable shape while the cable stays fully covered,
3. continuous erosion, where the layer is progressively worked away and the cable is exposed and can fail.

To translate the accepted amount of movement into a design criterion, Van Rijn (2019) introduces a calibration or reduction coefficient r , applied directly to the theoretical Shields threshold,

$$\theta_{cr} = r \cdot \theta_{cr,shields} \quad (4.8)$$

with $\theta_{cr,shields} = 0.05$ and r between 0.4 and 1.0. The factor r scales the theoretical threshold to match the required performance. A low r , about 0.4 to 0.5, gives a conservative, static design in which almost no rock moves under the design condition, requiring a larger rock. A high r , towards 1.0, deliberately accepts limited transport, allowing a smaller rock. These choices define two distinct engineering philosophies,

- a static, or statically stable, design, in which the rock is sized so that the acting load stays below the chosen threshold ($\theta \leq \theta_{cr}$) and the profile is assumed not to change, and
- a dynamic, or reshaping, design, in which limited movement is accepted on purpose and the change of the profile is predicted with a deformation model (Chapter 5).

Both use the same Shields balance of Section 4.3 and differ only in the accepted damage, and therefore in the value of r . The value of r adopted for this study, and the way the criterion is applied, are set in the methodology.

4.5 Non-uniform flow and turbulence

The Shields criterion is derived for uniform flow over a flat, unbounded bed (Schierack and Verhagen, 2019, Section 3.2). A free-standing rock berm is not that situation. The berm protrudes above the seabed, so the flow accelerates over its crest and separates in its lee, which raises the local velocity and, above all, the turbulence (Schierack and Verhagen, 2019, Section 3.4). Schierack expresses this non-uniformity with a velocity factor on the load, close to one in accelerating flow but larger than one in the decelerating, separated flow behind an obstacle.

In the Van Rijn method this extra velocity and turbulence around the structure is carried by an enhancement factor γ_{str} on the acting velocity (Van Rijn, 2019),

$$\gamma_{str} = 1 + n \gamma \quad (4.9)$$

with n between 2 and 3 and γ between 0.2 and 0.3, which gives values between 1 and 2. The value derived for slender structures such as monopiles sits at the high end of this range. Which value is appropriate for a free-standing, low and wide rock berm is a parameter of this study and is set, with the other design parameters, in the methodology. A dedicated turbulence-based stability parameter also exists (Hofland, 2005), which replaces the mean bed shear stress by the peak of the turbulent fluctuations, but it is beyond the scope of this thesis.

Tell that γ_{str} depends on more factors, in methodology we try to optimize this parameter for every part in the nearshore.

4.6 From criteria to methods

The Shields balance and the damage factor of this chapter form the general basis of the static design approach. In Chapter 5 each stability method expresses the acting load and the accepted damage in its own measure, the damage factor r for Van Rijn, the damage level S_d for Van der Meer, the number of displaced rocks N_{od} for the toe methods, and a deformation class for the Deltares reshaping model. The near-bed load itself comes from the wave and current forcing of Chapter 3, through the combined bed shear stress for the shear-based methods and through the near-bed orbital velocity or the effective wave height for the wave-height-based methods. Placing these load and damage measures side by side, on the common basis established here, is the aim of the comparison in Chapter 5.

5. Rock Stability Assessment Methods

5.1 Introduction

Rock protection for a subsea cable can be sized with several established stability methods, but these methods were not developed for the same object. They differ in the structure they describe, in the hydraulic loading they were calibrated for, and in how they define damage. This chapter reviews the methods that are relevant to a free-standing rock berm and groups them by the structure they were made for:

- rubble-mound (breakwater) models, which size the armour on a wave-exposed slope: Hudson (1953/1959), Van der Meer (1988) and the shallow-water forms of Van der Meer and Van Gent;
- toe berm models, which size the small berm at the foot of an armour slope: Van Gent and Van der Werf (2014) and Herrera and Medina (2015);
- the flat-bed model, which sizes protection on a horizontal or mildly sloping bed under current and waves: Van Rijn (2019), built on the Shields concept of Chapter 4;
- the rock berm deformation model, developed for offshore cable-cover berms: Deltares HaSPro / RoBeD (2023).

Each method returns a different quantity (a required rock size, a damage number, or a deformation class) with a different damage definition. For this reason the methods are first described one by one, and then set side by side in the comparative summary of Section 5.7, which states which method suits which situation, which damage criterion it uses, and which inputs each formula needs. A widely used industry method, Thusyanthan et al. (2013), is discussed separately in Section 5.6 and explained why it is not carried into the comparison.

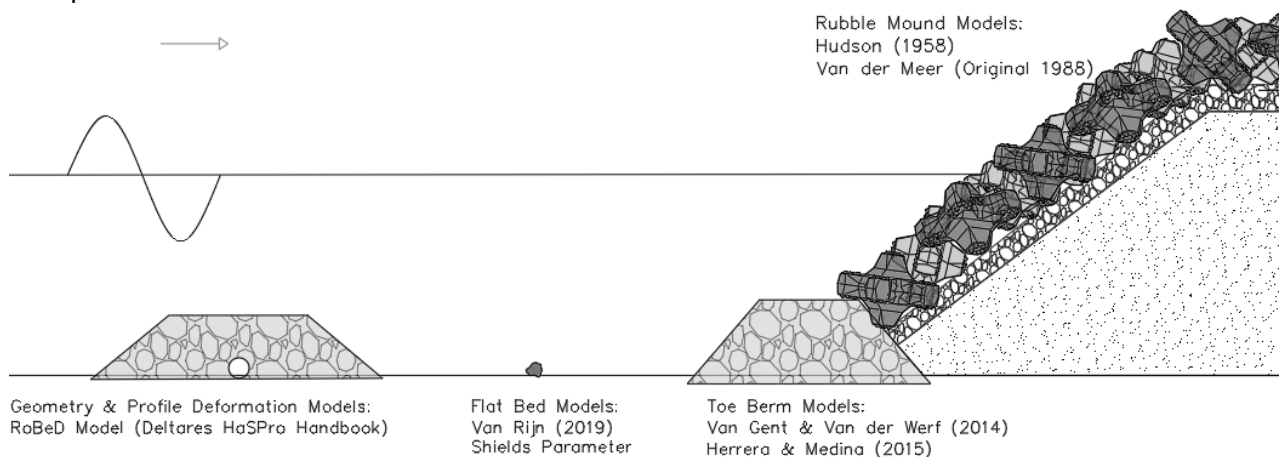


Figure 5.1: overview of the stability methods positioned on a single cross-section (rubble-mound slope, toe berm, flat-bed rock, rock berm). (W.Pol drawing)

5.2 Rubble-mound (breakwater) models

The rubble-mound models size the armour stone on a wave-exposed slope. Hudson is the original empirical formula, and Van der Meer extended it for random waves with an explicit damage level, later adapted for shallow, breaking water. All were developed for a structure slope and for waves only, without a current, so their role here is a slope-based, wave-only reference rather than a method for the free-standing cable berm.

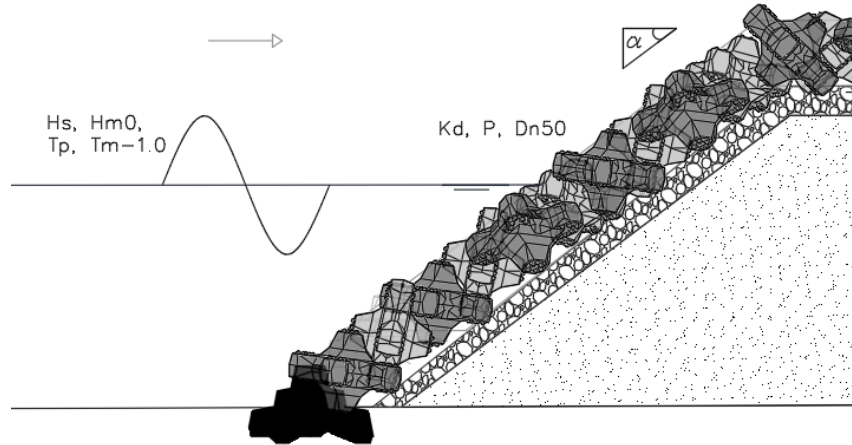


Figure 5.2: Rubble-mound armour slope with the governing inputs of Hudson and Van der Meer (H_s , H_{m0} , T_p , $T_m - 1.0$, K_d , P , D_{n50} , slope angle α). (W.Pol drawing)

5.2.1 Hudson (1953/1959)

Background and physics. Hudson was derived from regular-wave model tests at the US Waterways Experiment Station and is the classical first estimate for armour stability on a slope (Schierreck and Verhagen, 2019, Ch. 8). It balances the wave force on a stone against its submerged weight through a single stability coefficient KD. In nominal-diameter form it reads

$$D_{n50} = \frac{H_s}{\Delta (K_D \cot \alpha)^{1/3}} \quad (5.1)$$

KD is an empirical coefficient that absorbs the accepted damage, the armour shape and the placement method. For rough, angular natural rock in a double layer it is about 2 for waves breaking on the slope and about 4 for non-breaking waves (CEM, 2003, Table 5.22).

Symbol	Meaning	How to determine
Dn50	required nominal rock diameter (m)	output
Hs	significant wave height (m)	route data
Δ	relative buoyant density $(\rho_s - \rho_w)/\rho_w$	$\rho_s = 2650$ or 3000 kg/m^3 (variable)
KD	stability coefficient	table value, about 2 (breaking) to 4 (non-breaking)
$\cot \alpha$	slope gradient	structure or berm geometry

Table 5.1

Boundary conditions and validity. Slope armour on a permeable core, non-overtopped, with waves that do not break at the toe. The formula is valid for about $1.5 < \cot \alpha < 4$ (Schierreck and Verhagen, 2019); outside this range $\cot \alpha$ is insufficient to describe the equilibrium on the slope and the predicted stability becomes unrealistic. It applies from deep to shallow water but not to very shallow, depth-limited conditions. There is no wave period, no number of waves, no permeability parameter and no current in the formula.

Limitations for a cable berm. KD fixes an implicit damage level of roughly five per cent without an explicit scale, so the result cannot be moved to another damage level. For an impermeable core extra corrections to KD are needed. Because it represents a slope and not a bed or a free-standing berm, its use here is limited to a quick, conservative first estimate.

5.2.2 Van der Meer (1988)

Background and physics. Van der Meer was derived from a large series of random-wave model tests at Delft Hydraulics and extends Hudson to irregular waves (Schierack and Verhagen, 2019, Ch. 8). It adds the wave period, the number of waves N, and the notional permeability P, and splits the response into a plunging and a surging regime through the Iribarren number. The two forms are

$$\frac{H_s}{\Delta D_{n50}} = 6.2 P^{0.18} \left(\frac{S_d}{\sqrt{N}} \right)^{0.2} \xi_m^{-0.5} \quad (5.2)$$

$$\frac{H_s}{\Delta D_{n50}} = 1.0 P^{-0.13} \left(\frac{S_d}{\sqrt{N}} \right)^{0.2} \sqrt{\cot \alpha} \xi_m^P \quad (5.3)$$

The damage is explicit through the damage number S_d , the eroded cross-sectional area normalised by the square of the nominal diameter,

$$S_d = \frac{A_e}{D_{n50}^2} \quad (5.4)$$

with the start of damage at S_d of about 2 to 3 and failure at higher, slope-dependent values of roughly 8 to 17 (Schierack and Verhagen, 2019). The Iribarren number uses the mean period,

$$\xi_m = \frac{\tan \alpha}{\sqrt{\frac{H_s}{L_m}}} \quad (5.5)$$

Symbol	Meaning	How to determine
S_d	damage number A_e/D_{n50}^2	chosen damage level (start 2 to 3, failure 8 to 17)
N	number of waves in the storm	storm duration divided by T_m
P	notional permeability	0.1 (impermeable core) to 0.6 (homogeneous)
ξ_m	Iribarren number	slope and wave steepness at T_m
L_m	wavelength at T_m	linear wave theory

Table 5.2

Boundary conditions and validity. Deep water ($h > 3 H_{s,toe}$); slope armour; $\cot \alpha = 1.5$ to 6; notional permeability $P = 0.1$ to 0.6; N up to 7500 (equilibrium damage); wave steepness ξ_m up to about 0.06; Iribarren number roughly 0.7 to 7; relative density Δ about 1 to 2.1; separate plunging and surging branches; waves only, no current. The coefficients are fitted averages with relative standard deviations of 6.5 per cent (plunging) and 8 per cent (surging).

Limitations for a cable berm. Like Hudson, Van der Meer was developed and validated for a breakwater armour slope, not for a berm on a horizontal or mildly sloping bed, and it contains no current. It is valid in the deeper part of the domain and drops out as the waves approach breaking. Its relevance here is as a wave-only, slope-based reference.

5.2.3 Shallow water: why the Rayleigh assumption fails

Van der Meer (1988) is written in terms of the significant wave height H_s , but most of the damage is done by the highest waves, for instance $H_{2\%}$. The tests behind the formula were carried out mainly in deep water, where the individual wave heights follow a Rayleigh distribution and the ratio $H_{2\%}/H_s$ is fixed at about 1.4. That factor 1.4 is therefore built implicitly into the fitted coefficients $c_{pl} = 6.2$ and $c_s = 1.0$. The water can be treated as deep for this purpose while $h > 3 H_{s,toe}$, and becomes very shallow once $H_{s,toe} < 0.7 H_s$. Why Rayleigh no longer applies. As the waves propagate into shallow water the highest waves become depth-limited and break first. This clips the upper tail of the wave-height distribution, so the distribution is no longer Rayleigh and the fixed ratio $H_{2\%}/H_s = 1.4$ is no longer valid. Using the deep-water formula in this range designs for a heavier wave than actually reaches the structure, which is safe but oversizes the rock. The short-term wave heights are then described by the composite Weibull distribution of Battjes and Groenendijk (2000),

$$F(H) = 1 - \exp\left[-\left(\frac{H}{H_1}\right)^2\right] \quad \text{for } H \leq H_{tr} \quad (5.6)$$

$$F(H) = 1 - \exp\left[-\left(\frac{H}{H_2}\right)^{3.6}\right] \quad \text{for } H \geq H_{tr} \quad (5.7)$$

which is Rayleigh-shaped (exponent 2) up to a transition wave height H_{tr} and Weibull-shaped (exponent 3.6) above it, so the over-predicted high waves of the Rayleigh tail are truncated. The practical procedure is: first compute H_{tr} from the water depth and the foreshore slope, then compute H_{rms} from H_{m0} , and finally read the wave-height statistics (including $H_{2\%}$) from the table in the Rock Manual (Box 4.4, p. 359) or the fuller table in Battjes and Groenendijk (2000).

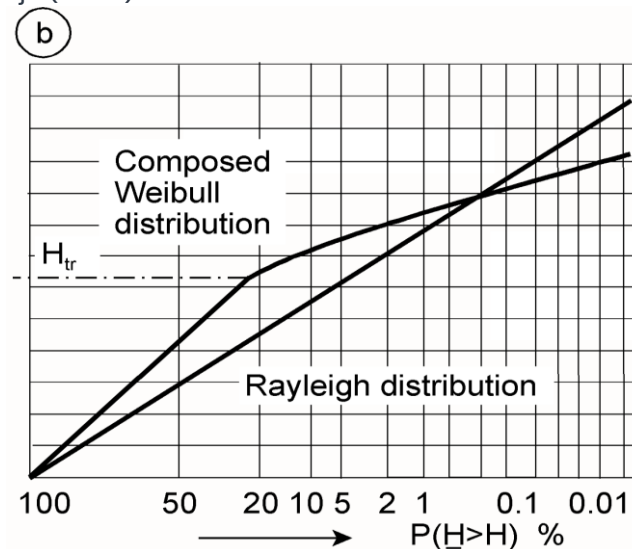


Figure 5.3: composite Weibull distribution versus the Rayleigh distribution on a wave-height exceedance plot; above the transition height H_{tr} the composite Weibull tail falls below Rayleigh, so the highest waves are chopped off. After the shallow-water stability. (Lectures TU Delft, Prof. Antonini)

5.2.4 Shallow-water stability formulas

Four formulas extend slope-armour stability into shallow, depth-limited water. Two are shallow-water forms of Van der Meer, and two are separate formulas by Van Gent that avoid the wave period and the $H_{2\%}/H_s$ ratio. All use the same damage number S_d as Van der Meer (1988).

1. Rewritten Van der Meer. The first Van der Meer form keeps the ratio $H_{2\%}/H_s$ explicit. Pulling the implicit 1.4 out of the deep-water coefficients gives $cpl = 6.2 * 1.4 = 8.7$ and $cs = 1.0 * 1.4 = 1.4$, with the Iribarren number ξ_m on the mean period unchanged,

$$\frac{H_s}{\Delta D_{n50}} = 8.7 \frac{H_s}{H_{2\%}} P^{0.18} \left(\frac{S_d}{\sqrt{N}} \right)^{0.2} \xi_m^{-0.5} \quad (5.8)$$

$$\frac{H_s}{\Delta D_{n50}} = 1.4 \frac{H_s}{H_{2\%}} P^{-0.13} \left(\frac{S_d}{\sqrt{N}} \right)^{0.2} \sqrt{\cot \alpha} \xi_m^P \quad (5.9)$$

In shallow water the actual $H_{2\%}/H_s$ is computed from the composite Weibull distribution above rather than taken as 1.4. Using H_s with the original deep-water formula instead is conservative.

2. Modified Van der Meer (Van Gent et al., 2004). The Rock Manual gives a second Van der Meer form that recalibrates the coefficients on the original dataset together with new shallow-water data and uses the spectral period $T_{m-1,0}$ instead of the mean period,

$$\frac{H_s}{\Delta D_{n50}} = 8.4 P^{0.18} \left(\frac{S_d}{\sqrt{N}} \right)^{0.2} \frac{H_s}{H_{2\%}} \xi_{s-1,0}^{-0.5} \quad (5.10)$$

$$\frac{H_s}{\Delta D_{n50}} = 1.3 P^{-0.13} \left(\frac{S_d}{\sqrt{N}} \right)^{0.2} \frac{H_s}{H_{2\%}} \sqrt{\cot \alpha} \xi_{s-1,0}^P \quad (5.11)$$

with $cpl = 8.4$ ($\sigma = 0.7$) and $cs = 1.3$ ($\sigma = 0.15$), valid for $0.25 \leq H_{s0}/h \leq 1.5$. These are the two shallow-water Van der Meer forms.

3. Van Gent et al. (2003). The Rock Manual also gives a separate formula by Van Gent et al. (2003) for the case where there is no or insufficient information about the wave period. It is independent of the wave period $T_{m-1,0}$ and independent of the ratio $H_{2\%}/H_s$, and it accounts for the structure permeability through the ratio of the core to the armour diameter,

$$\frac{H_s}{\Delta D_{n50}} = 1.75 \sqrt{\cot \alpha} \left(1 + \frac{D_{n50,core}}{D_{n50}} \right) \left(\frac{S_d}{\sqrt{N}} \right)^{0.2} \quad (5.12)$$

with a standard deviation $\sigma = 0.10$ for rock slopes with a permeable core and $\sigma = 0.12$ for an impermeable core; the same expression applies to plunging and surging conditions, valid for $1.25 < h/H_{s,toe} < 3$.

4. Modified Van Gent (Scaravaglione et al., 2025). Scaravaglione et al. (2025) proposed the most recent formula, aimed at very and extremely shallow water. It uses the spectral wave height H_{m0} instead of H_s , keeps a small dependence on the wave steepness $s_{m-1,0}$, and remains independent of the ratio $H_{2\%}/H_s$,

$$\frac{H_{m0}}{\Delta D_{n50}} = 3.3 \sqrt{\cot \alpha} \left(1 + \frac{D_{n50,core}}{D_{n50}} \right) s_{m-1,0}^{0.1} \left(\frac{S_d}{\sqrt{N}} \right)^{0.2} \quad (5.13)$$

with a standard deviation $\sigma = 0.36$ for rock slopes with a permeable core, valid for $0.4 \leq H_{m0,deep}/h \leq 5$.

Boundary conditions and validity. All four are slope-armour, wave-only formulas for depth-limited water. The Van der Meer forms use S_d and the spectral or mean period; the modified Van der Meer is valid for $0.25 \leq H_{s0}/h \leq 1.5$. The Van Gent (2003) form is period-independent and ratio-independent, valid for $1.25 < h/H_{s,toe} < 3$, with permeability through $D_{n50,core}/D_{n50}$. The modified Van Gent (Scaravaglione et al., 2025) form extends this to very and extremely shallow water, valid for $0.4 \leq H_{m0,deep}/h \leq 5$.

Relevance for a cable berm. Because these forms extend slope-armour stability into the depth-limited range, they remain usable inside the breaker zone and can be placed alongside the toe-berm method of Herrera and Medina (2015) for the same breaking condition. They remain slope-based, wave-only methods.

5.3 Toe berm models

The toe berm models size the small rubble berm at the foot of a rubble-mound structure that supports the armour slope. They are the closest in form to a free-standing rock berm, because they describe a low berm on the bed and use the damage number N_{od} , the number of displaced rocks within a strip as wide as one D_{n50} , as an explicit damage criterion. Two methods are considered: Van Gent and Van der Werf (2014) for general depths, and Herrera and Medina (2015) for very shallow water on a steep, breaking foreshore.

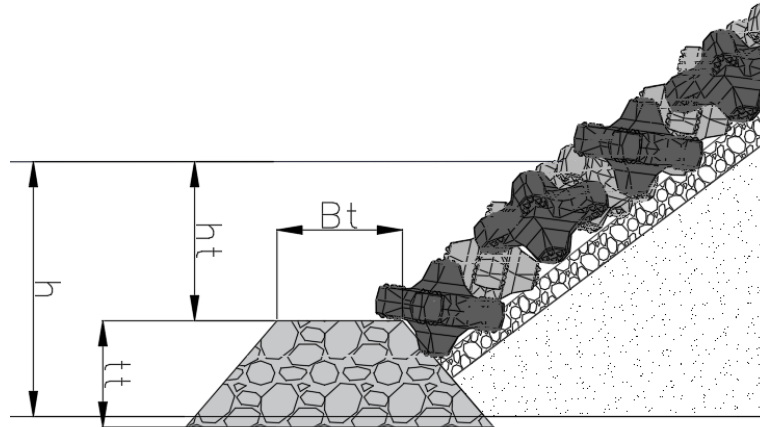


Figure 5.4: toe berm geometry defining the water depth h , the depth above the toe h_t , the berm width B_t and the berm thickness t_t . (W.Pol drawing)

5.3.1 Van Gent and Van der Werf (2014)

Background and physics. Van Gent and Van der Werf (2014) derived a toe stability formula from physical model tests with rock toe structures. Rather than the wave height directly, it uses a characteristic near-bed orbital velocity $\hat{u}$ based on H_s , the spectral period $T_{m-1,0}$ and the water depth above the toe h_t ,

$$\hat{u} = \frac{H_s}{T_{m-1,0} \sinh(k h_t)} \quad (5.14)$$

with k the wave number from the linear dispersion relation at $T_{m-1,0}$. This velocity is obtained from the same linear wave theory adopted throughout this thesis (Chapter 3): it uses the linear dispersion and the sinh depth-attenuation. It is, however, a deliberately simplified characteristic velocity rather than the physical peak near-bed orbital velocity $\hat{u}_b = \pi H_s / (T_p \sinh kh)$ used in Chapter 3 and by Van Rijn (Section 5.4). Van Gent and Van der Werf omit the factor π and use the spectral period $T_{m-1,0}$ and the depth above the toe h_t ; the resulting scale difference is absorbed into the calibration coefficient 0.32.

For this reason $\hat{u}$ must be evaluated with Van Gent's own definition and coefficient, and the peak orbital velocity of Chapter 3 cannot be substituted into this formula without recalibration. Damage increases with wave height, wave period, berm width and berm thickness, and decreases with water depth above the toe and with rock size. These influences give a damage formula and its inverse sizing form,

$$N_{od} = 0.032 \left(\frac{t_t}{H_s} \right) \left(\frac{B_t}{H_s} \right)^{0.3} \left(\frac{H_s}{D_{n50}} \right)^3 \left(\frac{\hat{u}}{\sqrt{g H_s}} \right) \quad (5.15)$$

$$D_{n50} = 0.32 \frac{H_s}{N_{od}^{1/3}} \left(\frac{B_t}{H_s} \right)^{0.1} \left(\frac{t_t}{H_s} \right)^{1/3} \left(\frac{\hat{u}}{\sqrt{g H_s}} \right)^{1/3} \quad (5.16)$$

Symbol	Meaning	How to determine
Dn50	nominal rock diameter (m)	output (or input for Nod)
Nod	damage number (displaced rocks per Dn50 width)	chosen design value
Hs	significant wave height (m)	route data
$\hat{u}$	characteristic near-bed orbital velocity (m/s)	linear wave theory, from H_s , $T_{m-1,0}$, h_t
h_t	water depth above the toe or berm (m)	bathymetry
Bt, tt	berm width and thickness (m)	geometry input
$T_{m-1,0}$	spectral wave period (s)	wave spectrum / NKT data

Table 5.3

Boundary conditions and validity. Armour slopes of 1:1.5 and 1:2 above the toe; toe within 10 to 30 per cent of the water depth in front of the structure ($h_t/h = 0.10$ to 0.30); water depth in front of the toe of 1.2 to 4.5 times the wave height ($h/H_s = 1.2$ to 4.5); wave steepness $sp = 0.015$ to 0.04 ; waves only, no current; non-overtopped; tested on a fixed bed with no toe scour. A roughness factor of about 1.5 is available for smoother armour slopes above the toe.

Limitations for a cable berm. It was developed for a toe that supports a breakwater armour slope, with wave run-down from that slope acting on the toe, so it assumes a slope above the berm that a free-standing cable berm does not have. The formula does not contain the relative density Δ , so it cannot by itself distinguish between normal and high-density rock, which is an explicit design variable in this thesis. Its applicability here is indicative rather than direct.

5.3.2 Herrera and Medina (2015)

Background and physics. Herrera and Medina (2015) developed a formula specifically for toe berms in very shallow water on a steep sea bottom (slope $m = 1/10$) under breaking waves, the situation in which they show that existing formulas, including Van Gent and Van der Werf (2014), do not perform reliably. Their formula (Eq. 12) predicts Nod for a standard berm geometry ($B_t = 3 D_{n50}$, $t_t = 2 D_{n50}$) directly from the deep-water wave conditions and the water depth at the toe,

$$N_{od} = \left(\frac{(H_{s0} \cdot L_{0p})^{1/2}}{\Delta D_{n50}} - 5.5 \right) \left(\left(-0.2 \frac{h_s}{D_{n50}} + 1.4 \right) \exp \left(0.25 \frac{h_s}{D_{n50}} - 0.65 \right) \right)^{1/0.15} \quad (5.17)$$

with $L_{0p} = g T_p^2 / 2\pi$ the deep-water wavelength at the peak period, accompanied by 90 per cent confidence intervals. Following the CIRIA Rock Manual (2007), three damage levels are used: Nod = 0.5 (start of damage), 2.0 (moderate, berm still supporting the armour) and 4.0 (failure).

Symbol	Meaning	How to determine
Nod	damage number (output of Eq. 12)	computed for a candidate D_{n50}
H_{s0}	deep-water significant wave height (m)	offshore wave data
L_{0p}	deep-water wavelength $g T_p^2 / 2\pi$ (m)	from T_p
h_s	water depth at the toe (m)	bathymetry
Δ	relative buoyant density $(\rho_s - \rho_w) / \rho_w$	$\rho_s = 2650$ or 3000 kg/m^3 (variable)
D_{n50}	nominal rock diameter, used as input	candidate value to test

Table 5.4

Boundary conditions and validity. Standard berm geometry ($B_t = 3 D_{n50}$, $t_t = 2 D_{n50}$) and slope $m = 1/10$ only; $-0.15 < h_s/H_{s0} < 1.5$ (negative meaning an emerged berm); $-0.5 < h_s/D_{n50} < 5.01$; wave steepness $0.02 < s_{0p} < 0.07$ for Eq. 12 (the experimental data span 0.008 to 0.08); $Nod \leq 4.0$; breaking waves only, no current. The formula returns Nod for a chosen D_{n50} , so it is used to explore the trade-off rather than to size the rock in one step.

Limitations for a cable berm. The geometry is fixed and only waves are considered. Because it is one of the few methods explicitly validated for breaking waves on a steep bottom, it is directly relevant to the bedrock case in this thesis, in particular where the cable route lies within or close to the wave-breaking zone of Chapter 2.

5.4 Flat-bed model (Van Rijn, 2019)

Van Rijn (2019) extends the classical Shields approach of Chapter 4 to bed and berm protection with large rocks, cobbles, boulders and rocks in the range $D_{50} = 0.03$ to 3 m, under a current, waves, or both together. Among the core sources it is the only method that explicitly covers combined wave-and-current loading on a horizontal or mildly sloping bed or berm rather than a sloped armour layer, which makes it the most directly applicable to the free-standing rock berm.

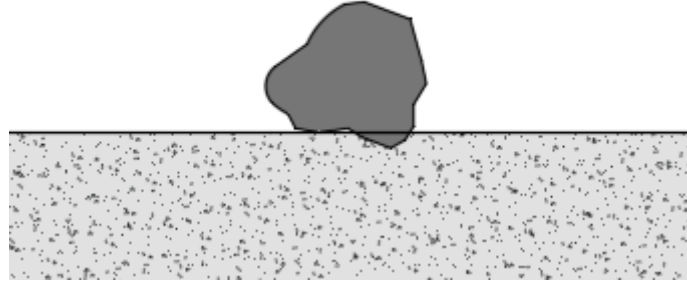


Figure 5.5

Background and physics. The method keeps the Shields form but replaces the fixed threshold by a design threshold, the critical mobility number,

$$\theta_{cr} = r \cdot \theta_{cr,shields} \quad (5.18)$$

with $\theta_{cr,shields} = 0.05$ for coarse material and the reduction coefficient r in the range 0.4 to 1.0. The coefficient r sets the accepted degree of movement, so it supports both a static design (no movement) and a dynamic design (some accepted movement), which is directly useful for the objective to avoid overdesign.

Stability criterion and required size. The rock is stable while the acting bed-shear stress stays below the critical value,

$$\theta = \frac{\tau_{b,cw}}{\Delta \rho_w g D_{50}} \leq \theta_{cr} \quad (5.19)$$

$$D_{50} = \frac{\tau_{b,cw}}{\Delta \rho_w g \theta_{cr}} \quad (5.20)$$

Van Rijn is written in terms of the sieve diameter D_{50} rather than the nominal diameter D_{n50} used by the slope methods ($D_{50} \approx 1.2 D_{n50}$). The current- and wave-related bed-shear stresses are added,

$$\tau_{b,cw} = \tau_{b,c} + \tau_{b,w} \quad (5.21)$$

each following the quadratic friction law, enhanced by the structure factor γ_{str} ,

$$\tau_{b,c} = 0.125 \rho_w f_c (\gamma_{str} u_c)^2 \quad (5.22)$$

$$\tau_{b,w} = 0.25 \rho_w f_w (\gamma_{str} U_w)^2 \quad (5.23)$$

The wave contribution uses the near-bed orbital velocity and amplitude from linear wave theory,

$$U_w = \frac{\pi H_s}{T_p \sinh\left(\frac{2\pi h}{L_s}\right)} \quad (5.24)$$

$$A_w = \left(\frac{0.5 T_p}{\pi}\right) U_w \quad (5.25)$$

The rough-regime friction factors, and Van Rijn's explicit power-law approximations that allow a closed-form solution, are

$$f_c = \frac{0.24}{[\log(12h/k_s)]^2} \quad (5.26)$$

$$f_w = \exp\{-6 + 5.2 \left(\frac{A_w}{k_s}\right)^{-0.19}\} \quad (5.27)$$

$$f_{c,a} = 0.11 \left(\frac{h}{k_s}\right)^{-0.3}, \quad f_{w,a} = 0.1 \left(\frac{A_w}{k_s}\right)^{-0.3} \quad (5.28)$$

Symbol	Meaning	How to determine
D50	required median rock diameter (m)	output
$\theta_{cr,shields, r}$	critical Shields (0.05); reduction coefficient	$r = 0.4$ (static) to 1.0
u_c	depth-averaged current velocity (m/s)	current data (Chapter 3), if applicable
U_w, A_w	near-bed peak orbital velocity and amplitude	linear wave theory, from H_s, T_p, h
h, L_s	water depth and wavelength (m)	bathymetry; linear wave theory
k_s	Nikuradse bed roughness (m)	$k_s = \alpha D50, \alpha = 1.5$ to 2
γ_{str}	velocity and turbulence enhancement	1 without structure, up to about 2 with a berm

Table 5.5

Boundary conditions and validity. D50 = 0.03 to 3 m; horizontal or mild slope; relative roughness $h/(\alpha D50)$ and $A_w/(\alpha D50)$ in the practical range 10 to 300; static design via $\theta_{cr} = r \cdot 0.05$, dynamic design via the bed-load equation of Cheng (2002); the sizing form shown is the horizontal-bed case, the full formula adds longitudinal and lateral slope factors $K\alpha_1, K\alpha_2$ (equal to one for a horizontal bed). Field calibration is comparatively sparse for the largest sizes.

Limitations for a cable berm. The structure factor γ_{str} carries a real uncertainty, since a value between one and two changes the shear stress by up to a factor of four ($\gamma_{str} = 1 + n \cdot \gamma$ with $n = 2$ to 3, $\gamma = 0.2$ to 0.3). The critical Shields parameter itself scatters between about 0.01 and 0.12 in the data, so the method is best used with r as an explicit, honestly reported damage choice rather than a single deterministic value.

5.5 Rock berm deformation model (Deltares HaSPro/ RoBeD, 2023)

The Deltares HaSPro Handbook (2023) is the newest guideline for offshore scour and cable protection. Of the core sources, it is the only one made especially for rock berms that cover cables. It looks at three things a berm must do. External stability means the berm resists the wave and current loads. Interface stability means the sand under the berm is not washed out through the rock layer. Flexibility means the berm can follow changes in the seabed and edge scour without losing its function (Handbook, Section 5.1).

For external stability the handbook gives a rock berm deformation model, called RoBeD, that predicts how much the berm will deform. It uses four deformation classes, from no deformation (class 0) to full wash-away (class 3). In class 3 the rock moves even on the flat seabed. The main number is the mobility. It is the acting Shields number divided by the critical Shields number (Handbook, Appendix I),

$$MOB = \frac{\theta}{\theta_{cr}} \quad (5.29)$$

The critical Shields number follows from the parameterization of the Shields curve by Soulsby (1997, Eq. A.6, which contains the factor 1.2 on the dimensionless grain size), and the roughness is taken as $ks = 2.5 D_{50}$. The recommended threshold is $MOB_{thresh} = 0.95$. A mobility of about 1 corresponds to a Shields number of roughly 0.06, which is why the complete flattening of the profile (class 3) coincides with Shields numbers larger than about 0.06. This ties the RoBeD classes directly to the Shields framework of Chapter 4 and to Van Rijn (Section 5.4).

A berm has a flat top and side slopes, so the mobility is checked at three places, each against the same threshold. On a side slope the rock moves more easily, because gravity already pulls it down the slope, so less load is needed to start the motion. The handbook uses a slope factor for this. It depends on the angle of repose of the rock and the slope angle (Handbook, Eq. 35, Appendix A.7),

$$Ks = \sin(\varphi - \alpha) / \sin(\varphi) \quad (5.30)$$

Here φ is the angle of repose of the rock, usually about 40 degrees, and α is the slope angle. The mobility on the slope is the flat-bed mobility divided by this factor. So the mobility on the side slope $MOB_{top;slope}$ is higher than the mobility on the flat top $MOB_{top;flat}$, and both are compared with the mobility of the seabed around the berm, $MOB_{bed;flat}$. A steeper side slope gives a smaller slope factor, so a higher mobility on the slope. This is why the side of a berm starts to move before the flat top. The classes look like sudden steps, but that is only because one fixed threshold is used to mark a motion that in reality starts slowly. So the class is a simple label put on a gradual process.

The four classes follow from which position first exceeds the threshold (Handbook, Section 5.3, Figure 5.4).

Class	Mobility criterion	Behaviour of the berm	Cover on the cable
0	$MOB_{top}; slope \leq MOB_{thresh}$	No movement. The berm keeps its installed trapezoidal shape and no deformation is expected.	Kept, no change
1	$MOB_{top}; slope > MOB_{thresh}$ $MOB_{top}; flat \leq MOB_{thresh}$	The side slopes reshape toward the parabolic equilibrium profile. Lowering of the top is not needed, but regression of the slope can lower the top if the berm is too narrow.	Kept if the berm is wide enough
2	$MOB_{top}; flat > MOB_{thresh}$	The flat top also becomes mobile, so a lowering of the top is required to reach a stable profile.	At risk, the crest drops
3	$MOB_{bed}; flat > MOB_{thresh}$ ($\theta > \text{about } 0.06$)	Even the surrounding seabed is mobile, so the profile will eventually erode away.	Lost

Table 5.6

The handbook gives two ways to work with the mobility. The first is a set of fixed mobility-limit tables (Section 4.4.1.4, Tables 4.8 to 4.12). These were made for scour protection around monopiles, one table for each pile diameter from 6 to 14 m. In these tables the deformation is scaled by the pile diameter and the Keulegan-Carpenter number, and it is driven by the flow speed-up and the horseshoe vortex around the pile. A rock berm over a cable has no pile and no pile diameter to scale with. So the pile method is not only less exact for a berm, it does not even apply to it.

The second way is the Rock Berm Deformation model (RoBeD, Section 5.3). It was tested against physical model tests in the Delta Flume and the Atlantic Basin. For these tests the handbook advises $MOB_{thresh} = 0.95$, because that value fitted the rock-berm data best. So for a rock berm, RoBeD with $MOB_{thresh} = 0.95$ is the right method.

Next to the classes, the handbook gives a clear way to find the needed Top of Product Cover (TOPC). This is the rock cover you must install so that, after the berm reshapes in the design storm, the cover left over the cable is still above the Minimum Top of Cover (MTOC). Rock berms reshape into a parabola shape. The ratio of its bottom width to its height depends on the Shields number (Roulund et al., 2018, Handbook Eq. 20),

$$\frac{B_{berm}}{h_{berm}} = f(\theta) = 6000 \theta^2 + 50 \theta + 2.4 \quad (5.31)$$

Here θ is the combined wave-current Shields number for the design load at the seabed,

$$\theta = \frac{\tau_{cw}}{(\rho_s - \rho_w) g d_{50}} \quad (5.32)$$

where τ_{cw} is the combined wave-current bed shear stress. Note that the fit of Eq. 20 was made only from data with clear deformation. The handbook says the design equations (Eq. 20 to 26) hold for berms that reshape, so for high Shields numbers. For lower Shields numbers the class check of Section 5.3 gives extra insight.

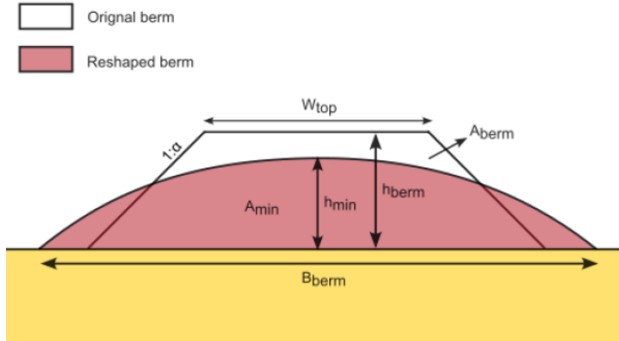


Figure 5.6: installed (trapezoidal) rock berm and reshaped (parabolic) profile, defining W_TOP , the side slope $1:\alpha$, the berm heights h_berm and h_min , the reshaped bottom width B_berm and the areas A_berm and A_min (Deltares, Handbook of Scour and Cable Protection Methods)

The installed and minimum berm heights relate to the cover requirements through a cable-position offset, denoted Δ_{cable} to avoid confusion with the relative density Δ ,

$$h_{berm} = TOPC + \Delta_{cable}, \quad h_{min} = MTOC + \Delta_{cable} \quad (5.33)$$

with $\Delta_{cable} = D_{cable}$ for a surface-laid cable, or $\Delta_{cable} = Z_{DOL} - Z_{SBL}$ for a buried or free-spanning cable. The installed trapezoidal area and the minimum reshaped parabolic area are

$$A_{berm} = W_{TOP} h_{berm} + \alpha h_{berm}^2 \quad (5.34)$$

$$A_{min} = \frac{2}{3} B_{berm} h_{min} \quad (5.35)$$

Setting the installed area equal to the minimum reshaped area gives a quadratic for h_{berm} , of which the maximum root and the relations above yield the installed berm height and the required TOPC (Handbook Eq. 26),

$$h_{berm} = 0.5 \left[\sqrt{\left(\frac{W_{TOP}}{\alpha} \right)^2 + \frac{4}{\alpha} \frac{2}{3} f(\theta) h_{min}^2} - \frac{W_{TOP}}{\alpha} \right] \quad (5.36)$$

$$TOPC = 0.5 \left[\sqrt{\left(\frac{W_{TOP}}{\alpha} \right)^2 + \frac{4}{\alpha} \frac{2}{3} f(\theta) (MTOC + \Delta_{cable})^2} - \frac{W_{TOP}}{\alpha} \right] - \Delta_{cable} \quad (5.37)$$

RoBeD works with waves and current together. The mobility follows the DNV Recommended Practice (DNV-RP-0618). The combined bed shear stress is found by a vector addition of the current shear τ_c and the wave shear τ_w (Handbook, Appendix A.6, Eq. A.31 and A.32). This is different from Van Rijn (Section 5.4), who adds the two in a straight (linear) way as $\tau_{b,cw} = \tau_{b,c} + \tau_{b,w}$. The current shear comes from the depth-averaged current, and the wave shear from the near-bed orbital speed found with linear wave theory. If there is only current, $\tau_w = 0$ and θ becomes the current Shields number. The waves and current are taken as perpendicular to the berm, which the handbook calls the normal case.

RoBeD is made for the reshaping case with a stable parabola shape. The mobility uses $_{MOB} = \theta / \theta_{cr}$, with the critical Shields number from the Soulsby (1997) fit and the roughness $k_s = 2.5 D_{50}$. The method is set for offshore conditions, and the waves and current are taken as perpendicular to the berm. Above a Shields number of about 0.06 the berm flattens fully, which is the upper limit of the method.

RoBeD covers berm shape, interface stability and flexibility together, and it is the only core source made especially for cable-cover berms. So it is the most useful design reference for the stability framework of this thesis, together with Van Rijn (2019) for the stone-size calculation itself.

Symbol	Meaning	How it is determined
θ	Combined wave-current Shields number at the seabed	$\theta = \tau_{cw} / [(\rho_s - \rho_w) g d_{50}]$
$_{MOB}$	Mobility number, with threshold 0.95	$_{MOB} = \theta / \theta_{cr}$
Ks	Slope factor on the flat-bed mobility	$Ks = \sin(\varphi - \alpha) / \sin(\varphi)$
$_{TOPC}, M_{TOC}$	Installed and minimum top of cover (m)	TOPC is an output. MTOC is a project requirement
W_{TOP}, α	Top width and side-slope parameter 1: α	Geometry input
$\Delta_{cable}, D_{cable}$	Cable-position offset and cable diameter (m)	D_{cable} for a surface-laid cable, $Z_{DOL} - Z_{SBL}$ for a buried cable
φ	Angle of repose of the rock	Typically 40 degrees

Table 5.7

5.6 A method considered but not used: Thusyanthan et al. (2013)

Thusyanthan, Jegandan and Robert (2013) is a paper on rock berm stability that is often used by companies and regulators in industry practice. Because it is well known, it is discussed here, even though it is not one of the core methods carried into the comparison of Section 5.7.

The paper gives two methods for rock berm stability, one based on Soulsby (1997) and one based on the CIRIA Rock Manual (2007), plus design charts for a flat seabed and an amplification factor for berms with 1:3 or 1:4 side slopes. The method is left out of the comparison for three reasons.

No scalable damage level. Both methods are based on the threshold of motion, not on a parameter that lets the designer choose how much movement is acceptable, such as S_d (Van der Meer), r (Van Rijn) or Nod (Van Gent and Van der Werf, Herrera and Medina). The method is therefore calibrated for little to no movement, which tends to give larger, more conservative rock sizes.

Breaking waves not properly addressed. The word breaking only appears as a label on the design charts, with no discussion of how the formulae behave near or after breaking, or up to what point they stay valid.

Inconsistent wave-current direction. The Soulsby-based method includes an angle between the wave and current direction in its formula (Eq. 11), but the design charts do not vary or report this angle, so they likely assume the most conservative case of waves and current in the same direction. The CIRIA-based method has no directional term at all.

Together, these points mean the method is more conservative than the methods used in this thesis, fitting its purpose as a quick, safe check for early-stage engineering rather than a detailed design tool. Since avoiding unnecessary over-design is a key goal of this thesis (Chapter 1), the method is discussed for completeness but not used further.

5.7 Comparative summary

The methods are set side by side below. The table states the situation each method was made for, the inputs it needs, the quantity it returns, its damage criterion, its boundary conditions and validity, and how it fits a nearshore cable berm. All entries are verified against the cited sources. The applicability indications follow from the methods' own stated validity. The actual method selection and calculations for this thesis are made in the methodology, not here.

Method	Situation it was made for	Required inputs	Output	Damage criterion	Boundary conditions / validity	Fit for cable berm
Hudson (1953/59)	Slope armour, permeable-core breakwater, regular waves	Hs, KD, Δ , $\cot \alpha$	Dn50 / W50	Implicit fixed (~5%), no scale	Slope armour, non-overtopped, permeable core; KD = 2 breaking / 4 non-breaking (rock); $1.5 < \cot \alpha < 4$ (Schierreck); deep to shallow but not very shallow; no Tp, N, P or current	Low: first estimate only
Van der Meer (1988/98)	Slope armour, random waves, deep water	Hs, Tm, Δ , Dn50, $\cot \alpha$, P, N, Sd	Dn50 / Ns at chosen Sd	$S_d = Ae/Dn50^2$ (start 2-3, failure 8-17)	Deep water ($h > 3 H_{s,toe}$); slope armour; $\cot \alpha = 1.5-6$; $P = 0.1-0.6$; $N < 7500$; som up to 0.06; $\xi m \sim 0.7-7$; $\Delta \sim 1-2.1$; plunging/surging split; no current	Low-med: slope, not berm
Shallow-water slope formulas (rewritten & modified Van der Meer; Van Gent 2003; modified Van Gent / Scaravaglione 2025)	Slope armour, random waves, shallow depth-limited (breaking)	Hs or Hm0, H2%, Tm-1,0, Δ , Dn50, core, $\cot \alpha$, P, N, Sd	Dn50 / Ns at chosen Sd	$S_d = Ae/Dn50^2$ (as Van der Meer)	Depth-limited breaking foreshore; uses composite Weibull H2% (Battjes-Groenendijk), not Rayleigh 1.4; modified VdM $0.25 \leq H_s/h \leq 1.5$; Van Gent 2003 $1.25 < h/H_{s,toe} < 3$; mod. Van Gent $0.4 \leq H_{m0,deep}/h \leq 5$; no current	Low-med: usable in breaker zone, still a slope
Van Rijn (2019)	Horizontal / mild-slope bed protection, current +/- waves	uc, Hs, Tp, h, Δ , slope, $\alpha(ks)$, r, γ_{str}	required D50 (+ dynamic via Cheng 2002)	$r = 0.4-1.0 \rightarrow \theta_{cr} = r \cdot 0.05$ (static); Cheng bed-load (dynamic)	D50 = 0.03-3 m; horizontal / mild slope; $h/(\alpha D50)$ and $Aw/(\alpha D50) = 10-300$; D50 is sieve size (~1.2 Dn50); slope factors $K_{\alpha 1,2} = 1$ for flat bed; sparse field data at largest sizes	High: only one with combined wave+current; static and dynamic
Van Gent & Van der Werf (2014)	Toe of breakwater armour slope, waves	Hs, Tm-1,0, ht, Bt, tt, Dn50, $\cot \alpha$, roughness	Nod or Dn50	Nod (displaced rocks in a Dn50-wide strip)	$ht/h = 0.10-0.30$; $h/H_s = 1.2-4.5$; slopes 1:1.5 & 1:2; $sp = 0.015-0.04$; waves only; non-overtopped; fixed bed (no toe scour); no Δ in formula	Med: berm geometry fits, but assumes armour slope above + down-rush
Herrera & Medina (2015)	Toe berm, very shallow + steep ($m = 1/10$), breaking	hs, Hs0, L0p (Tp), Δ , Dn50 (std Bt = 3Dn50, tt = 2Dn50)	Nod (Eq. 12) + 90% CI	Nod = 0.5 start, 2.0 moderate, 4.0 failure (CIRIA 2007)	$-0.15 < hs/Hs0 < 1.5$ (neg. = emerged); $-0.5 < hs/Dn50 < 5.0$; $0.02 < s_{0p} < 0.07$ for Eq. 12 (data span 0.008-0.08); $m = 1/10$; standard berm; breaking waves only	Med-high for bedrock / steep breaking; fixed geometry, waves only
Deltares HaSPro / RoBeD (2023)	Loose-rock scour protection & cable-cover berms, offshore, current + waves	Shields θ ($\rightarrow$ uc, Hs, Tp, h, Δ , D50), WTOP, MTOC, Dcable, side slope	berm geometry (Bberm/hberm, TOPC) + deformation class; external / interface / flexibility	MOBthresh = 0.95 recommended; classes 0-3; wash-away (class 3) at $\theta > \sim 0.06$	Reshaping regime; MOB = θ/θ_{cr} (App. I); θ_{cr} Soulsby Eq. A.6 (incl. 1.2 factor); $ks = 2.5 D50$; perpendicular wave+current assumed; offshore-calibrated	High: only one explicitly for cable berms + 3 requirements

Table 5.8

The following table indicates, from the literature, which method suits which situation in this thesis. It is an applicability overview, not the applied method selection of the methodology.

Situation	Primary method	Supporting / check	Why
Bedrock, steep nearshore, breaking waves	Herrera & Medina (2015)	Van Gent (range check); Hudson (first estimate)	Only method for a steep bottom with breaking waves; shows that gentle-slope formulas fail here.
Mobile sand, combined waves + current (crossing / obstacle)	Van Rijn (2019)	Deltares RoBeD (deformation + interface / flexibility)	Only method with genuine wave+current on a bed; Deltares covers winnowing and falling apron on sand.
Cable berm geometry and reshaping / deformation (any)	Deltares RoBeD (2023)	Van Rijn (D50 sizing)	Provides berm geometry and a deformation class rather than only D50.
Quick first sizing / sanity check	Hudson / Van der Meer	—	Classical and fast, but without wave period or current.

Table 5.9

6. Extra Stability Checks

The rest of this thesis, and the design tool, size the rock berm itself. This chapter covers the checks around it: how the seabed next to the berm erodes (scour), how the sand under the berm can wash out and how the layers stop it (winnowing), and how the berm copes with anchors and trawl gear. These are indications only. Each one needs its own study per project. For scour, a project uses a 3D model to see how the hole grows. Sources: Schiereck and Verhagen (2019); Deltares (2023); Joustra (2013); Van de Sande (2012); Van Rijn (2024); Sumer et al. (2011).

6.1 Scour

Chapter 5 checks whether the armour stones stay in place. It does not say how the seabed next to the berm erodes. There are few edge-scour rules for long structures, so this section gives a simple estimate for scour driven by currents. It follows the plane-jet theory of Schiereck and Verhagen (2019, pp. 82–86). It is a first estimate only and has not been tested for rock berms.

The berm as a jet. The berm narrows the water column, so the current speeds up over the crest. From continuity, the crest speed is

$$u_0 = u \frac{h}{h - h_{berm}} \quad (6.1)$$

Here u is the normal current, h the water depth, h_{berm} the berm height, and $B = h - h_{berm}$ the water left above the crest. Past the crest the flow spreads out and slows down with the distance x , like a plane jet:

$$u_c(x) = u_0 \sqrt{\frac{2B}{x}} \quad (6.2)$$

Why only currents. This estimate uses currents only. Waves move the sand back and forth, so what a wave crest lifts, the trough brings back. Short waves mainly stir the sand up. They do not dig a deep, lasting hole. Steady tidal and background currents do, because they carry the sand away for good.

Final scour depth. Erosion stops when the spreading jet drops below the speed the bed can take. From plane-jet lab tests (Schiereck and Verhagen, 2019, Eq. 4.7), the final scour depth is

$$\frac{h_{se}}{2B} = 0.008 \left(\frac{u_0}{u_{*c}} \right)^2 \quad (6.3)$$

Here h_{se} is the depth below the natural bed, u_0 the crest speed, and u_{*c} the critical shear velocity of the bed material (from Shields). The $2B$ is the mirror height of a jet that runs along the bed.

How fast it forms. A full hole can take days or weeks of storm or tide. The depth grows over time as

$$h_s(t) = h_{se} \left[1 - \exp\left(-\frac{t}{T}\right) \right] \quad (6.4)$$

with T the time scale. T comes from the Breusers time scale (Schiereck and Verhagen, 2019, Eq. 4.5):

$$T_* = \frac{\sqrt{g \Delta d_{50}^3}}{(2B)^2} T \quad (6.5)$$

Here t is how long the event lasts, Δ the relative density of the bed sand, d_{50} its median grain size, and B is the water column above the crest of the berm. The value of T^* is read from a graph in Schiereck and Verhagen (2019), set by the ratio u_0/u^*c .

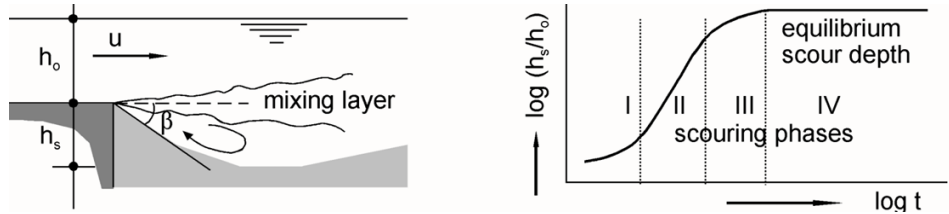


Figure 6.1: the mixing layer (h_o , u , h_s , angle β) and the four scour phases I-IV on a $\log(h_s/h_o)$ versus $\log(t)$ plot up to the final depth. After Schiereck and Verhagen (2019).

This plane-jet model is our own adaptation for a berm. It has not been validated for this shape. We use it only to flag where scour may reach the berm edge, and roughly how fast. A project confirms it with a 3D model.

6.2 Winnowing

Winnowing is the sand under the berm washing up and out through the gaps between the stones. It is driven by the water pressure and by the turbulence near the bed. If the sand under the cable washes out, the berm and the cable lose their support and can sink unevenly. This is a different failure than the stone stability of Chapter 5. It is about the contact between the coarse stones and the finer sand, the interface.

One layer where we can, more where we must. One armour layer over the cable is the simplest option. Two things force us to add layers. First, large armour stones cannot be dropped straight onto the cable without risk of damage, so a layer of smaller stones goes down first to protect it. Second, if the sand can wash out through the armour, a finer filter layer is needed between the armour and the sand. In both cases a single layer becomes an armour layer plus one or more underlayers, and interface stability comes into play.

(The largest stone that can still be dropped safely on the cable, which sets when a protective underlayer is needed, is an installation limit (fall-pipe or side-dumping), not a filter result. Ask to Shamsa NKT installation guidance)

Two kinds of filter. The interface can be a closed or an open filter (Schierreck and Verhagen, 2019, Ch. 6). A closed filter uses the classic Terzaghi rules: each layer is fine enough to hold back the layer below it. In numbers this is a stability, a permeability, and an internal-stability rule:

$$\frac{d_{15,F}}{d_{85,B}} < 5 \quad (6.6)$$

$$\frac{d_{15,F}}{d_{15,B}} > 5 \quad (6.7)$$

$$\frac{d_{60,B}}{d_{10,B}} < 10 \quad (6.8)$$

F is the coarser filter layer, B the finer base layer. The first rule keeps the base grains in place. The second stops pressure building up at the interface. The third keeps each layer stable in itself. One step of these rules only spans a small size jump, so going from fine sand to coarse armour can take several layers, sometimes four or five.

An open filter (Schierreck and Verhagen, 2019, after de Graauw, 1983) uses coarser stone with bigger gaps. It accepts that a little sand may move, as long as the flow stays below a critical value. It needs fewer, thicker layers and is often cheaper for a berm on sand.

How to size an open filter. For an open filter the key number is the critical filter velocity $u_{f,c}$. The sand stays put as long as the flow speed inside the filter stays below $u_{f,c}$. This speed is higher for coarser filter stone and a thicker layer, and lower when the flow is more turbulent (Hoffmans, 2012; Den Adel, 1992). Van de Sande (2012) tested this in the flume and tuned it: the layer thickness fits better to the median stone size d_{f50} , and Hoffmans' turbulence factor needed adjusting.

How thick the layer must be. Joustra (2013) studied the thickness of open filters. Older rules (de Graauw et al., 1984; Klein Breteler, 1989; Bakker et al., 1994) give about two to three times the median stone size d_{f50} in steady, even flow. Turbulence makes it thicker, by a factor Joustra puts between about 1.2 and 2.5, but the data are limited. Van Rijn (2024) gives general guidance for such underlayers.

Rule of thumb for a rock berm. For rock berms the Deltares handbook (2023) says winnowing is less of a problem than around a monopile, because there is no pile to speed up the flow. About four to five stone layers usually stop it (Verheij et al., 2012). Some sinking of the berm into the bed can even help.

One more risk: liquefaction. On silty or fine sandy beds there is one more risk. Sumer et al. (2011) found that a berm that is stable under big waves can still fail if the soil under it liquefies. The liquefied soil moves with the waves and drags the stones, and the berm can sink into it. Stability is then set by a mobility number that uses the density of the liquefied soil, against the Keulegan–Carpenter number (their Fig. 14). This is a separate check, only on beds that can liquefy.

What	Rule	Source
Filter stability	$d_{15,F} / d_{85,B} < 5$	Schiereck & Verhagen (2019), Eq. 6.1
Filter permeability	$d_{15,F} / d_{15,B} > 5$	Schiereck & Verhagen (2019), Eq. 6.1
Internal stability	$d_{60,B} / d_{10,B} < 10$	Schiereck & Verhagen (2019), Eq. 6.1
Open filter	stable while flow $u_f < \text{critical filter velocity } u_{f,c}$	Hoffmans (2012); Van de Sande (2012)
Open-filter thickness	about 2 to 3 x d_{f50} ; more with turbulence (1.2 to 2.5)	Joustra (2013); de Graauw et al. (1984)
Berm filter thickness	about 4 to 5 stone layers	Deltares (2023); Verheij et al. (2012)
Liquefaction	mobility number (with p_{liq}) vs KC number	Sumer et al. (2011)
Largest droppable stone	installation limit for adding an underlayer	NKT guidance (to add)

Table 6.1

Like scour, this is an indication. The number of layers, the stone sizes, and the largest stone that can be dropped safely on the cable are fixed per project.

6.3 Anchor and Trawling Protection

A berm may also need to take hits from bottom-trawl gear and dragging anchors. We do not compute these in the main tool. Designing for them by default would make the berm far too big, which goes against the goal of not over-building. We treat them as extra conditions, checked per project, because the risk changes a lot by location:

- High-risk zones: shipping lanes, anchorages, and busy fishing grounds, where heavy gear hits often.
- Low-risk zones: deeper or restricted waters with little human activity, where the loads of Chapter 5 govern the design.

Where people are active, standard guidelines help:

- Trawl gear: guidance (Van Kessel and Stam, 2010; CIRIA, 2007) shows that about 0.50 m of rock cover over the cable is usually enough. The sloped berm deflects dragging trawl boards (500 to 2000 kg), so they dig in less than about 0.30 m.
- Dragging anchors: to force a 3-tonne stockless anchor to break out and come loose you need a large berm, with a bottom width of about 15 m (Van Kessel and Stam, 2010).

Building a berm to stop a large anchor gives a huge structure. So each project does a Cable Burial Risk Assessment (CBRA). Where the anchor risk is very high, the usual advice is to route the cable around the danger area instead of over-building the rock. This keeps anchors and trawling as conditions on the route and the minimum cover, not as loads in the tool.

Part B – Methodology

7. Methodology

7.1 Introduction

A rock berm over a subsea cable is not designed with one calculation. The route is long and the conditions change along it. The seabed changes, the water depth changes, and the waves and the current change. The methods in Chapters 4 and 5 were each made for a different structure and a different situation, and they describe damage in different ways. For that reason the tool does not use one formula for the whole route. It looks at every point, chooses the methods that fit that point, sizes the smallest stone that is still safe, and then shapes a berm that keeps enough cover on the cable after a storm.

The tool answers two questions and keeps them apart. The first question is how big the stone must be to stay stable under the load. The second question is what shape the berm must have so the cover on the cable is still there after the berm has reshaped in a storm. These are two different questions, so the tool splits them into two branches. Branch A gives the stone size. Branch B gives the berm shape and checks how much it deforms. Keeping them apart makes the design clear. Branch A gives only an indicative stone size, and Branch B then sizes its own stone from the accepted deformation before it builds the berm.

The two branches. Branch A and Branch B are shown in Figure 7.1 and Figure 7.2. Branch A classifies each point, picks the methods that apply, checks each one against its valid range, and gives one governing stone size with the other methods next to it for comparison. Branch B sizes its own stone from the accepted deformation, builds the berm with the RoBeD model, and gives the deformation class of every point. The Branch A stone is kept next to it only as an indication. In Branch B the stone is tuned by hand until the whole route sits at the accepted deformation.

The two branches do not carry the same weight. Most methods in Branch A were made for breakwaters and their toes, not for a low, free berm on the open bed. Their results are therefore only used as a comparison, and whether they can be used for a berm is checked in the results chapter. RoBeD, used in Branch B, is the only reviewed model made for a rock berm that reshapes, so it carries the design. This chapter first sets the shared inputs and the load at the bed (Section 7.3), then describes Branch A (Section 7.4) and Branch B (Section 7.5), and ends with the sensitivity analysis, the case study and the limitations.

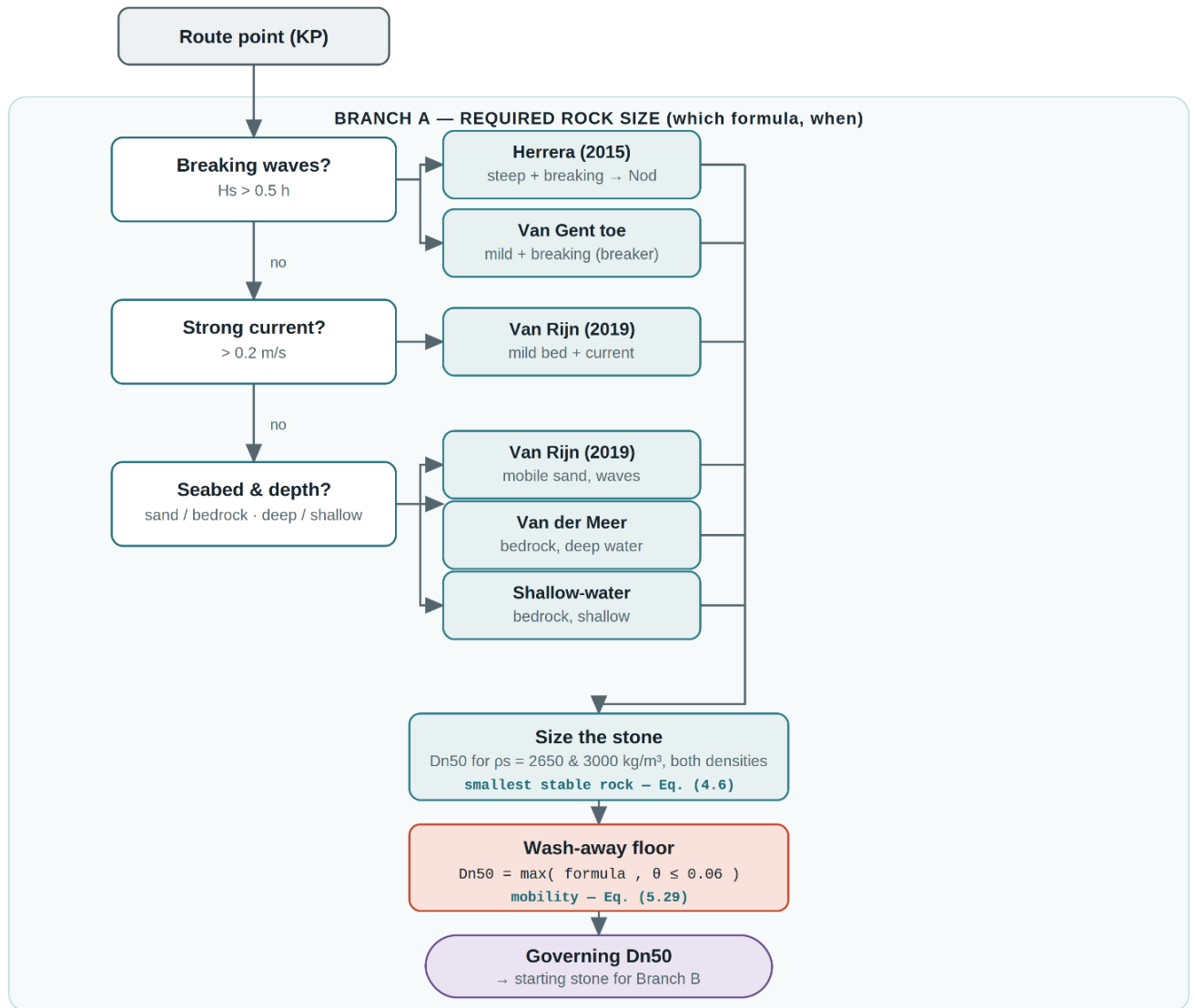


Figure 7.1: Branch A

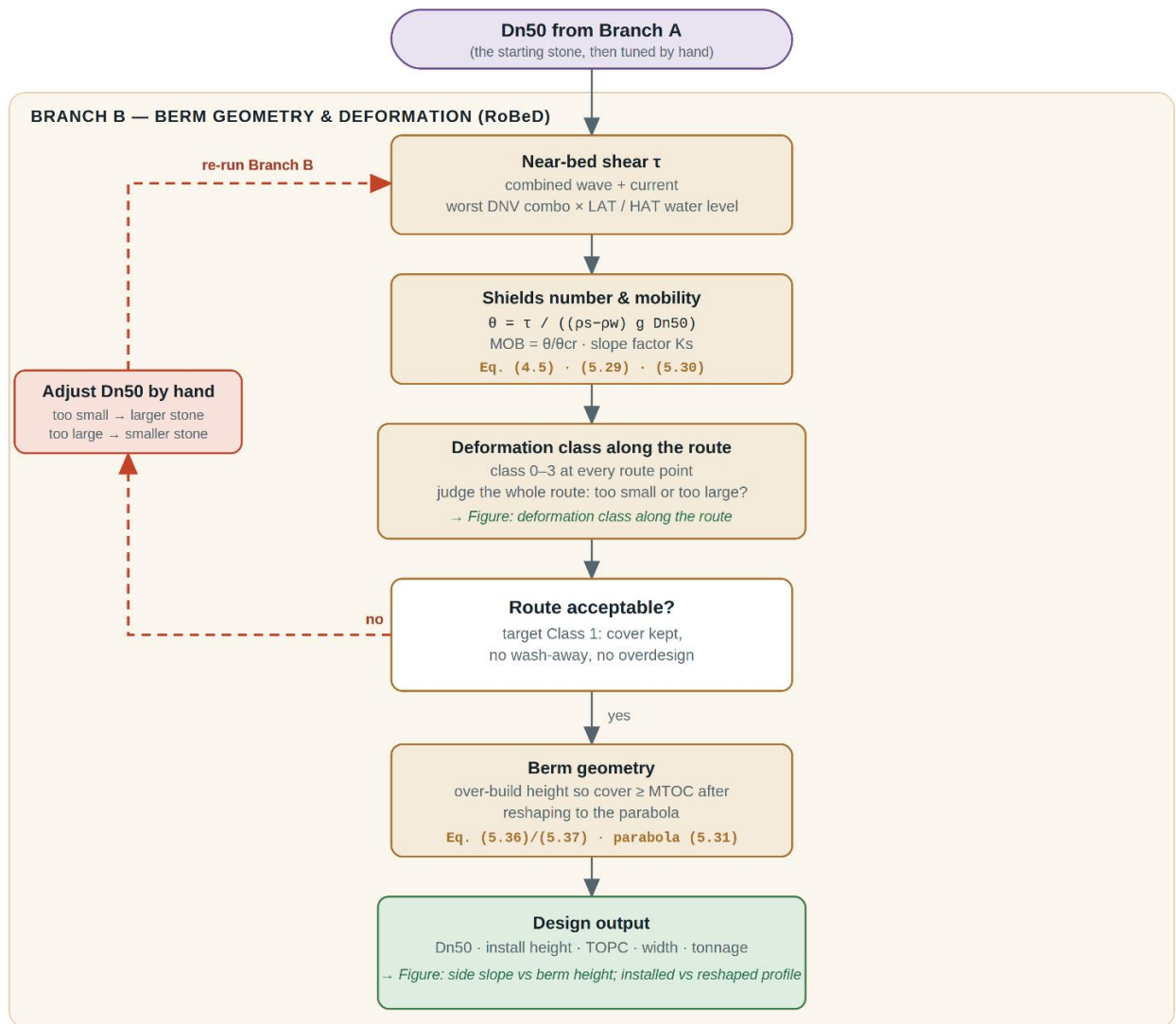


Figure 7.2: Branch B

7.2 Scope

The design object is a free rock berm that lies over an unburied cable. It is not a rock layer placed against another structure. Because it stands alone on the seabed, it is loaded from both sides and can reshape to both sides, and its stability comes from the bed shear rather than from run-up on a slope. The berm is looked at in the nearshore zone, up to just past the breaker line, where the load on the cable protection is largest.

The seabed along a route can be many soil types, not only rock and sand. The tool groups them into two design cases. The first case is a hard seabed (bedrock), where the bed does not erode, so it is only a stability question for the rock. The second case is a mobile seabed (sand and other loose sediment), which occurs mostly at crossings where the cable cannot be buried. There the bed can erode under the berm, so the interface with the sediment and the risk of sinking are extra points. The two cases are kept apart, so a result is never carried from one bed to the other.

Waves are the main load and come from the route data. The current and the combined wave-current case are handled in the shared load basis (Section 7.3.3). Anchor and trawling loads are kept as a boundary condition on the minimum cover, not as a calculated load, following the common minimum cover of about half a meter. The fully broken inner surf zone and settlement in soft subsoil are out of scope.

7.3 Design framework and computational approach

Every route point goes through the same two branches. First the inputs and the load at the bed are set, because both branches use them. Then Branch A sizes the stone and Branch B shapes the berm. The inputs come from two places, the route data file and the tool itself, so they are described apart.

7.3.1 Route input data

The tool reads a data file with one row per route point (KP). The columns are:

- SR and SP: the seabed slope in two directions, in degrees. The tool uses the steepest of the two.
- ST: the seabed type, the soil at that point.
- WL: the water depth at the point.
- DOC: the required cover on the cable, the depth of cover.
- WHS10 and Tp10: the significant wave height and the peak period for the 10-year return period.
- WHS100 and Tp100: the same two values for the 100-year return period.
- CU10 and CU100: the current velocity for the 10-year and the 100-year return period.
- Angle between current and waves (optimization): the angle from the wave and current roses. This is used only in the second version of the tool (Section 7.3.4).

The seabed columns (SR, SP, ST, WL, DOC) come from the survey department, and the wave and current columns (WHS, Tp, CU) come from the metocean department. The angle between the current and the waves also comes from the metocean roses.

For this study the route did not have enough points. A set of representative points was therefore made to show the full process on a realistic route. The point that has real data was set as the most critical point, so the governing design uses a real point while the made points show how the design changes along the route. This is stated openly and comes back in the case study and the limitations.

Column	Meaning	Unit	Source
point_ID	Route point identifier (KP)	-	route
SR, SP	Seabed slope in two directions, the steepest is used	deg	survey
ST	Seabed type, the soil at that point	-	survey
WL	Water depth at the point	m	metocean
DOC	Required cover on the cable	m	client
WHS10, WHS100	Significant wave height, 10-year and 100-year	m	metocean
Tp10, Tp100	Peak wave period, 10-year and 100-year	s	metocean
CU10, CU100	Current velocity, 10-year and 100-year	m/s	metocean
Angle	Angle between waves and current, Version 2 only	deg	metocean
Hs0 (optional)	Offshore wave height, used when needed near breaking	m	metocean
LAT, HAT (optional)	Water depth at low and high astronomical tide	m	metocean

Table 7.1

7.3.2 Design inputs in the tool

Next to the data file, the engineer sets the design inputs in the tool. These are our choices, not route data:

- Rock density: a normal density of about 2650 kg/m³ and a high density of about 3000 kg/m³. Every result is made for both.
- Berm geometry: the side slope, the top width and the cable diameter.
- Accepted damage per method: Van Rijn r, Van der Meer Sd, and the Van Gent and Herrera Nod.
- Accepted RoBeD deformation: the accepted mobility on the flat top of the berm.
- Load factors: the turbulence factor γ_{str} and the current threshold.
- Client limits: the required cover MTOC when it is set as one value instead of per point, and limits on the cover and the berm width.
- Water levels: the low and high water level (LAT and HAT), either per point in the file or as an offset set here.

In the tool these inputs are grouped as general inputs, Branch A inputs and Branch B inputs, so it is clear which input belongs to which branch.

Rock Berm Design Tool **NKT**
Made by W. Pol

GENERAL INPUT - used by both branches

Normal density	2650	High density	3000
LAT offset (m, - = lower)	0	HAT offset (m, + = higher)	0

BRANCH A - stone sizing (Van Rijn shear + formula damage)

Ka1	1.0	Ka2	1.0
γ_{str} (turbulence)	1.6	Current sig (m/s)	0.2
Van Rijn r	1.0	Van der Meer Sd	2.0
Van Gent Nod	0.5	Herrera Nod	2.0

BRANCH B - RoBeD geometry & deformation

Berm side slope 1:a	3.0	Top width (m)	3.0
Cable dia (m)	0.2	RoBeD accept MOBtop;flat	0.7
Req. cover MTOC (m, 0=data)	0.0	Berm height (0=auto)	0.0
Max cover (m, 0=off)	0.0	Max width (m, 0=off)	0.0

BRANCH A - load the route and assess every point

☒ Diagnostic mode (full per-point breakdown)

theta_cr=0.055 SHORT mode (Van Rijn ks fixed; geometry ks=2.5 D50)

route report appears here after Start Calculation ...

BRANCH B - RoBeD design figures (pick a stone, check deformation, build the geometry)

Standard Dn50 (m, 0=auto) density:

Figure 7.3: GIU

7.3.3 Shared load basis

Both branches need the load at the bed. The tool brings the surface wave down to the bed through the near-bed velocity, so the effect of the depth is already in it (Chapter 4). The load follows the DNV combinations, and the worse of the two pairs is used, the 10-year wave with the 100-year current and the 100-year wave with the 10-year current. Each point is checked at both LAT and HAT, and the water level with the largest bed shear is taken. This is usually LAT, but the tool checks both instead of assuming it.

The near-bed wave shear and current shear are the same for both branches, but the way they are combined is not. Branch A follows Van Rijn and adds the two bed shear stresses directly, $\tau_c + \tau_w$, without using the direction (Eq. 5). Branch B follows RoBeD and combines them as vectors, which does use the angle between the waves and the current (Section 7.5.1). The two combinations are set out further in Section 7.3.4.

Because the tool checks every point, it also finds the most critical point of the route by itself, which is the point that needs the largest stone. This point is the basis for the design figures and the geometry checks of Branch B. The most critical point of the route, the point that needs the largest stone, is always taken as the design point, so that every point along the route is covered and safe.

7.3.4 Wave-current interaction angle (optimization)

The simple sum of the two shear stresses does not use the angle between the waves and the current. In reality the combined load at the bed depends on that angle. As shown in Section 3.3, the correct way is to add the two as shear velocities and not as shear stresses, because adding the stresses directly only holds when the two are perpendicular. The combined value is

$$u_{*cw} = \sqrt{u_{*c}^2 + u_{*w}^2 + 2 u_{*c} u_{*w} \cos \theta}$$

with u_{*c} from the current, u_{*w} from the wave, and theta the angle between them (Section 3.3). The load is largest when the two are aligned and smallest when they oppose. The angle changes the combined bed shear wherever a current is present, so it works through the shared load basis and reaches both branches.

Branch B already uses the angle in RoBeD's own combination (Section 7.5.1), so the baseline is not angle-free. The optimization is to replace RoBeD's Soulsby combination with the more detailed shear-velocity combination of Section 3.3, $\tau = \tau_c + \tau_w + 2 \cdot \sqrt{(\tau_c \cdot \tau_w)} \cdot \cos \alpha$, in a second version of the tool. Both versions use the real angle from the roses, so the difference between them is the combination method, and how much it changes the design is examined in the results chapter.

7.4 Branch A: Rock sizing

Branch A answers the first question; how big the rock must be. It sizes the smallest stable stone at every route point and gives one governing value, which is kept as an indication next to the Branch B design. The steps below follow the Branch A figure (Figure 7.1).

7.4.1 Classification and the decision tree

Branch A first reads every route point and puts it in a class before any stone is sized. The tool makes four checks at each point: is the seabed in scope, do the waves break at that depth, is the bed steep or mild, and is there a strong current. These checks use the shared load basis of Section 7.3 and do not repeat the theory, which is in Chapters 4 and 5.

The result of the checks goes into a decision tree, which picks the methods that fit that point. A tree is used instead of one formula, because the methods were each made for a different situation, so not every method is valid at every point. The tree keeps each method inside the situation it was made for. It also checks each chosen method against its own valid range. If a point falls outside all ranges, it is marked out of scope with the reason, instead of giving a number that cannot be trusted.

Order of the tree (Figure 7.1). If the waves break, a steep bed uses Herrera and a mild bed uses the Van Gent toe method. If the waves do not break but there is a strong current, Van Rijn is used. If there is no strong current, the seabed and the depth decide: mobile sand uses Van Rijn, bedrock in deeper water uses Van der Meer, and bedrock in shallow water uses the shallow-water formulas. Hudson runs next to the non-breaking branches as an indicative check only. For each situation the tree gives one governing method and reports the other methods next to it for comparison, not as the design value.

7.4.2 Native and transferred methods

The methods in Branch A do not all describe the same object. Only the flat-bed method of Van Rijn (2019) was made for loose rock on a bed under waves and current, which is the closest to a free berm, so Van Rijn is the native method for the berm. The other methods were made for other structures. Hudson and Van der Meer are for the armour of a breakwater slope, Van Gent and Herrera are for the toe at the foot of such a slope, and the shallow-water formulas are for wave-exposed emergent slopes.

The tree still lets each method govern in the situation it fits best, because in its own range each is the best option available. So on many points the governing value still comes from a method that was made for another structure. For that reason the whole of Branch A is treated as indicative and not as the final design. The native method and the transferred methods are kept apart, and whether the transferred values can be used for a berm is not decided here but is checked against the case studies in the results chapter. The comparison methods are always shown next to the governing value so this check can be made.

7.4.3 Fitting the formulas to the rockberm

Because most methods come from other structures, Branch A makes a few adjustments so that the load they see is the load at the berm on the bed, not at the surface or at a breakwater.

Turbulence factor. Van Rijn adds the extra bed turbulence around a structure with the factor γ_{str} , which multiplies the near-bed velocities in the bed shear. It is not a fixed value. Van Rijn writes it as $\gamma_{str} = 1 + n$ gamma, so it grows with the local turbulence and is larger where a structure, a current or wave breaking is present, and smaller on a calm flat bed. The baseline uses one value in the range 1 to 2. A further optimization sets γ_{str} per part of the nearshore, for example a lower value in the shoaling zone and a higher value in the breaker zone, instead of one value for the whole route. The effect of this on the design is examined in the results chapter. γ_{str} is used in the Branch A sizing only, so Branch B does not count the current turbulence twice.

Roughness without a loop. The bed shear needs the stone roughness $k_s = \alpha D_{50}$, but D_{50} is the value being solved. To avoid a loop the sizing uses a fixed reference stone for k_s , while Branch B uses the real $k_s = 2.5 D_{50}$ once the stone is known.

Sieve to nominal. Van Rijn is written in the sieve diameter D_{50} , which is about 1.2 times the nominal D_{n50} used everywhere else, so the tool converts its result back to D_{n50} .

Near-bed wave load. The methods that use the near-bed velocity, Van Rijn and the two toe methods, already act at the bed, because the near-bed velocity weakens with depth through the term $1/\sinh(k h)$ (Chapter 3 and 4), so they are not reduced again. Hudson has no depth term of its own, so it is given the effective near-bed wave height

$$H_{s,b} = \frac{H_s}{\cosh(k h)}$$

Which is the surface H_s scaled down to the bed with the dynamic-pressure response factor of linear wave theory (Chapter 3, Eq. 3.10). The shallow-water forms and Herrera use the local depth-limited wave height for which they were calibrated, so they keep their own wave input. This reduction is why the surface-wave value, which is far too heavy for a low berm on the bed, is not used directly.

7.4.4 Governing stone, wash-away floor and density

After the tree has run the methods, Branch A picks the governing stone size for the point. This is done for both rock densities, the normal 2650 kg/m³ and the high 3000 kg/m³. A denser stone can be smaller for the same load.

A lower bound is then put on the stone, the wash-away floor. This is the smallest stone that keeps the RoBeD mobility below the class-3 limit, which matches a Shields number of about 0.06 (mobility, Eq. 5.29). The governing stone is the larger of the formula value and this floor (Figure 7.1), so the stone can never be so small that the whole berm would wash away. The wash-away floor is the one RoBeD value that Branch A uses, and it links Branch A to Branch B through the Shields number.

7.4.5 Output and diagnostics

Branch A gives two kinds of output. The short report gives, per point, the governing stone size, the method used, the other methods for comparison, and any warning, such as a point that is out of scope or an input that is outside a method's range. The diagnostic mode gives the full breakdown per point, with every step and every intermediate value, so a result can be traced and checked. The diagnostic mode is part of Branch A. Nothing is hidden, and a point that cannot be trusted is flagged rather than given a number.



Figure 7.4

7.5 Branch B: berm geometry and deformation (RoBeD)

Branch B answers the second question, what shape the berm must have. It sizes its own stone from the accepted deformation and, with the RoBeD deformation model, builds the berm and checks how much it deforms along the whole route. The Branch A stone is kept next to it only as an indication. Branch B is the main design of this study, because RoBeD is the only reviewed model made for a rock berm that reshapes. The steps below follow the Branch B figure (Figure 7.2).

7.5.1 Shear, Shields number and deformation class

Branch B first works out the load on the stone at the point. It uses the same near-bed wave and current shear as Branch A, but combines them in RoBeD's own way. It leaves out the turbulence factor γ_{str} , which belongs to the Van Rijn sizing, so the current turbulence is not counted twice.

From the shear the tool computes the Shields number and the mobility, $MOB = \theta / \theta_{cr}$ (Chapter 4 and 5). Because a berm has a flat top and side slopes, the mobility is checked at three places, with a slope factor that lets the side slope move before the flat top. The mobility sets the deformation class, from class 0 (no movement) to class 3 (wash-away), following the RoBeD classes (Chapter 5).

Branch B combines the wave and current shear with RoBeD's own vector method, which includes the angle between them (Deltares 2023, Appendix A.6, Eq. A.31, after Soulsby 1997). A mean combined shear is formed first, $\tau_m = \tau_c \cdot [1 + 1.2 \cdot (\tau_w / (\tau_c + \tau_w))^{3.2}]$, and then a maximum combined shear, $\tau_{max} = \sqrt{(\tau_m + \tau_w \cdot \cos \alpha)^2 + (\tau_w \cdot \sin \alpha)^2}$, with α the angle between the current and the waves. This differs from Branch A, where the two stresses are added directly with no angle. The angle comes from the wave and current roses, and a collinear angle (the largest load) is used as the safe default when it is not known.

7.5.2 Reshaping and required cover

RoBeD assumes the berm reshapes under a design storm from its installed trapezoidal shape to a parabolic shape, with a width-to-height ratio that depends on the Shields number (Chapter 5, Eq. 5.31). The berm is built higher than the minimum, so that after it has reshaped the cover on the cable is still at least the required minimum cover (MTOC).

The installed height and the resulting cover follow from setting the installed area equal to the reshaped area (Eq. 5.36 and 5.37). In the tool the side slope and the top width are inputs, and the installed height is solved from these equations. The over-build is applied smoothly with the Shields number, so the berm height is a smooth function of the stone and the slope, without a jump at a class boundary.

7.5.3 Accepted deformation

The design does not force the berm to stay fully static. It accepts a controlled and limited deformation, because this allows a smaller stone, which matters over a long route for material, quarrying and transport. The accepted amount is set by the mobility on the flat top of the berm, for which a safe-side value of 0.7 is used. At this level the side slopes reshape (class 1) while the top is held and the cover is kept. This is RoBeD's own design basis (Chapter 5), so the choice rests on the source and not on the author. The accepted level is an input, so it can be tightened or relaxed and discussed with the supervisors.

7.5.4 One grading over the whole route, turned by hand

Branch B is applied over the whole cable route, not at one section. One rock grading is chosen for the route, and each point gets its own installed height, the over-build, so that after it reshapes under its local load the cover is still kept. The tool then shows the deformation class at every point. The engineer judges the whole route from these classes and tunes the stone by hand: a larger stone if any point reaches class 2 or 3, and a smaller stone if the whole route is over-built. Branch B is then run again with the new stone. This loop is repeated until the route sits at the accepted class (Figure 7-B). Keeping the tuning in the hands of the engineer, rather than hiding it in an automatic rule, keeps the design under control and easy to explain.

By default one rock grading is used for the whole route. The deformation-class figure shows, per project, which points are safe with that grading and which are not. When the route falls into clear groups, as in Figure 7.6 where three families of points can be seen, the input file can be split into these subsections and a separate berm designed for each family. This avoids over-dimensioning the milder families to the size that only the most heavily loaded family needs.

7.5.5 The design figures

Branch B produces the figures that support the design decision. Each of them can be made for the normal and for the high rock density, so the effect of the density is visible.

- Step 1, pick the stone. The Shields number for a range of stone sizes at the governing point, so the smallest safe stone can be read off. Figure 7.5 shows an example.
- Step 2, deformation class along the route. The class at every route point for the chosen stone. This is where the route is judged too small or too large.
- Step 3, geometry. The side slope against the before-storm berm height, so the effect of a steeper or a milder slope is visible. Figure 7.7.
- Step 3b, cross-section. The installed berm against the reshaped berm at the most critical point, on a true scale, with the cable and the minimum cover. Figure 7.8 shows the cross-section at the most critical point, with the installed berm in blue (as-built, aanleg) and the reshaped berm in red (after the storm), together with the cable and the minimum cover.

These figures are made both for the engineer, to weigh the trade-off between stone size, berm volume and deformation, and for the client, to explain how the berm behaves during and after a storm.

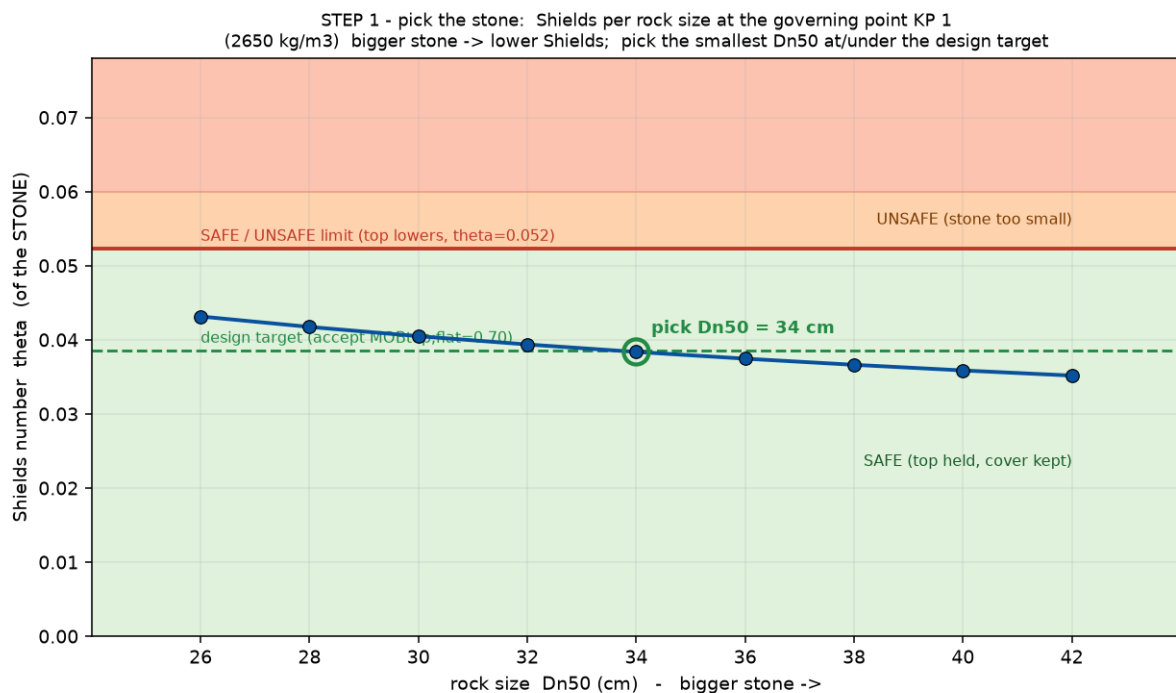


Figure 7.5:

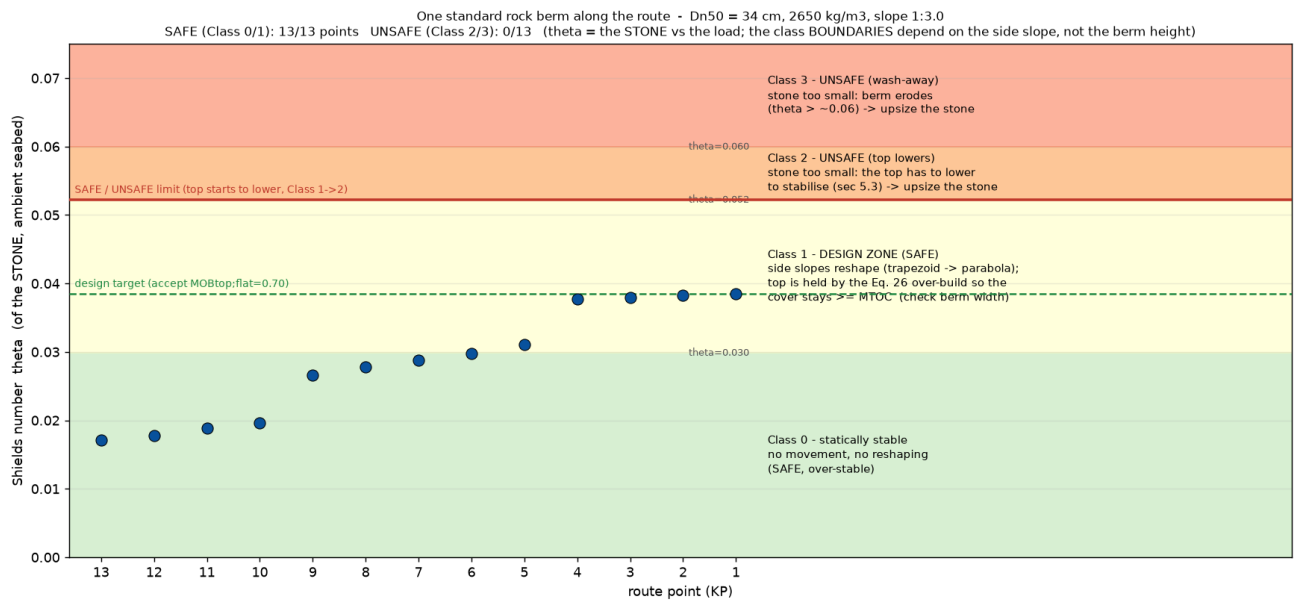


Figure 7.6:

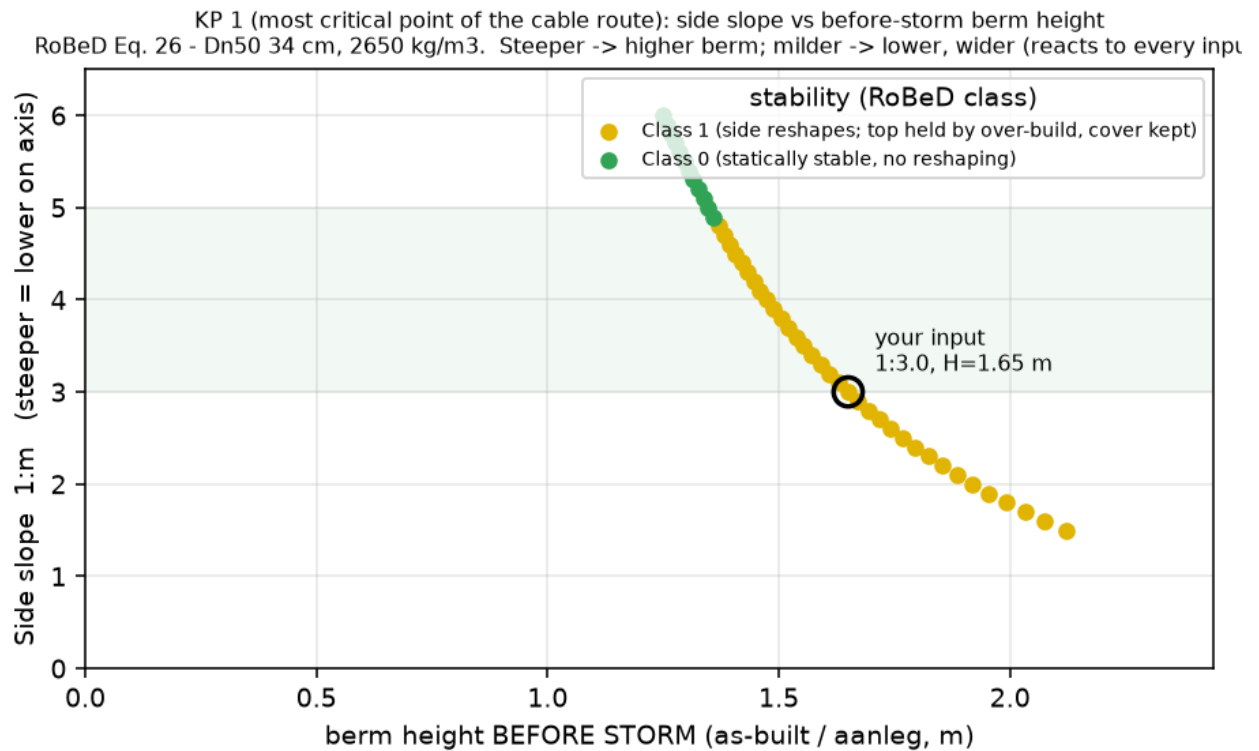


Figure 7.7:

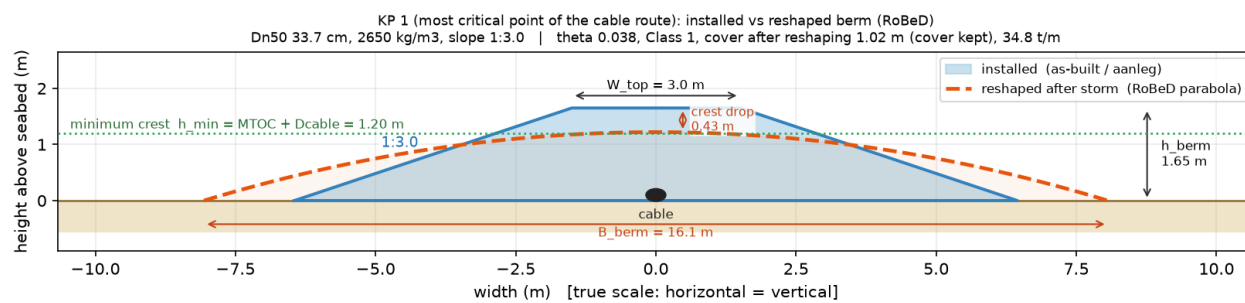


Figure 7.8:

7.5.6 Why the design is built this way

Branch B is built to give one clear picture that can be used and explained. It keeps the RoBeD design as the main design, because it is the only model made for a berm that reshapes, and it keeps Branch A separate as an indicative check. In the results chapter the RoBeD design is placed next to the physical case studies. If the case studies use larger stones, this shows whether the RoBeD design is enough on its own or whether the stone should be increased toward the Branch A values. Keeping the two branches apart makes this comparison possible and honest, and it is why the tool reports both.

7.6 Case studies: applying the framework to real projects

The framework is applied to three real projects from the past. The three sites were chosen because they are different: a sandy seabed with a strong current, and two rocky seabeds where the waves break. This lets the case study show the main point of the framework, in Branch A the decision tree picks a different method for each situation and that each method stays inside its valid range. Next to that Branch B will give a complete rock berm design.

The available measured data is limited. For each route only a small number of points was available, so a set of representative points was constructed around them to show how the design develops along a real route. The measured point is always set as the most critical, governing point, so all the calculations are anchored to a real point. This is stated openly. A further gain of this approach is that the tool automatically finds the most critical point along the route, and it runs each point at both LAT and HAT to see which water level gives the highest shear, which is most often LAT but is always checked.

The three projects. The inputs below are taken from the project data sheets (100-year and 10-year conditions):

	Project A	Project B	Project C
Seabed	rippled sand	bedrock	bedrock
Water depth h (m)	22.0	22.7	13.34
H_s , 100-yr (m)	9.9	12.6	7.9
T_p , 100-yr (s)	14.1	~19	13.8
Current, ~100-yr (m/s)	1.47	0.73	low / n.a.
H_s/h (breaking if >0.5)	0.45 (no)	0.55 (yes)	0.59 (yes)

Table 7.2

Branch A, the minimum stone per project. The tool first classifies each site from the raw numbers and follows the Branch A decision tree to the sizing method that fits it. This gives the minimum stone size per project:

- Project A, sandy seabed, waves do not break, strong current. The tree sends this to Van Rijn, the only method that handles a combined wave and current load on a bed.
- Project B, rocky seabed, waves break, moderate current. The tree sends this to the shallow-water slope formulas, the breaker zone. Because these are waves-only, Van Rijn and Van Gent are also shown for comparison, with a note that breaking and current cannot both be captured by one method, so the result is indicative.
- Project C, rocky seabed, waves break, almost no current. The tree again sends this to the shallow-water slope formulas.

So the same tool, on the same run, uses three different sizing methods for the three sites. This is exactly what Branch A is meant to do, and it is the direct answer to the first weakness of Section 7.1.

Branch B, the design stone and the berm per project. Branch B is the leading design and it finds its own stone. For each project it determines the best stone size directly from the accepted deformation with the RoBeD model, and this is the stone the design is calculated with. With that stone it returns the deformation class, the installed geometry (berm height, cover TOPC, reshaped width and side slope) and the tonnage, for the normal and the high rock density.

The Branch A stone is kept next to it only as an indication. So both branches run on every project, but the design stone and the berm come from Branch B, while Branch A gives an indicative stone for comparison. For each project the Branch B result is reported as a small set of buildable berm designs, a low-cost design with the smallest rock and the most reshaping, a compact design with a bigger and more stable stone, and a high-density design with a smaller stone.

These design results are presented and discussed in the results chapter. The case study is the input for the benchmark of Branch B against the physical study cases (Section 7.7) and for the sensitivity study (Section 7.8).

7.7 Benchmarking Branch B against the reference cases

Branch B is the leading design of this study. Branch A gives only the minimum stone size, while Branch B gives a full berm design, the stone together with the geometry and the accepted reshaping. RoBeD is the only reviewed model that was made for a rock berm that is allowed to reshape, so the berm design rests on Branch B. The benchmark uses three real cases, and because these cases are themselves complete berm designs, Branch B is the branch that compares directly with them, with Branch A added on the stone-size axis.

The three cases are not equally strong. One case was tested in a physical model test, which is the strongest reference. A second case is a real design that was independently reviewed by a specialist party but not physically tested, so it is a credible reference on paper. The third case is a future project, so there the tool is used to make a design rather than to check one. The benchmark therefore rests mainly on the one physically tested case, with the reviewed case as support and the future project as an application.

Why a physical model test is a good reference. In a physical model test the real situation is reproduced and the berm is allowed to reshape by a controlled and limited amount, which is the same behaviour that this framework accepts. Because the tested berm behaved well, it comes close to the ideal design for its conditions, so it is a solid reference to check the tool against.

How the benchmark is done. For each case the tool is run at the case conditions. Branch B produces a full berm design, the stone size together with the geometry and the accepted reshaping, so for the physically tested case it can be compared with the tested berm on the whole berm and not only on the stone. Branch A gives only the minimum stone size, so it is compared on the stone-size axis. In this way the stone values of Branch A, Branch B and the real design are read on one axis, while the geometry is compared on the physically tested case.

What the benchmark can show. RoBeD is calibrated for offshore conditions, not for the nearshore. It is therefore possible that Branch B gives a smaller stone than the real design in the nearshore. If that is the case, the true best design would sit between the smaller Branch B stone and the larger, indicative Branch A stone. Whether this happens, and where the real design falls between the two, is examined in the results chapter. The benchmark is what tells us whether Branch B is enough on its own or whether the stone should be increased toward Branch A.

7.8 Parameters and sensitivity analyses for both branch A and B

The tool uses many inputs. Some are measured and carry uncertainty, such as the waves, the current and the seabed slope. Some are design choices, such as the damage level and the rock density. Some are modelling parameters that are not measured directly, such as the turbulence factor g_{str} . This section lists every input with its default value and range, and then studies which of them matter most. The sensitivity is run for each branch on its own, because the two branches have different outputs and are driven by different inputs. The output of Branch A is the governing stone size D_{n50} . The outputs of Branch B are the deformation class, the installed berm height and the tonnage.

Environmental inputs. These are read per route point from the metocean and geotechnical survey. They are not tuned. They define the load on the berm.

Symbol	Meaning	Unit	Source	Branch
Hs	significant wave height	m	metocean survey	A and B
Tp	peak wave period	s	metocean survey	A and B
h	water depth	m	bathymetric survey	A and B
uc	near-bed current speed	m/s	metocean survey	A and B
theta	wave-current angle	deg	wave and current roses	B
ib	seabed slope	deg	geophysical survey	A
ST	soil type	-	geotechnical survey	A

Table 7.3:

Rock and modelling parameters. These are set once in the interface and hold for the whole route. Defaults and ranges:

Symbol	Meaning	Default	Range	Source	Branch
rho_s	rock density (normal / high)	2650 / 3000	2650 – 3000 kg/m ³	design choice	A and B
rho_w	sea-water density	1025	fixed, kg/m ³	physics	A and B
gstr	turbulence amplification	1.6	1.0 – 2.0	Van Rijn (2019)	A
r	Van Rijn damage level	1.0	0.4 – 1.0	design choice	A
Sd	Van der Meer damage	2.0	2 – 8	design choice	A
Nod	Van Gent / Herrera damage	0.5 / 2.0	0.5 – 2.0	design choice	A
P	notional permeability	0.4	0.1 – 0.6	Van der Meer	A
N	number of waves in the storm	3000	up to 7500	design choice	A
cot alpha	berm side slope	2.0	1.5 – 6.0	design choice	B
a	roughness factor, $k_s = a \cdot D_{50,ref}$ (fixed reference)	2.0 / 2.5	2.0 Van Rijn, 2.5 RoBeD	literature	B
theta_cr	critical Shields number	0.055	fixed	explicit choice (Ch. 5)	B
theta_wa	wash-away Shields number	0.06	fixed	RoBeD	B
l:m	steepest allowed side slope	1:3	buildability	design choice	B
Ka1, Ka2	Van Rijn slope factors	1.0	up to 1.0	Van Rijn	A and B

Table 7.4:

Client and geometry inputs. These come from the project requirements.

Symbol	Meaning	Default	Source
MTOC	required cover over the cable	site value, ≥ 0.5 m (CIRIA)	client
Btop	berm crest (top) width	3.0 m	design choice
Dcable	cable diameter	0.2 m	client
max cover / max width	buildability limits (0 = off)	0 / 0	client / vessel

Table 7.5:

7.8.1 Branch A sensitivity

For Branch A the output studied is the governing stone size D_{n50} . The inputs that are expected to drive it most are the significant wave height H_s , the turbulence factor g_{str} , the accepted damage level of the governing method, the near-bed wave height used in Hudson, and the rock density. These are the levers introduced in Section 7.4, and they are the ones varied here. The other inputs are held at their default.

g_{str} . The turbulence factor g_{str} is not measured. Van Rijn gives a range of about 1 to 2. A higher g_{str} raises the bed shear and so the stone size, so it is a strong lever in Branch A. In this thesis it is kept fixed, but it is a clear candidate for a future improvement with a flow model, for example OpenFOAM, which could give a site-specific value instead of a range. The sensitivity shows how much a better g_{str} would be worth.

7.8.2 Branch B sensitivity

For Branch B the outputs studied are the deformation class, the installed berm height and the tonnage. The inputs that are expected to drive these most are the stone size that comes out of Branch A, the accepted mobility on the flat top, the side slope, the top width, and the wave-current angle used in the RoBeD combination. Running the sensitivity for Branch B on its own shows which of these the geometry is most sensitive to, which is useful because these are the levers the engineer tunes by hand in the route loop of Section 7.5.4.

7.8.3 Rock density across both branches

The rock density is treated as a design variable and is run at the normal 2650 and the high 3000 kg/m³ value in both branches, so it appears in the sensitivity of Branch A and of Branch B. A denser rock is stable at a smaller stone size, so the stone diameter drops. For a berm whose size is set by the required cover, the volume stays almost the same, so the denser rock mainly makes the berm heavier in tonnes per meter. Both effects are reported, so the trade-off between a smaller stone and a heavier berm can be weighed per project instead of fixed in advance.

7.8.4 Method

The sensitivity analysis is done in two steps, for each branch. The first step is a screening with the Morris method. It changes one input at a time from many different starting points and measures how much the output changes each time. This gives a simple ranking of the inputs, from the ones that matter most to the ones that hardly matter, and it needs only a small number of runs. The second step is a variance based analysis with the Sobol method. For each input it gives two numbers between 0 and 1. The first order index is the share of the spread in the output that the input explains on its own. The total index also includes the effect of that input working together with the other inputs. Morris is used first to remove the inputs that do not matter, and Sobol is then used on the inputs that remain to get the exact shares. Both steps use the ranges from the tables above and are run with the SALib library. The sampling settings and the resulting index values are given with the results.

7.9 Refined with the shear-velocity combination

Branch B combines the wave and the current shear with RoBeD's own Soulsby vector method (Section 7.5.1). This last section describes an extra refinement step, which is our own addition to the method and is used only after Branch B has been benchmarked against the physical study cases.

Once Branch B is shown to sit close to the tested design, a second version of the tool replaces the RoBeD combination with the shear-velocity combination of Section 3.3,

$$u_{*CW} = \sqrt{u_{*c}^2 + u_{*w}^2 + 2 u_{*c} u_{*w} \cos \theta}$$

Which combines the wave and the current as velocities before squaring rather than as stresses. The aim is to see whether this combination brings Branch B even closer to the tested design. The two combinations are run on the same route and the same conditions, so the only change is the way the wave and the current are combined. The difference in the stone size and in the geometry along the route is quantified in the results chapter. The step is kept separate and is done only after the benchmark, so the leading design stays on RoBeD's own basis and the refinement is a clearly marked improvement on top of it, not a change to the main method.

7.10 Limitation and open points

The framework is open about what it does not yet solve.

- Damage measure. The methods use different damage quantities: r for Van Rijn, Nod for Van Gent and Herrera, S_d for Van der Meer, and a deformation class for RoBeD. There is no established way to convert one into another. The tool keeps the methods separate and uses a common damage level only to compare stone sizes, not to force a conversion. A defensible mapping is left as an open question for the supervisors.
- RoBeD validity range. RoBeD is calibrated on an offshore monopile test database, so its validity is set by three ranges (Deltares 2023, Table 4.1): the relative mobility on top MOB_{top} from 0.24 to 1.17, the total Keulegan–Carpenter number KC_{tot} from 1.1 to 7.8, and the wave–current dominance U_{rel} from -0.56 to 0.63 . The tool checks each point against these ranges and reports the result. On the mobility, which is the parameter RoBeD depends on most, the accepted design stays inside the range. The clearest open point is KC_{tot} , which is defined relative to a pile diameter and is therefore not defined for a free berm, so the tool reports it as not applicable rather than inventing a value. RoBeD also uses linear wave theory with no breaking term, so breaking points are flagged as non-conservative. These caveats are why Branch B is benchmarked against the physical study cases in Section 7.7 rather than trusted on its own.
- Turbulence factor. g_{str} is kept fixed and is probably too high for a free-standing berm. The right value should be agreed, or found with a flow model, not assumed.
- Edge scour. There is no validated way to predict edge–scour depth for a rock berm (Chapter 6), so it is kept as a qualitative boundary condition, not a false precise number.

Part C – Results

8. Results

This chapter checks whether the tool works on real cases. It first gives the real designs of the projects, then compares the Branch A stone size against them, and then makes a fuller comparison with the Branch B berm design. The chapter ends with a short conclusion. The interpretation and the recommendations follow in the Discussion chapter, and the sensitivity analysis and the shear-velocity refinement are in Chapter 9. The three projects are the ones introduced in Section 7.6 (Table 7.2).

8.1 The real designs

For each project a real design is available, and this is the reference the tool is compared against. The designs differ in how strong the evidence is. Project A was independently reviewed by a specialist party but not physically tested. Project B was tested in a physical model. Project C is a future project, so its design is a proposed design. What was designed, the rock grading and the berm geometry, is given below.

The rock is delivered as a standard grading, defined by its mass limits under EN 13383. The nominal diameter of a stone of mass M is

$$D_{n50} = (M_{50} / \rho_s)^{(1/3)}$$

So a grading with a lower and an upper mass limit gives a range of nominal diameters. The single value used for the comparison is the median nominal diameter D_{n50} , which follows from the mass M_{50} at which 50 percent of the class is lighter (EN 13383), read from the grading curve of the class. For the 60/300 kg grading this gives a D_{n50} of about 0.36 m at the high density and about 0.375 m at the normal density, within a range of roughly 0.27 to 0.48 m. For the 10/60 kg grading it gives about 0.21 m, within a range of roughly 0.15 to 0.27 m. The comparison uses the median D_{n50} , and it holds across this range, because the conclusions do not change within the width of the grading.

Benchmark input data. The real design of each project:

Project	Grading	Density (kg/m ³)	D_{n50} (grading)	Side slope	Crest width W_{top}	Cable dia	Min cover
A Reviewed	Armor HD LMA 60/300 kg	HD (3100)	~0.36 m	1:3	1.2	0.2 m	1 m
B Tested	Armor HD LMA 60/300 kg	ND (2670)	~0.36 m	1:6	4.0 m	0.15 m	1.8 m
C Future	HD 10/60 kg	HD (3000)	~0.21 m	1:4	1.0 m	0.15 m	0.9 m

Table 8.1: The real designs of the three projects, and the stone size from the grading.

8.2 Branch A: comparison on the stone size

Branch A gives only a rock size, so it is compared with the real design on the rock-size axis. Every sizing method is computed at each point, and the decision tree then selects the method that is valid for that situation. Each method is checked against its own validity range, and a method that falls outside its range is marked as not valid and is not used. The valid methods that the tree does not select are kept next to the selected one for comparison.

The water depths are given relative to LAT, the lowest astronomical tide, which is the chart datum used in the source reports. The tool evaluates each point at the low water level, because that gives the highest near-bed velocity and so the largest rock. Each project is computed at its own design density, so the tool and the real design are compared on the same density within a project. The design density is 3100 kg/m³ for Project A, 2670 kg/m³ for Project B and 3000 kg/m³ for Project C, and every rock size in this section is at that density. The governing point of every route is KP 1, the shallowest and most heavily loaded point, and this is the point reported below.

8.2.1 Project A

KP 1 of Project A is a mobile sandy seabed in 22 m of water below LAT, with a current of 1.35 m/s at the 10-year return period and 1.6 m/s at the 100-year return period. The design wave is $H_s = 9.9$ m with $T_p = 14.1$ s, so $H_s/h = 0.45$, the waves are not depth-limited and the bed is mild. The decision tree selects Van Rijn, the flat-bed method for a combined wave and current load.

Method	Status	Reason (technical validity)	Dn50 (m)
Van Rijn (2019), flat bed	Valid, selected	made for a bed under a combined wave and current load, which is the situation at this point	0.393
Hudson (1953/1959)	Valid	the waves are not depth-limited, so the formula applies. It is used on the near-bed height $H_{s,b} = H_s / \cosh(kh)$ (Eq. 3.10), which lowers it from 2.14 m on the surface H_s	1.68
Van Gent and Van der Werf (2014), toe	Valid, uncertain	the toe geometry $h_t/h = 0.95$ lies outside the 0.1 to 0.3 range the formula was fitted on, so its value cannot be trusted here	0.74
Van der Meer (1988)	Not valid	the water is too shallow, $h/H_s = 2.2$ against the required 3. The formula assumes a Rayleigh wave-height distribution, which no longer holds on a shallow foreshore	(2.30)
Shallow-water formulas (4 forms)	Not selected	these were derived for a depth-limited, wave-breaking foreshore. At this point the waves are not depth-limited, so this situation is not the one they describe	1.96 to 3.13
Herrera and Medina (2015), toe	Not valid	derived for a steep bottom with breaking waves. This point has a mild bed and no depth-limited breaking	–

Table 8.2: All Branch A methods at KP 1 of Project A, at the design density of 3100 kg/m³.

The selected method is Van Rijn, which gives $D_{n50} = 0.393$ m. The real design uses a 60/300 grading with a median rock of about 0.356 m, so the selected method is 10 percent above the real design, on the safe side. Van Rijn is also the smallest valid method at this point. Hudson, which is valid here but was made for breakwater armour, gives 1.68 m, which is 372 percent above the real design.

8.2.2 Project B

KP 1 of Project B is a bedrock seabed in 22.7 m of water below LAT. The design wave is $H_s = 12.59$ m with $T_p = 19.75$ s, the same condition as the 100-year test in the physical model, so $H_s/h = 0.55$ and the waves are depth-limited and breaking on a mild bed, with a moderate current. The tree selects the shallow-water formulas, which are the forms derived for this situation.

Method	Status	Reason (technical validity)	Dn50 (m)
Shallow-water formulas (4 forms)	Valid, selected	derived for a depth-limited, breaking foreshore, which is the situation at this point	2.48 to 3.55
Van Rijn (2019), flat bed	Valid, comparison	it applies to a bed under waves and current, but it contains no term for wave breaking, so in the surf zone it does not capture the full load	0.58
Van Gent and Van der Werf (2014), toe	Valid, uncertain	the toe geometry $h_t/h = 0.91$ lies outside the 0.1 to 0.3 range the formula was fitted on	0.91
Van der Meer (1988)	Not valid	the water is too shallow, h/H_s below 3, so the Rayleigh distribution behind the formula no longer holds. This is exactly the case the shallow-water forms were derived for	(2.90)
Hudson (1953/1959)	Not valid	the side slope $\cot(\alpha) = 6.0$ lies outside its 1.5 to 4.0 range, and the formula uses a surface wave height that is not representative once the waves are depth-limited	(3.03)
Herrera and Medina (2015), toe	Not valid	derived for a steep bottom, while this point is mild, and the wave steepness $s_{0p} = 0.016$ and $h_s/H_{s0} = 1.80$ fall outside its range	–

Table 8.3: All Branch A methods at KP 1 of Project B, at the design density of 2670 kg/m³.

The selected shallow-water formulas give 2.48 to 3.55 m, against a real design with a median rock of about 0.374 m. The smallest valid method at this point is Van Rijn with 0.58 m, which is 55 percent above the real design. So even the most favourable valid method is well above what was actually built and tested, and the method that the tree selects is far above it.

8.2.3 Project C

Project C is a future project on a bedrock seabed, and the route is mixed. KP 1, the design reference point, is the shallowest point at 13.34 m below LAT, where the design wave is $H_s = 7.9$ m with $T_p = 13.8$ s, so $H_s/h = 0.59$ and the waves are depth-limited and breaking. The tree selects the shallow-water formulas. Along the deeper part of the route, from about 16 to 21 m below LAT, the waves are no longer depth-limited and those points are selected for Van Rijn, which asks for a larger rock.

Method	Status	Reason (technical validity)	Dn50 (m)
Shallow-water formulas (4 forms)	Valid, selected	derived for a depth-limited, breaking foreshore, which is the situation at this point	1.73 to 2.62
Van Rijn (2019), flat bed	Valid, comparison	it applies to a bed under waves and current, but it contains no term for wave breaking	0.46
Van Gent and Van der Werf (2014), toe	Valid, uncertain	the toe geometry $h_t/h = 0.92$ lies outside the 0.1 to 0.3 range the formula was fitted on	0.71
Van der Meer (1988)	Not valid	the water is too shallow, h/H_s below 3, so the Rayleigh distribution behind the formula no longer holds	(2.02)
Hudson (1953/1959)	Not valid	the formula uses a surface wave height that is not representative once the waves are depth-limited (on the near-bed height it would give 1.95 m)	(1.95)
Herrera and Medina (2015), toe	Not valid	derived for a steep bottom, while this point is mild, and $h_s/H_{s0} = 1.68$ falls outside its range	–

Table 8.4: All Branch A methods at KP 1 of Project C, at the design density of 3000 kg/m³.

The selected shallow-water formulas give 1.73 to 2.62 m, against a proposed grading with a median rock of about 0.208 m. The smallest valid method at this point is Van Rijn with 0.46 m, which is 121 percent above the proposed design. Along the deeper non-breaking part of the route Van Rijn asks for up to about 0.40 m.

8.2.4 How far is Branch A from the real designs

Putting the three projects next to each other shows a clear pattern:

Project	Situation	Selected method	Dn50 selected	Smallest valid method	Dn50 smallest	Real design	Difference on the smallest
A	not depth-limited, sand, current	Van Rijn	0.393 m	Van Rijn	0.393 m	0.356 m	+10 percent
B	depth-limited, breaking	shallow-water formulas	2.48 to 3.55 m	Van Rijn	0.58 m	0.374 m	+55 percent
C	depth-limited, breaking	shallow-water formulas	1.73 to 2.62 m	Van Rijn	0.46 m	0.208 m	+121 percent

Table 8.5: The Branch A result against the real design. The last column compares the smallest valid method, which is the most favourable value Branch A can give at that point.

Branch A sits above the real design at all three points, but how far above depends on the situation. Where the waves are not depth-limited, at Project A, the best valid method is 10 percent above the real design. That is a workable result. Where the waves are depth-limited, at Projects B and C, the best valid method is already 55 and 121 percent above. And if the design follows the method that the decision tree actually selects, which is the shallow-water formulas, the result becomes six to twelve times the real design. So Branch A moves further away from reality as soon as the waves become depth-limited. That is exactly the situation a nearshore cable berm is built for

8.2.5 Why the Branch A stones stay indicative

Across the three projects only Van Rijn (2019) was made for loose rock on a bed under waves and current, so it is the native method and the closest to a free berm. All the other sizing methods come from rubble-mound structures, as set out in Chapter 5. Hudson (1953/1959), Van der Meer (1988) and the shallow-water formulas are breakwater-armour methods, and Van Gent and Van der Werf (2014) and Herrera and Medina (2015) are toe methods. A breakwater slope and a toe are loaded differently from a low berm lying on the seabed, so even where these methods are technically valid for the point, their rock size is an indication for a free berm and not a design value. This is why they are reported next to Van Rijn and not used to set the design.

Hudson is applied on the near-bed wave height $H_{s,b} = H_s / \cosh(kh)$ (Eq. 3.10), which lowers its rock size and brings it closer to a real bed load. Even so it stays far above the native Van Rijn value and far above the real designs. The consequence is that Branch A on its own cannot give the design rock for a nearshore berm in the breaker zone. The berm design has to come from Branch B, which is treated next.

8.3 Branch B: the berm design

Branch B gives the full berm design, the rock together with the geometry and the accepted reshaping, so here the tool is compared with the real design on the whole berm and not only on the rock. Every project is treated in the same two steps. The first step asks which rock the tool would recommend for the conditions and the required cover, and places it next to the rock that was actually chosen. The second step enters the design that was chosen, with its own rock and its own geometry, and shows how that berm deforms under the design storm. All runs use the geometry of the real design and the design density of that project.

8.3.1 How Branch B is used

The geometry is the input, the rock is the result. The required cover over the cable, the crest width and the side slope are entered first. With that geometry fixed, the tool finds the rock size that keeps the berm within the accepted deformation. The berm height is not fixed at the minimum. The model builds the berm higher than the minimum, so that the cover over the cable is still met after the berm has reshaped under the design storm. The amount of over-build grows with the Shields number, so a smaller rock leads to a higher installed berm.

The rock size as a single value. In reality the rock in a berm falls within a grading range. In the tool this range is brought back to one nominal value, the Dn50. For stability, which is the focus of this thesis, this is enough, because the median rock governs the stability. A controlled loss of a small part of the rock by reshaping is accepted, as long as the cover on the cable is kept, and the washing out of finer material through the berm is controlled by the underlayer.

The design condition. Every point is evaluated at LAT and HAT, and the tool takes the condition that gives the largest rock. For a given wavelength the depth at LAT gives the highest near-bed orbital velocity, so LAT normally governs.

8.3.2 Project A

Project A is a mobile sandy seabed at KP 1 in 22 m of water below LAT. The significant waveheight is 9.9 m with a peak period of 14.1 s and the current reaches 1.6 m/s at the 100-year return period. With a ratio of wave height to water depth of 0.45 the waves are not depth-limited, so the berm is loaded by a combined wave and current action on an open bed. The berm has a side slope of 1:3, a crest width of 1.2 m and a required cover of 1.0 m over a cable of 0.2 m, at a rock density of 3100 kg/m³. The route runs from 22 m at KP 1 to 30 m at the far end, so KP 1 is both the shallowest and the most heavily loaded point.

1. Step 1, the rock. The tool recommends a rock of 0.393 m for these conditions. The design that was built uses a 60/300 grading with a median rock of about 0.356 m. The tool is therefore 10 percent above the choice that was made, on the safe side, and both values fall within the same grading.
2. Step 2, the berm. Entering the design that was built, with its own rock and its own geometry, gives deformation class 0 at KP 1. The Shields parameter is 0.0275 and the mobility on the flat top is 0.50, well below the threshold of 0.95, while the mobility on the 1:3 side slope reaches 0.87 and therefore also stays below it. The berm stays in place, the crest does not lower and the cover of 1.0 m is kept in full. The installed berm is 1.47 m high with a base of about 10 m, at about 26 tonnes per meter.

The second view shows how much room this design has. It gives the deformation class for every combination of side slope and installed berm height, so the whole design space can be read in one picture. With this rock the berm stays in class 0 over a wide range of slopes and heights, and the design that was built lies well inside that range and not close to its edge.

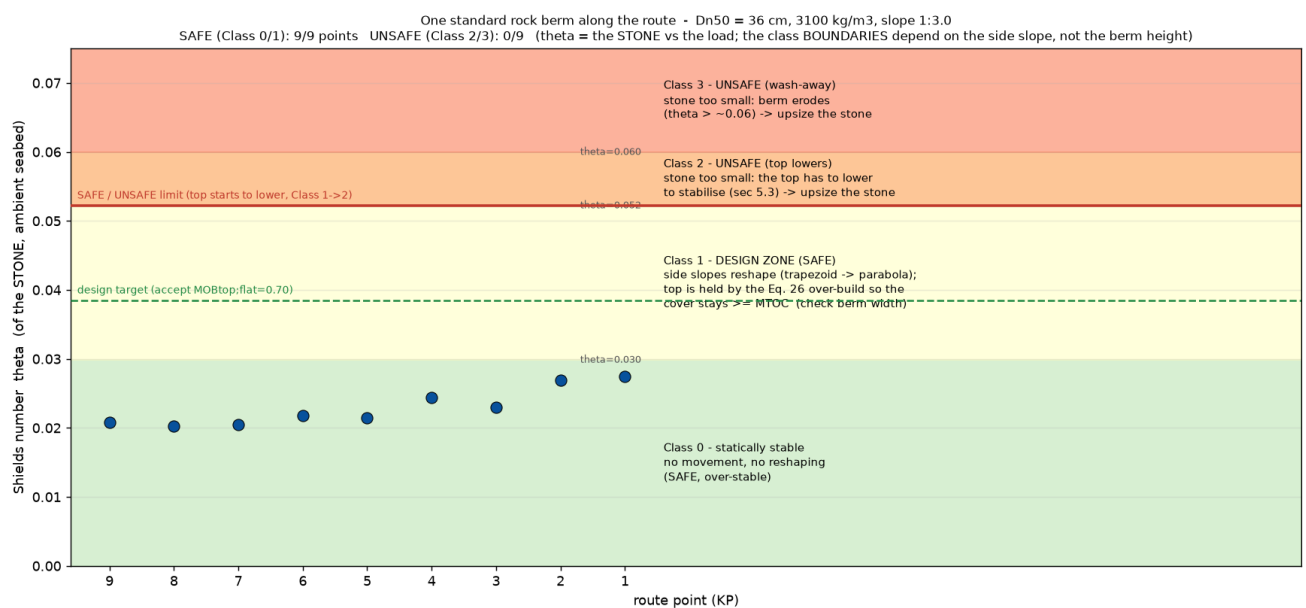


Figure 8.1: deformation class along the route for Project A, with the rock of the built design.

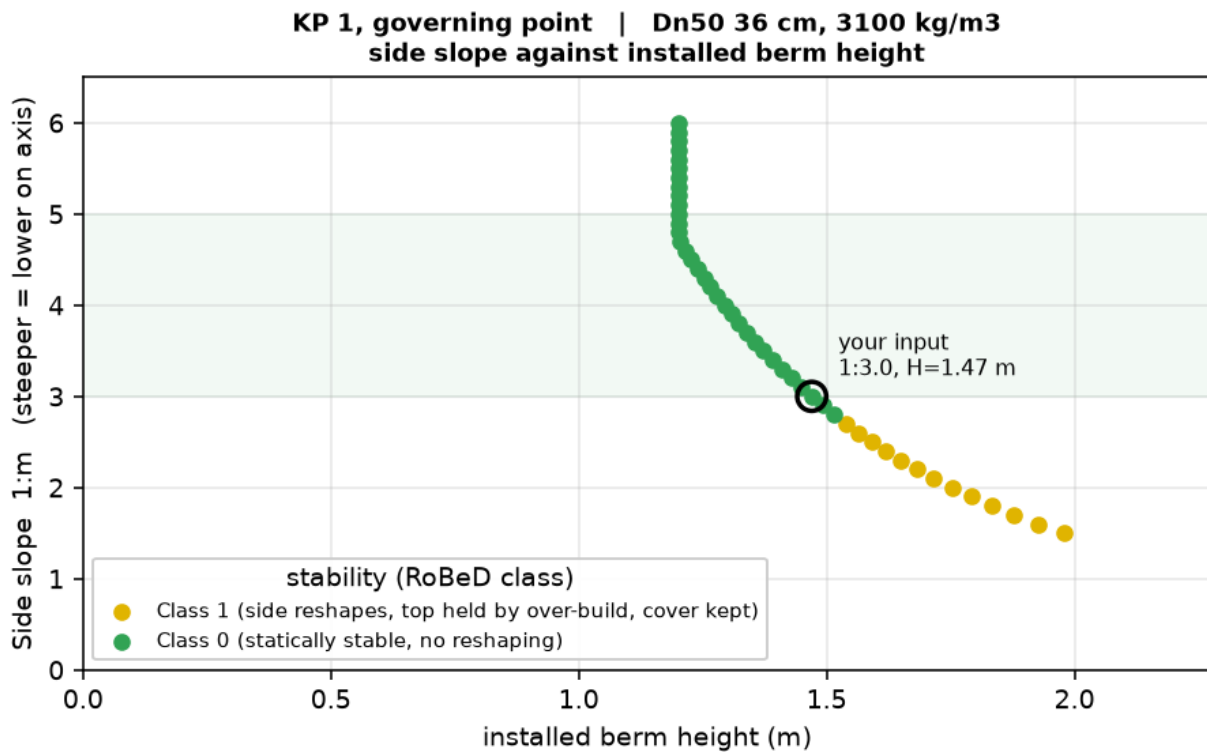


Figure 8.2: side slope against installed berm height for Project A, with the built design marked.

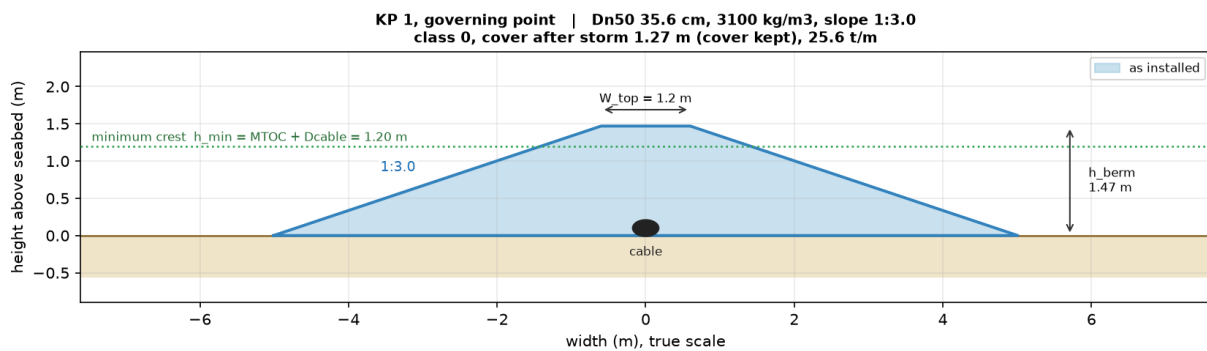


Figure 8.3: cross-section before and after the storm for Project A, at KP 1

	Rock Dn50	Class	Installed height	Cover after storm	Tonnage
Recommended by the tool	0.393 m	0	1.44 m	1.24 m, kept	24.5 t/m
Design that was built	0.356 m	0	1.47 m	1.27 m, kept	25.6 t/m

Table 8.6: Project A, the recommendation of the tool and the verification of the design that was built.

8.3.2.1 Comparison with the specialist report

The design of Project A was checked by a specialist engineering company before it was built. That report is a commercial document, so it is not reproduced here and only the comparison is given.

Two things about it matter for this thesis. The first is that the report sizes the rock with the same approach that is used here, a Shields parameter compared against an accepted level, and that it takes damage level 2 as acceptable, which it expresses as a critical Shields parameter of 0.04. The second is that it includes a worked calculation of the load at its critical section, with all intermediate quantities listed. That makes it possible to run the tool on exactly the same input and compare the two step by step, rather than only comparing the final rock size.

Quantity	Specialist report	This tool	Difference
Bed roughness k_s	0.52 m	0.52 m	identical
Wave friction factor f_w	0.079	0.079	identical
Current bed shear stress	8.01 Pa	7.7 Pa	-4 percent
Representative bed shear stress	162.56 Pa	147.0 Pa	-10 percent
Shields parameter	0.039	0.0349	-10 percent

Table 8.7: The load calculation of the tool against the calculation in the specialist report, on the same input.

This thesis is aligned with the specialist report. The bed roughness and the wave friction factor come out identical, so the roughness model and the friction model are the same. The bed shear stress and the Shields parameter agree within 10 percent. The small remaining difference is in the definition of the wave velocity. Both use the same wave shear stress relation, the tool writing it with a quarter of the density and the peak orbital velocity and the report with half the density and a representative orbital velocity. These are the same expression, because the peak velocity is a factor root two above the representative one. What is left is only the way that representative velocity is obtained for an irregular sea, the report taking it from the wave spectrum and the tool from the significant wave height with linear wave theory.

There is evidence that the tool is correct. The tool was built from the literature, without access to this calculation, and it reproduces an independent professional calculation of the bed load to within 10 percent on the same input, with two of the intermediate quantities identical. Predictions of bed shear stress from different methods commonly differ by much more. The load model in the code is therefore implemented correctly, and the results of this thesis rest on a sound calculation.

The tool and the specialist report differences. On the load the tool is about 10 percent lower, which is the less conservative direction. On the acceptance level the tool is stricter, at 0.0385 against 0.04. Combining the two gives the rock each method would require at this section, 0.200 m for the report and 0.188 m for the tool. The tool therefore asks for about 6 percent less rock, so it is marginally on the less conservative side, by an amount that is small compared with the width of a rock grading. Both methods land inside the same grading class.

8.3.3 Project B

Project B is a bedrock seabed at KP 1 in 22.7 m of water below LAT. The significant wave height is 12.59 m with a peak period of 19.75 s and the current is moderate at about 0.73 m/s at the 100-year return period. With a ratio of wave height to water depth of 0.55 the waves are depth-limited and breaking, so this berm sits in the breaker zone. The berm has a side slope of 1:6, a crest width of 4.0 m and a required cover of 1.8 m over a cable of 0.15 m, at a rock density of 2670 kg/m³. The side slope of 1:6 is much milder and the crest of 4.0 m much wider than on Project A, which makes this a low and broad structure. The route runs from 22.7 m at KP 1 to 25.1 m at the far end and the wave height falls from 12.59 m to about 7.3 m along it, so KP 1 is again the most heavily loaded point.

1. The rock. The tool recommends a rock of 0.369 m for these conditions. The design that was built uses a 60/300 grading with a median rock of about 0.374 m. The difference is 1 percent, so on this project the tool and the design that was built give practically the same rock.
2. The berm. Entering the design that was built, with its own rock and its own geometry, gives deformation class 0 at KP 1. The Shields parameter is 0.0383 and the mobility on the flat top is 0.70, below the threshold of 0.95, while the mobility on the 1:6 side slope reaches only 0.88 because the mild slope keeps the slope factor high. The berm stays in place, the crest does not lower and the cover of 1.8 m is kept in full. The installed berm is 2.07 m high with a base of about 29 m, at about 91 tonnes per meter.

The mild side slope is what carries this design. The mobility on the flat top is the same 0.70 that Project C also reaches, but there the steeper 1:3 slope pushes the mobility on the slope past the threshold. Here the 1:6 slope keeps it at 0.88 and the whole berm stays static. Along the route the design stays in class 0 at every point, and because the wave height falls steadily away from KP 1 the margin only grows.

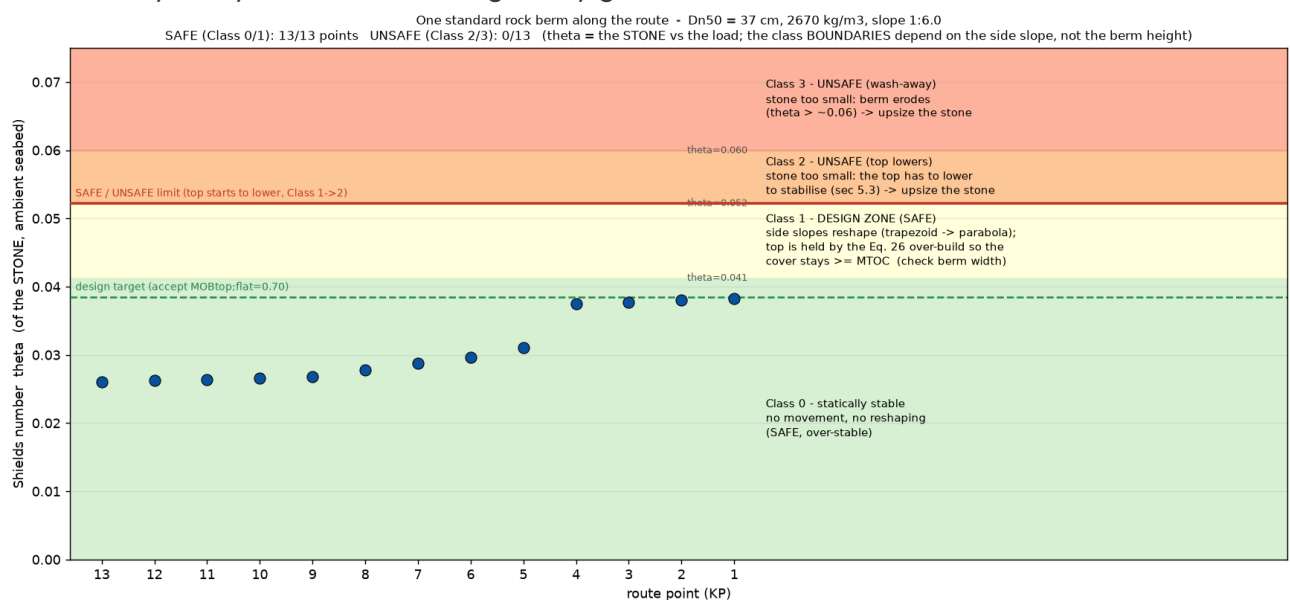


Figure 8.4: deformation class along the route for Project B, with the rock of the built design.

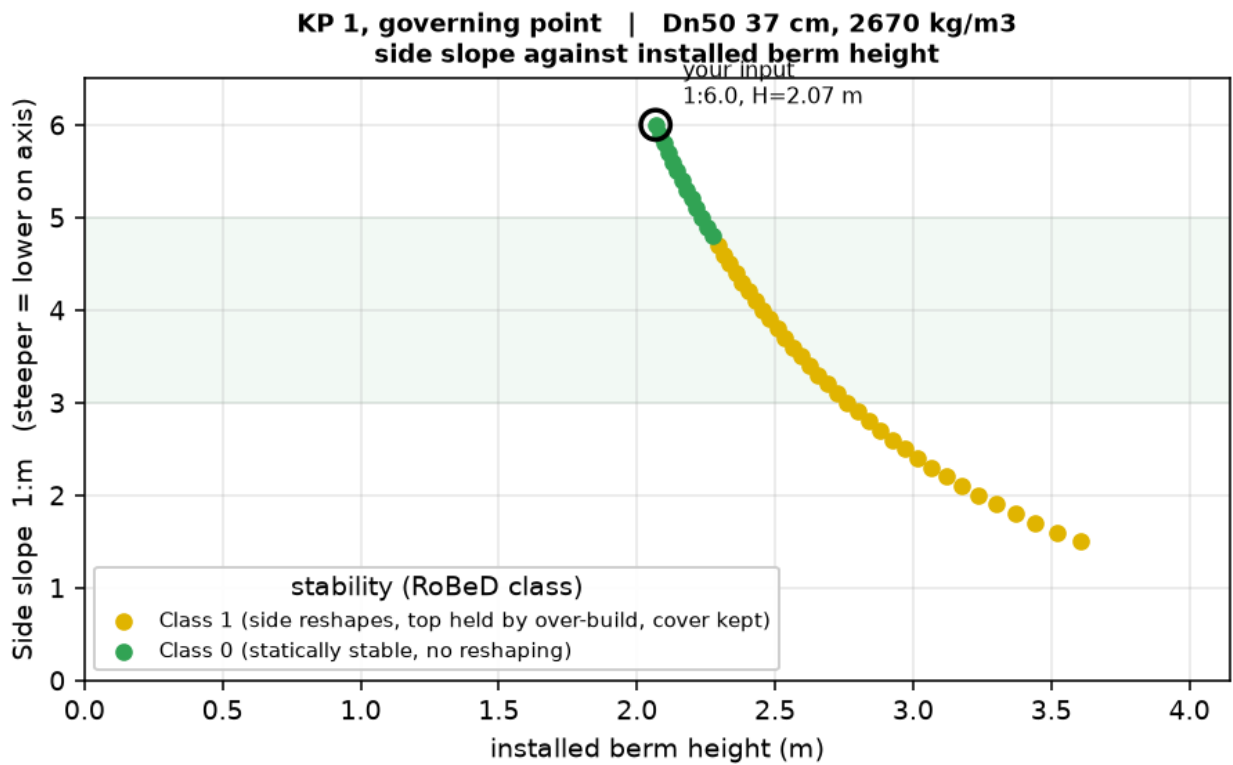


Figure 8.5: side slope against installed berm height for Project B, with the built design marked.

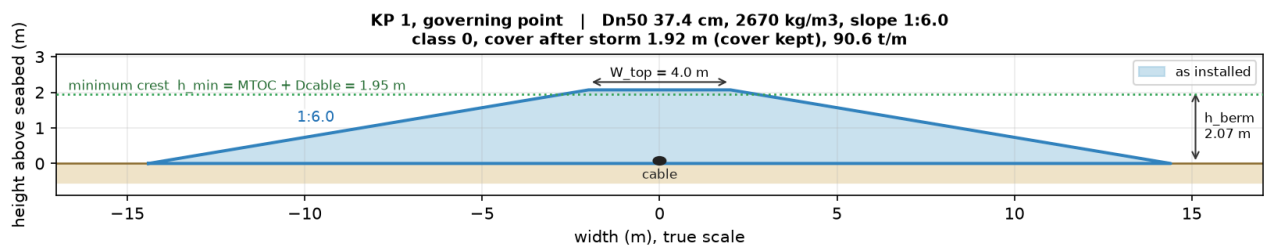


Figure 8.6: cross-section before and after the storm for Project B, at KP 1.

	Rock Dn50	Class	Installed height	Cover after storm	Tonnage
Recommended by the tool	0.369 m	0	2.08 m	1.93 m, kept	91.3 t/m
Design that was built	0.374 m	0	2.07 m	1.92 m, kept	90.6 t/m

Table 8.8: Project B, the recommendation of the tool and the verification of the design that was built

8.3.3.1 Comparison with the physical model test

The design of Project B was tested in a physical model before it was built. That test report is a commercial document, so it is not reproduced here and only the comparison is given. Three things about the test matter for this thesis. The berm was tested in a two-dimensional wave flume at a scale of 1:55. A series of conditions of increasing severity was applied to the same structure without rebuilding it in between, so the damage accumulated through the series, with the early conditions at LAT and the later ones at higher water levels and for longer durations. The damage criterion was the cover that remains over the cable rather than the movement of individual rocks, which is the same criterion the tool uses.

The tool designs for the 100-year wave at LAT. In the test program that condition was run on its own before the later ones were applied, so the measured profile after it can be compared directly with the prediction.

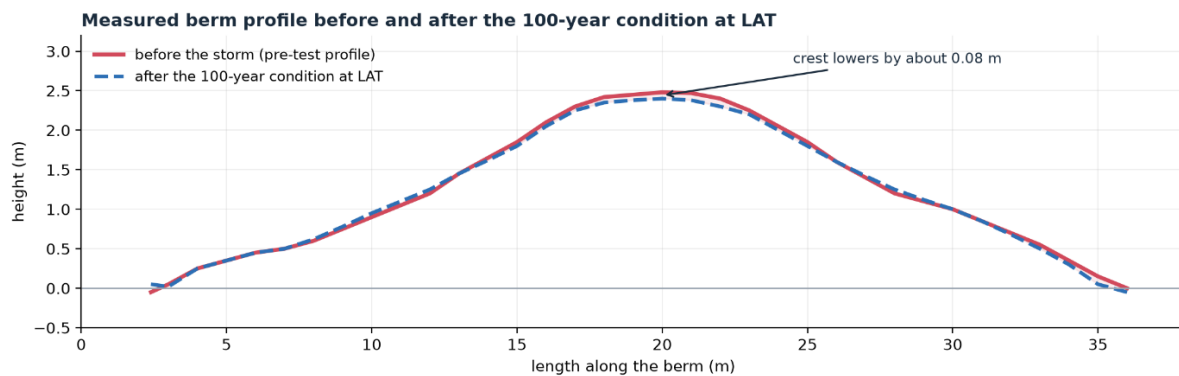


Figure 8.7: measured berm profile before and after the 100-year condition at LAT.

This thesis is aligned with the physical model test on both counts. On the rock the tool recommends 0.369 m against the 0.374 m that was tested, a difference of 1 percent. On the behaviour the tool predicts deformation class 0, a berm that does not reshape and a crest that does not lower. After the 100-year condition at LAT the berm was almost unchanged, the crest lowering by about 0.08 m, which is roughly one fifth of a rock diameter on a berm of more than 2 m high, while the seaward and side slopes kept their shape.

The comparison is also evidence that the tool is correct, and it is independent of the evidence from Project A. There the load model was shown to reproduce a professional calculation to within 10 percent on the same input. Here the design that follows from that load model is shown to behave as predicted when it is built and loaded in a laboratory. The two checks test different parts of the chain, the load and the resulting design, and they point the same way.

After the design condition the test and the tool no longer agree. The test went on with the same wave, but at higher water levels and for much longer. Over the whole series the crest lost about 0.5 m. That is much more than the small change after the design condition, and more than the tool would give. There are two reasons for this.

The first reason is the water level. At LAT the water is too shallow for such a high wave, so the wave breaks before it reaches the berm and is smaller when it arrives. At a higher water level the same wave does not break and hits the berm in full. The tool uses the same wave height at every water level. It only lowers the water speed at the bed when the water gets deeper. So it always takes LAT as the worst case. For a berm where the waves break, that is wrong.

The second reason is how long the storm lasts. The conditions that caused the damage ran five times longer than the design condition. The tool only looks at the strongest moment of the storm. It does not count how long the storm goes on, so it cannot see damage that grows slowly over time.

The tool has neither of these two things. Both are picked up in the Discussion chapter, and together they are the clearest way to make the tool better.

8.3.4 Project C

Project C is a future project on a bedrock seabed. KP 1 has a water depth of 13.34 m at LAT. The significant wave height is 7.9 m with a peak period of 13.8 s and the current is about 1.0 m/s at the 100-year return period. With a ratio of wave height to water depth of 0.59 the waves are depth-limited and breaking, so this berm sits in the breaker zone. The cable has an outer diameter of 0.15 m and the required cover over it is 0.9 m. The berm has a side slope of 1:4 and a crest width of 1.0 m. This is the smallest berm of the three and it lies in the shallowest water.

The route of Project C is also the most varied of the three. It runs from 13.34 m at KP 1 out to about 21 m at the far end, while the wave height stays close to 7.9 m along it. The ratio of wave height to water depth therefore drops from 0.59 to about 0.38. The waves break over the shallow part of the route but no longer break over the deeper part, so the two halves of the route are in different regimes. Because this project has not been built yet, the tool is used here to advise instead of to check.

Two designs are on the table. They differ in the rock density, and that choice sets everything else. A denser rock is more stable, so a smaller grading can be used, and a smaller grading gives a thinner armour layer and a lower berm. The berm height is built up from the cable, the filter layer and the armour layer, where the armour layer is 2.5 times the nominal rock size.

	High-density design	Normal-density design
Rock density	3100 kg/m ³	2650 kg/m ³
Grading	LMA 10/60	LMA 40/200
Rock Dn50	0.24 m	0.36 m
Cable	0.15 m	0.15 m
Filter layer	0.30 m	0.30 m
Armour layer, 2.5 Dn50	0.60 m	0.90 m
Berm height	1.05 m	1.35 m

Table 8.9: Project C, the two designs that are on the table.

1. The rock.

For the high-density design the tool recommends a rock of 0.223 m, against the 0.24 m of the LMA 10/60 grading, so the proposed rock is 8 percent above what is needed. For the normal-density design the tool recommends 0.345 m, against the 0.36 m of the LMA 40/200 grading, so that rock is 4 percent above what is needed. Both proposed gradings are therefore on the safe side of the tool.

2. The berm. The two designs behave differently.

The high-density design comes out as deformation class 0. The Shields parameter is 0.0345, the mobility on the flat top is 0.63 and on the 1:4 side slope it reaches 0.92, so both stay below the threshold of 0.95. The berm does not reshape, the crest does not drop and the cover of 0.90 m is kept in full at its minimum height of 1.05 m. No extra height is needed. The berm is about 16.9 ton per meter.

The normal-density design comes out as deformation class 1. The Shields parameter is 0.0380 and the mobility on the flat top is 0.69, but on the 1:4 side slope it reaches 1.01 and passes the threshold, so the side slopes reshape. Built at its height of 1.35 m the cover drops from 1.20 m to 0.85 m during the storm, which is 5 cm short of the required 0.90 m. Along the route two of the twelve points lose their cover. As proposed, this design does not quite meet the requirement, KP 1 is one of them.

	Rock Dn50	Class	Installed height	Cover after storm	Tonnage
High-density design	0.24 m	0	1.05 m	0.90 m, kept	16.9 t/m
Normal-density design	0.36 m	1	1.35 m	0.85 m, short of the 0.90 m	22.9 t/m

Table 8.10: Project C, the two designs as the tool sees them.

8.3.4.1 High-density design

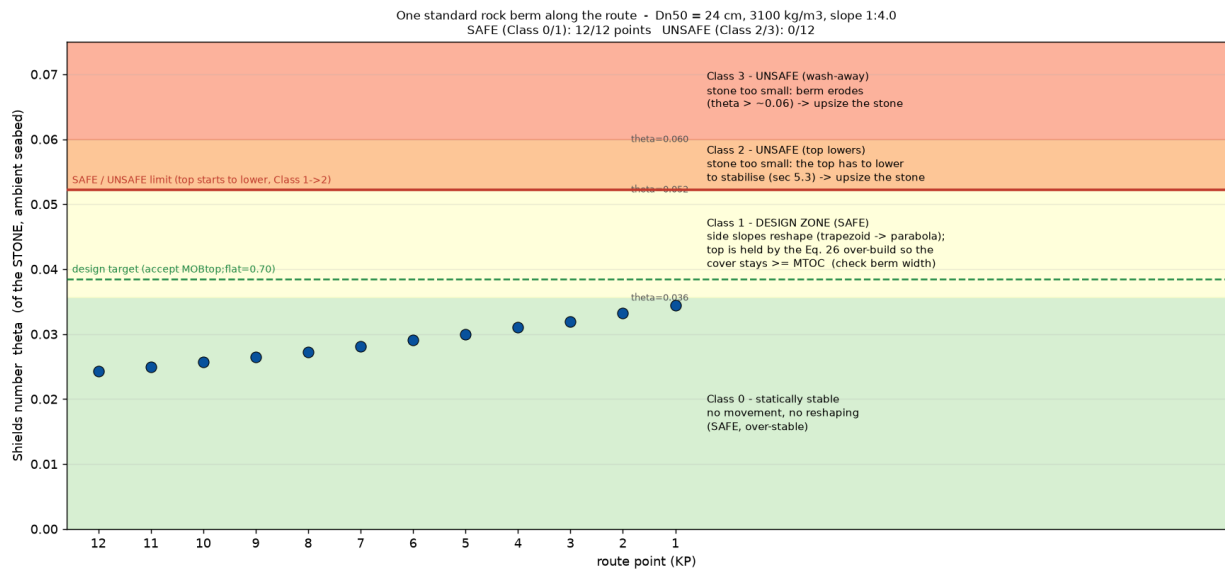


Figure 8.8: deformation class along the route for the high-density design of Project C.

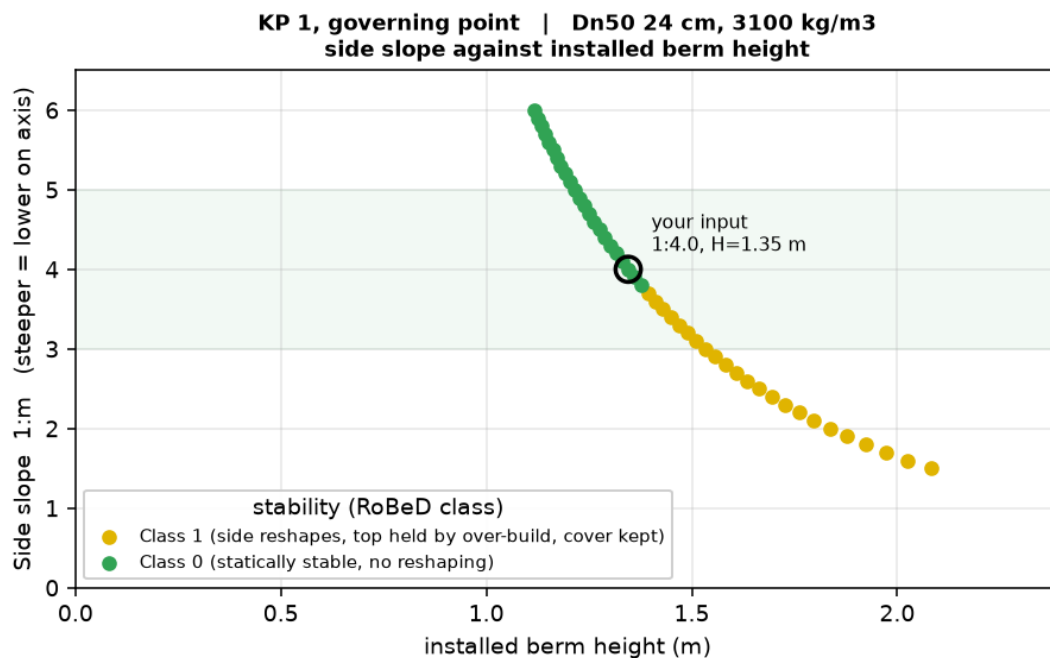


Figure 8.9: side slope against installed berm height for the high-density design of Project C, with the proposed design marked.

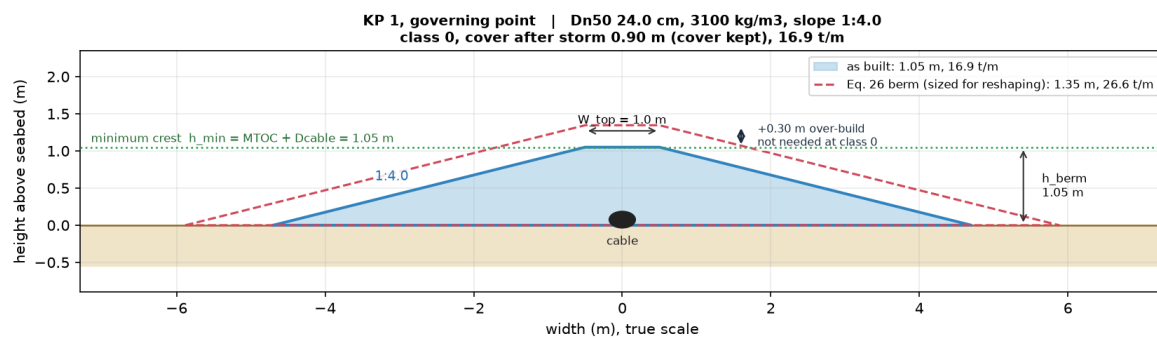


Figure 8.10: cross-section of the high-density design of Project C at KP 1. Eq. 5.36 returns 1.35 m, because it sizes the berm on the assumption that it reshapes. The class check says this berm does not reshape, so the over-build is not needed and the berm is built at 1.05 m.

8.3.4.2 Normal-density design

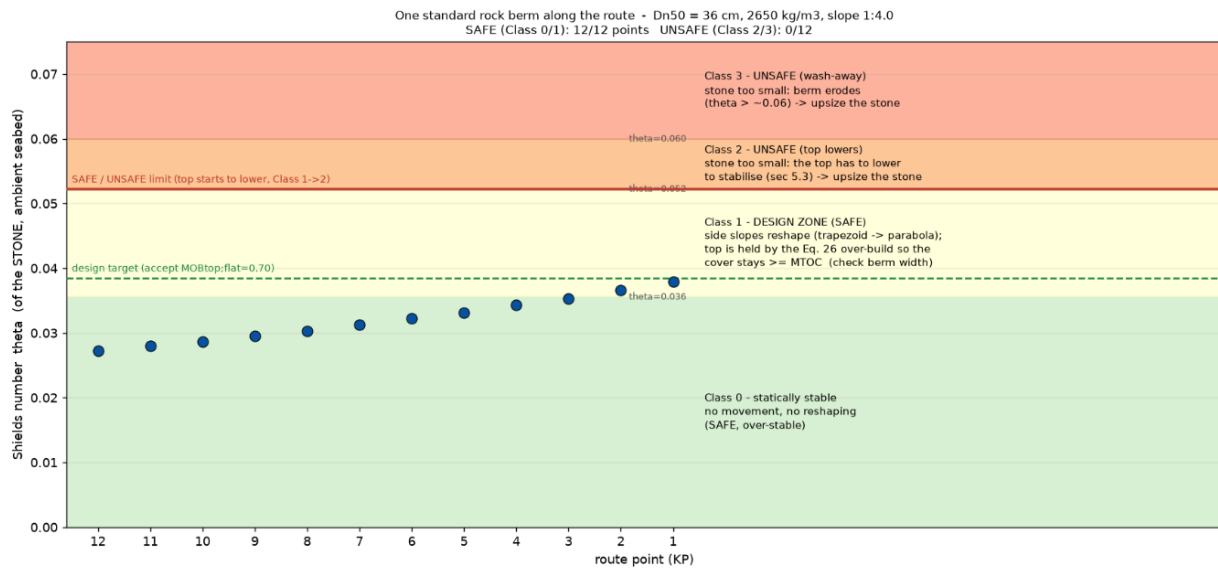


Figure 8.11: deformation class along the route for the normal-density design of Project C.

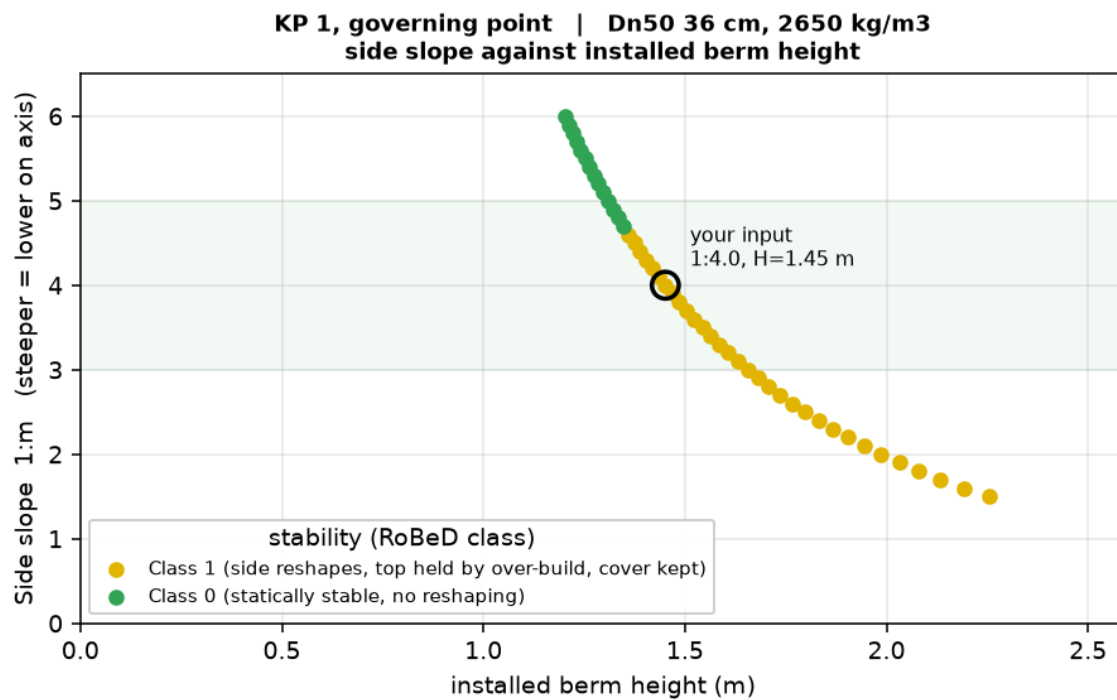


Figure 8.12: side slope against installed berm height for the normal-density design of Project C, with the proposed design marked.

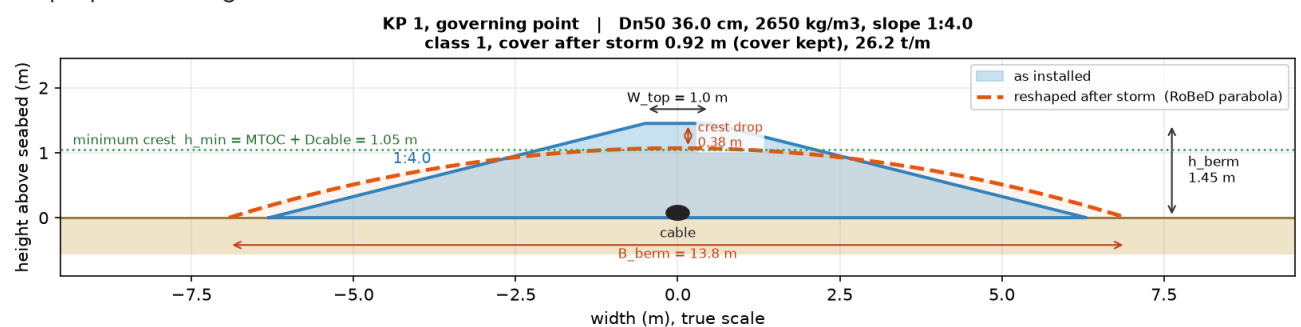


Figure 8.13: cross-section before and after the storm for the normal-density design of Project C, at KP1.

8.3.4.3 Optimisation of design project C

Only two gradings are available, so the rock is fixed. What is still open is the side slope, the crest width, and whether the berm can be built lower than Eq. 5.36 gives. Class 1 is accepted (Section 7.5.3), so the best design is the one with the fewest tonnes per meter that still keeps the 0.90 m cover at every point of the route.

The normal-density design must be built 0.10 m higher.

At the proposed 1.35 m it ends the storm with 0.85 m of cover, 5 cm short. Eq. 5.36 gives 1.45 m at this slope. Built to that height the crest still drops 0.38 m, but it ends at 0.92 m and the requirement is met everywhere. The volume goes from 22.9 to 26.2 tonnes per meter. Those 3.3 tonnes are the price of the reshaping.

The shape only changes the volume once the berm is stable.

Side slope	LMA 10/60 h_berm	class	built	LMA 40/200 h_berm	class	built
1:3	1.53 m	1	1.53 m, 26.6 t/m	1.66 m	1	1.66 m, 26.2 t/m
1:3.5	1.43 m	1	1.43 m, 26.6 t/m	1.54 m	1	1.54 m, 26.2 t/m
1:4	1.35 m	0	1.05 m, 16.9 t/m	1.45 m	1	1.45 m, 26.2 t/m
1:4.5	1.27 m	0	1.05 m, 18.6 t/m	1.37 m	1	1.37 m, 26.2 t/m
1:5	1.21 m	0	1.05 m, 20.3 t/m	1.31 m	0	1.35 m, 27.7 t/m
1:6	1.12 m	0	1.05 m, 23.8 t/m	1.20 m	0	1.35 m, 32.6 t/m

Table 8.11: the berm at KP 1 for both gradings, crest width 1.0 m. The middle columns give the height that Eq. 5.36 returns, the last column gives the berm that is built after the class check. The Eq. 5.36 area is 8.58 m² for LMA 10/60 and 9.88 m² for LMA 40/200 at every side slope.

Before the class check. The height from Eq. 5.36 falls as the slope gets milder, but the tonnage does not move. That follows from the equations themselves. A_{min} in Eq. 5.35 holds only $f(\theta)$ and h_{min} , so neither the side slope nor the crest width is in it, and Eq. 5.36 then solves h_{berm} so that A_{berm} of Eq. 5.34 equals A_{min} . Physically the berm reshapes to one parabola that depends on the Shields number alone, so whatever trapezoid is built must hold the rock of that same parabola. The slope only sets how far the crest falls to get there. The tool takes h_{min} 2 cm larger than Eq. 5.33 so the reshaped cover lands just above MTOC instead of exactly on it, which raises every Eq. 5.36 berm by about 4 percent.

After the class check. The last column is a different berm. Where the class is 0 the berm does not reshape, so the over-build is not needed and it is built at the minimum buildable height instead. That is where the tonnage moves, and that is where the whole saving of this section sits.

So the two do not contradict each other. While the berm reshapes the shape is not worth optimizing, because Eq. 5.35 fixes the volume. The moment it no longer reshapes the height becomes a real design variable, and then a steeper slope is genuinely lighter. A lighter berm is therefore not found by changing the shape. The slope and the crest width should be set on the footprint, on what the vessel can place, and on the placing tolerance the crest must absorb. The crest width of 1.0 m follows from that tolerance and is kept.

**Project C: rock needed per metre for the two available gradings
crest width 1.0 m, cover of 0.90 m kept at every route point**

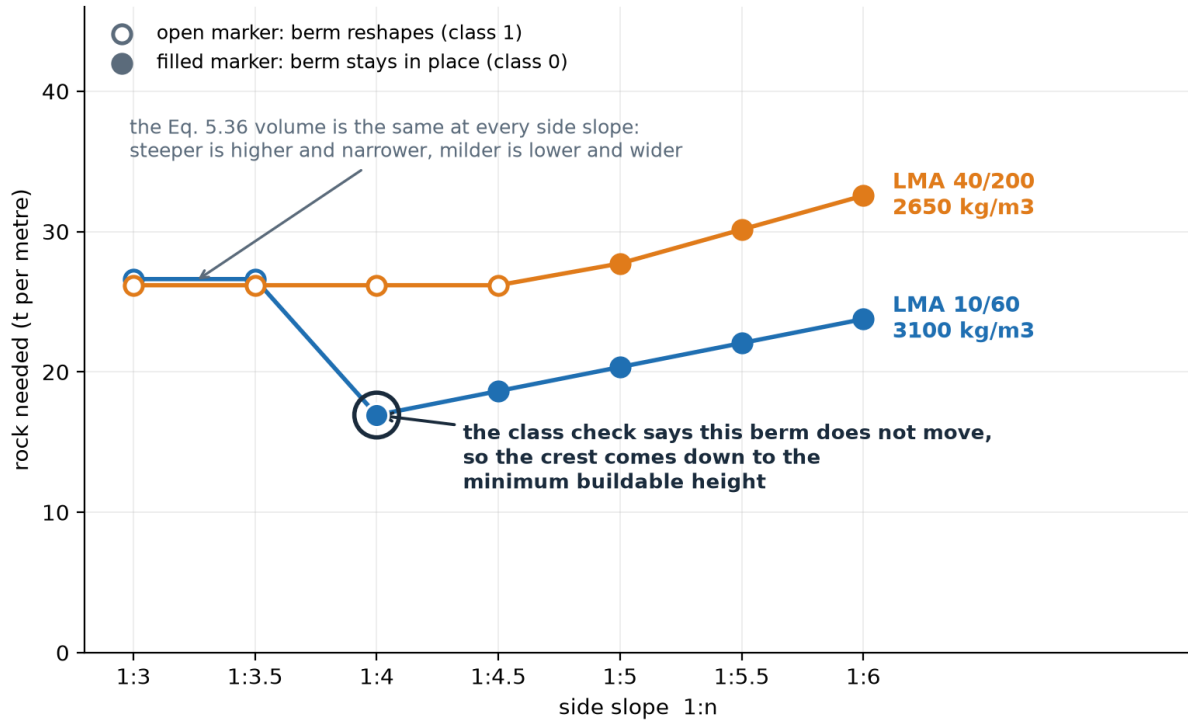


Figure 8.14: rock needed per metre of route against the side slope, for both gradings, at a crest width of 1.0 m. The flat parts are the volume from Eq. 5.36, which does not depend on the slope. The step down in the high-density curve is the height released by the class check.

	Becomes stable at	h_{berm} there	Minimum buildable	Released
High-density	1:3.75	1.385 m	1.05 m	0.335 m
Normal-density	1:4.64	1.355 m	1.35 m	0.005 m

Table 8.12: the height released by the class check, at the slope where each berm first becomes stable.

Eq. 5.36 gives a lower berm as the slope gets milder, but the minimum buildable height stays where it is. That minimum is set by the rock itself, because the armour layer is 2.5 Dn50. For LMA 10/60 that is 0.60 m and the floor is 1.05 m. For LMA 40/200 the armour layer alone is 0.90 m and the floor is 1.35 m, so this berm cannot physically be built lower than that.

The high-density berm becomes stable at 1:3.75, while Eq. 5.36 still asks for 1.385 m, so 0.335 m of over-build comes off. The normal-density berm becomes stable only at 1:4.64, where Eq. 5.36 has already fallen to 1.355 m against its own floor of 1.350 m. There is nothing left to release. Milder still and Eq. 5.36 drops below the floor, so the berm has to be built higher than the equation asks while its base keeps widening, and the volume rises to 27.7 t/m at 1:5 and 32.6 t/m at 1:6. That is the upward kink in Figure 8.14.

While both berms reshape they need almost the same mass, 26.6 against 26.2 tonnes per metre. The larger rock needs a 15 percent larger berm, but its rock is 15 percent lighter per cubic metre, and the two cancel. This is also why the two curves lie on top of each other on the left of Figure 8.14. The difference between the gradings appears only once the class check lets one of them come down.

In short, the large rock is too big to build the small berm that its own stability would allow. This is what the higher density buys. At 3100 kg/m³ the berm becomes stable without the rock having to grow, so the floor stays at 1.05 m while Eq. 5.36 stays well above it.

The design rule.

Two steps follow, and they are the part of this section that carries over to other projects.

1. Find the steepest side slope at which the berm is still class 0. While it reshapes there is no point in optimizing the shape, because Eq. 5.35 fixes the volume.
2. At that slope, build at the minimum buildable height, not at the height Eq. 5.36 gives. The over-build in that height is for reshaping that no longer happens.

The height is then fixed, so a steeper slope really is lighter and the two steps pull the same way. On Project C this gives 1:4 at 1.05 m and 16.9 tonnes per meter.

How close to the threshold to sit is a project choice

Side slope	MOB _{top} ;slope	Class	Rock at 1.05 m	Margin to 0.95
1:3.75	0.950	1	not feasible, cover 0.68 m	0.000
1:3.8	0.944	0	16.2 t/m	0.006
1:3.9	0.931	0	16.6 t/m	0.019
1:4	0.920	0	16.9 t/m	0.030
1:4.5	0.873	0	18.6 t/m	0.077

Table 8.13: material against margin for the high-density design at 1.05 m.

The rule points at the class boundary, but a design should not sit on it. Going from 1:4 to 1:3.8 saves 0.7 tonnes per meter, 4 percent, and cuts the margin by a factor of five. That is a poor trade and 1:4 is kept. Where to sit on this ladder is not for the tool to decide. It depends on how much risk the project accepts, on how well the wave and current data at that location are known, and on what a repair would cost if the berm did move. A long route through well-surveyed water can sit closer to the threshold than a single crossing with a short hindcast record.

Angular rock

Even at 1:4 the design sits at 0.92 against the limit of 0.95. That is close. If the berm did move after all it has no spare height left, so the crest would drop to about 0.85 m and the cover would go to 0.70 m instead of the required 0.90 m.

There is one way to add safety here that does not add rock. RoBeD does not count the interlocking between the stones. The only stone-to-stone resistance in the model is the friction in the slope factor (Eq. 5.30), and the model treats the rock as loose. Angular, well-graded rock hooks into itself, so the berm is stronger than the number says. Rounded rock does not do that.

Asking for angular quarry rock in the specification therefore costs nothing in tonnes, changes nothing in the design, and gives back part of the margin that is thin here. It works because it is a procurement choice, made before the vessel sails. How the rock lands on the seabed is not a choice in the same way, because that follows from the water depth, the vessel and the weather on the day, so it cannot be counted on as a design measure.

High density comparison

High-density rock costs more per ton, about 30 percent. But you need much less of it, 16.9 against 26.2 tonnes per meter, so 35 percent fewer tonnes. Placing fewer tonnes is usually cheaper as well, because the vessel places rock by the load and the time follows the tonnage on board and not the rock size. Fewer tonnes means fewer loads and fewer trips.

So the two work against each other and there is no general answer. On tonnage alone the high-density design stays the cheaper one to buy until high-density rock costs more than $26.2 / 16.9 = 1.55$ times the normal price, so 55 percent more. At the 30 percent quoted here it is already the cheaper option on material, and the installation saving comes on top of that.

That number covers only the rock. The full comparison has to be made per project and together with the subcontractors, because it needs the real price per ton and the real placing rate of the vessel in tonnes per day, and those are confidential or not known yet. What the tool gives is the tonnage of every design that works, and that is the number both the price and the vessel rate are multiplied by. The tool does the engineering part, the project does the price part.

Recommended design

Both alternatives are already close to the sweet spot, which is a berm that is just stable and uses as few tonnes as possible.

The high-density design could even go a little lower. Table 8.13 shows that at a side slope of 1:3.8 it is still class 0 and needs 16.2 tonnes per meter instead of 16.9. That saves 4 percent, but it cuts the margin to the limit from 0.030 to 0.006, so it is not advised. At 1:4 and 1.05 m the design sits where it should.

The normal-density design must go up a little. At the proposed 1.35 m it is 5 cm short of the required cover, so it has to be built at 1.45 m, which brings it to 26.2 tonnes per meter. On the side slope and the crest width there is little left to gain, as this section has shown, so both designs are fine as they stand for construction.

On the side slope and the crest width there is little left to gain, as this section has shown, so both designs are fine as they stand for construction. Their cross-sections are given in Figure 8.10 and Figure 8.13.

Which of the two is finally built should be decided on the cost picture. That needs the price of both rock gradings and the price of the subcontractors who will place the berm. Until both are known the comparison cannot be closed, and it should be made per project rather than settled once for all projects.

One opportunity for follow-up

The design uses the wave height and the current speed without their direction, so the load is applied as if the waves always come in head-on. The hindcast record behind the route already holds the direction of every sea state. Filtering it for the large storm events and plotting a wave rose and a current rose would show where the severe loading comes from. Where the route can still be shifted it can be laid closer to that direction, and where the alignment is fixed the angle can still lower the design load. That is the same saving taken through the load instead of through the rock, and it would buy back exactly the margin that Table 8.13 shows is tight.

8.4 Sensitivity analyses

8.5 Code with angles

Calculate with all the different angles to how much it is different

9. Knowledge Gaps & Research Questions

- Based on linear wave theory, when we implement nonlinear terms our near-bed orbital velocity would increase, and in addition the excess water pressure would be more critical because the wave-crest would be steeper. We have to compensate this in our damage and friction numbers.
- No common damage axis. The methods express damage in different quantities, a reduction coefficient for Van Rijn, a damage number for Van Gent and Herrera and Medina, and a deformation class for Deltares. There is no validated relation that converts one into another. Agreeing or deriving such a mapping is the central open point of this thesis.
- RoBeD calibrated offshore, applied nearshore. The RoBeD reshaping model is validated for offshore conditions and assumes a wave and current direction perpendicular to the structure. Its use in the nearshore breaking zone, and for cable berms specifically, has not been validated. Nearshore validation is needed before it can be relied on.
- Winnowing threshold not verified. The rule that the berm height should be at least four rock layers to prevent winnowing is a rule of thumb from Verheij et al. (2012) that the handbook notes still needs field verification for rock berms.
- Scour, not calculated with 3D model.
- RoBeD polynomial from a limited dataset. The reshaping relation $B_{\text{berm}}/h_{\text{berm}}$ as a function of the Shields parameter comes from Roulund et al. (2018b) and a limited set of physical model tests. Its applicability across the full range of nearshore cable berm conditions is not established.
- Anchor and trawling loads not computed. These loads are treated as a boundary condition only, following a per project risk assessment rather than a universal figure. Bringing them into the calculation where they govern is left for future work.
- The core sustainable philosophy of this thesis is to permit controlled deformation (Class 1 or 2) in order to optimise material quantities and reduce the carbon footprint. In the current framework, the Step 1 stability check and the Step 2 deformation model are coupled through a single stone size, and the two steps apply different loads. Step 1 sizes the stone against a conservative load, and for Van Rijn this includes the turbulence factor g_{str} under currents up to 1.68 m/s, while Step 2 (RoBeD) applies a milder load without this factor. Because the Step 1 stone is already large enough to resist the conservative peak shear, its mobility in Step 2 stays well below the wash away limit and the RoBeD result falls into Class 0. The accepted damage inputs of Step 1 have only weak leverage on this, because they enter the stone size through a power of about 0.2 and the reduction coefficient r is bounded at 1.0. To reach genuine sustainability optimisation, a future improvement to the code should decouple the two steps with a feedback loop. The stone diameter from Step 1 would be deliberately reduced step by step into the active

reshaping regime (Class 1 or 2), and the tool would then check whether the resulting wider, flatter parabolic footprint uses less material overall than the Class 0 installation, while still keeping the required cover over the cable after reshaping.